```
src/context-selector.ts:1   export type SelectorSummary = {
src/context-selector.ts:2     selectorKind: "global.task" | "global.user_profile" | "global.user_profile.path" | "global.human_review.current" |
                            "global.human_review.current.comment" | "global.human_review.current.round" | "global.human_review.current.previous_output" |
                            "global.human_review.current.previous_output.path" | "direct" | "direct.data.path" | "source" | "unsupported";
src/context-selector.ts:3     sourceRoleId?: string;
src/context-selector.ts:4     validPath: boolean;
src/context-selector.ts:5     optional?: boolean;
src/context-selector.ts:6   };
src/context-selector.ts:7
src/context-selector.ts:8   export function listContextSelectorCandidates(args: {
src/context-selector.ts:9     incomingRoleIds?: string[];
src/context-selector.ts:10    joinMode?: string;
src/context-selector.ts:11    joinMin?: number;
src/context-selector.ts:12  } = {}): string[] {
src/context-selector.ts:13    const selectors = new Set([
src/context-selector.ts:14      "global.task", "global.user_profile", "global.human_review.current.comment",
src/context-selector.ts:15      "global.human_review.current.round", "global.human_review.current.previous_output.content"
src/context-selector.ts:16    ]);
src/context-selector.ts:17    const sources = Array.from(new Set(args.incomingRoleIds || []));
src/context-selector.ts:18    if (args.joinMode) {
src/context-selector.ts:19      if (args.joinMode !== "quorum_of" || Number(args.joinMin) >= sources.length) {
src/context-selector.ts:20        for (const source of sources) {
src/context-selector.ts:21          selectors.add(`source(${source}).content`);
src/context-selector.ts:22          selectors.add(`source(${source}).event`);
src/context-selector.ts:23          selectors.add(`source(${source}).data`);
src/context-selector.ts:24        }
src/context-selector.ts:25      }
src/context-selector.ts:26    } else if (sources.length) {
src/context-selector.ts:27      selectors.add("direct.content");
src/context-selector.ts:28      selectors.add("direct.event");
src/context-selector.ts:29      selectors.add("direct.data");
src/context-selector.ts:30    }
src/context-selector.ts:31    return Array.from(selectors).sort((left, right) => left.localeCompare(right));
src/context-selector.ts:32  }
src/context-selector.ts:33
src/context-selector.ts:34  export function isValidSelectorPath(path: string): boolean {
src/context-selector.ts:35    const isSegmentCharacter = (character: string) => {
src/context-selector.ts:36      const code = character.charCodeAt(0);
src/context-selector.ts:37      return (code >= 48 && code <= 57) || (code >= 65 && code <= 90) || (code >= 97 && code <= 122) || character === "_";
src/context-selector.ts:38    };
src/context-selector.ts:39    return Boolean(path) && path.split(".").every((segment) => segment.length > 0 && Array.from(segment).every(isSegmentCharacter));
src/context-selector.ts:40  }
src/context-selector.ts:41
src/context-selector.ts:42  export function isValidSourceRoleId(roleId: string): boolean {
src/context-selector.ts:43    const isCharacterAllowed = (character: string) => {
src/context-selector.ts:44      const code = character.charCodeAt(0);
src/context-selector.ts:45      return (code >= 48 && code <= 57) || (code >= 65 && code <= 90) || (code >= 97 && code <= 122) || "_.:-".includes(character);
src/context-selector.ts:46    };
src/context-selector.ts:47    return Boolean(roleId) && Array.from(roleId).every(isCharacterAllowed);
src/context-selector.ts:48  }
src/context-selector.ts:49
src/context-selector.ts:50  export function summarizeContextSelector(selector: string): SelectorSummary {
```

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src/context-selector.ts:51      const optional = selector.endsWith("?");
src/context-selector.ts:52      const value = optional ? selector.slice(0, -1) : selector;
src/context-selector.ts:53      if (optional && !value.startsWith("global.human_review.current")) return { selectorKind: "unsupported", validPath: false, optional };
src/context-selector.ts:54      if (value === "global.task" || value === "global.user_profile") return { selectorKind: value, validPath: true, optional };
src/context-selector.ts:55      if (value.startsWith("global.user_profile.")) return { selectorKind: "global.user_profile.path", validPath:
                                isValidSelectorPath(value.slice("global.user_profile.".length)), optional };
src/context-selector.ts:56      if (value === "global.human_review.current") return { selectorKind: value, validPath: true, optional };
src/context-selector.ts:57      if (value === "global.human_review.current.comment" || value === "global.human_review.current.round" || value ===
                                "global.human_review.current.previous_output") return { selectorKind: value, validPath: true, optional };
src/context-selector.ts:58      if (value.startsWith("global.human_review.current.previous_output.")) return { selectorKind:
                                "global.human_review.current.previous_output.path", validPath:
                                isValidSelectorPath(value.slice("global.human_review.current.previous_output.".length)), optional };
src/context-selector.ts:59      if (["direct.content", "direct.event", "direct.data"].includes(value)) return { selectorKind: "direct", validPath: true, optional };
src/context-selector.ts:60      if (value.startsWith("direct.data.")) return { selectorKind: "direct.data.path", validPath:
                                isValidSelectorPath(value.slice("direct.data.".length)), optional };
src/context-selector.ts:61      const sourceEnd = value.startsWith("source(") ? value.indexOf(").") : -1;
src/context-selector.ts:62      if (sourceEnd > 7) {
src/context-selector.ts:63        const sourceRoleId = value.slice(7, sourceEnd);
src/context-selector.ts:64        const sourceValue = value.slice(sourceEnd + 2);
src/context-selector.ts:65        const validSourceValue = ["content", "event", "data"].includes(sourceValue)
src/context-selector.ts:66          || (sourceValue.startsWith("data.") && isValidSelectorPath(sourceValue.slice("data.".length)));
src/context-selector.ts:67        if (isValidSourceRoleId(sourceRoleId) && validSourceValue) return { selectorKind: "source", sourceRoleId, validPath: true, optional
                                };
src/context-selector.ts:68      }
src/context-selector.ts:69      return { selectorKind: "unsupported", validPath: false, optional };
src/context-selector.ts:70    }
src/context-selector.ts:71
src/nl2mmd/catalog.ts:1       /**
src/nl2mmd/catalog.ts:2        * @fileoverview NL2MMD catalog/context loader from local OGSystem project metadata.
src/nl2mmd/catalog.ts:3        * File Set: nl2mmd-context
src/nl2mmd/catalog.ts:4        * Responsibilities:
src/nl2mmd/catalog.ts:5        * - Load runtime/law/role/model metadata into one prompt-ready context.
src/nl2mmd/catalog.ts:6        * - Resolve role mentions and expose supported DSL dictionary.
src/nl2mmd/catalog.ts:7        * Boundaries:
src/nl2mmd/catalog.ts:8        * - Does not call LLMs or generate Mermaid drafts.
src/nl2mmd/catalog.ts:9        */
src/nl2mmd/catalog.ts:10      import { readdir } from "node:fs/promises";
src/nl2mmd/catalog.ts:11      import { resolve } from "node:path";
src/nl2mmd/catalog.ts:12
src/nl2mmd/catalog.ts:13      import {
src/nl2mmd/catalog.ts:14        DEFAULT_ROLE_REPO,
src/nl2mmd/catalog.ts:15        resolveProjectRoleRootDir,
src/nl2mmd/catalog.ts:16        resolveTemplateRoleRootDir
src/nl2mmd/catalog.ts:17      } from "../runtime/bundled-repos.js";
src/nl2mmd/catalog.ts:18      import {
src/nl2mmd/catalog.ts:19        validateLawsConfig,
src/nl2mmd/catalog.ts:20        validateRuntimeConfig
src/nl2mmd/catalog.ts:21      } from "../runtime/config.js";
src/nl2mmd/catalog.ts:22      import { readJsonFile } from "../runtime/json-file.js";
src/nl2mmd/catalog.ts:23      import { chooseDefaultModelFromCatalog, loadModelCatalog } from "../runtime/model-catalog.js";
src/nl2mmd/catalog.ts:24      import { isDirectModelRef, loadModelSelection } from "../runtime/model-selection.js";
src/nl2mmd/catalog.ts:25      import { loadRolePackage } from "../runtime/role-repo.js";
src/nl2mmd/catalog.ts:26      import { logNl2MmdDebug } from "./logger.js";
src/nl2mmd/catalog.ts:27      import type { Nl2MmdContext, Nl2MmdRoleMention, Nl2MmdSupportedDictionary } from "./types.js";
src/nl2mmd/catalog.ts:28
src/nl2mmd/catalog.ts:29      function isFileNotFoundError(error: unknown): error is NodeJS.ErrnoException {
```

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src/nl2mmd/catalog.ts:30                  return (
src/nl2mmd/catalog.ts:31                    typeof error === "object" &&
src/nl2mmd/catalog.ts:32                    error !== null &&
src/nl2mmd/catalog.ts:33                    "code" in error &&
src/nl2mmd/catalog.ts:34                    (error as { code?: unknown }).code === "ENOENT"
src/nl2mmd/catalog.ts:35                  );
src/nl2mmd/catalog.ts:36                }
src/nl2mmd/catalog.ts:37
src/nl2mmd/catalog.ts:38                function isMissingPathError(error: unknown, path: string): boolean {
src/nl2mmd/catalog.ts:39                  return (
src/nl2mmd/catalog.ts:40                    isFileNotFoundError(error) &&
src/nl2mmd/catalog.ts:41                    typeof (error as { path?: unknown }).path === "string" &&
src/nl2mmd/catalog.ts:42                    (error as { path: string }).path === path
src/nl2mmd/catalog.ts:43                  );
src/nl2mmd/catalog.ts:44                }
src/nl2mmd/catalog.ts:45
src/nl2mmd/catalog.ts:46                async function readJsonFileIfExists(path: string): Promise<unknown | undefined> {
src/nl2mmd/catalog.ts:47                  try {
src/nl2mmd/catalog.ts:48                    return await readJsonFile(path);
src/nl2mmd/catalog.ts:49                  } catch (error) {
src/nl2mmd/catalog.ts:50                    if (isFileNotFoundError(error)) {
src/nl2mmd/catalog.ts:51                      return undefined;
src/nl2mmd/catalog.ts:52                    }
src/nl2mmd/catalog.ts:53                    throw error;
src/nl2mmd/catalog.ts:54                  }
src/nl2mmd/catalog.ts:55                }
src/nl2mmd/catalog.ts:56
src/nl2mmd/catalog.ts:57                function compact<T>(items: Array<T | undefined>): T[] {
src/nl2mmd/catalog.ts:58                  return items.filter((item): item is T => item !== undefined);
src/nl2mmd/catalog.ts:59                }
src/nl2mmd/catalog.ts:60
src/nl2mmd/catalog.ts:61                async function listDirectoryNamesIfExists(path: string): Promise<string[]> {
src/nl2mmd/catalog.ts:62                  try {
src/nl2mmd/catalog.ts:63                    return (await readdir(path, { withFileTypes: true }))
src/nl2mmd/catalog.ts:64                      .filter((entry) => entry.isDirectory())
src/nl2mmd/catalog.ts:65                      .map((entry) => entry.name)
src/nl2mmd/catalog.ts:66                      .sort();
src/nl2mmd/catalog.ts:67                  } catch (error) {
src/nl2mmd/catalog.ts:68                    if (isFileNotFoundError(error)) {
src/nl2mmd/catalog.ts:69                      return [];
src/nl2mmd/catalog.ts:70                    }
src/nl2mmd/catalog.ts:71                    throw error;
src/nl2mmd/catalog.ts:72                  }
src/nl2mmd/catalog.ts:73                }
src/nl2mmd/catalog.ts:74
src/nl2mmd/catalog.ts:75                function dedupeByKey<T>(items: T[], keyFor: (item: T) => string): T[] {
src/nl2mmd/catalog.ts:76                  const seen = new Set<string>();
src/nl2mmd/catalog.ts:77                  const deduped: T[] = [];
src/nl2mmd/catalog.ts:78                  for (const item of items) {
src/nl2mmd/catalog.ts:79                    const key = keyFor(item);
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src/nl2mmd/catalog.ts:80                    if (seen.has(key)) {
src/nl2mmd/catalog.ts:81                      continue;
src/nl2mmd/catalog.ts:82                    }
src/nl2mmd/catalog.ts:83                    seen.add(key);
src/nl2mmd/catalog.ts:84                    deduped.push(item);
src/nl2mmd/catalog.ts:85                  }
src/nl2mmd/catalog.ts:86                  return deduped;
src/nl2mmd/catalog.ts:87                }
src/nl2mmd/catalog.ts:88
src/nl2mmd/catalog.ts:89                function getDefaultRuntimeConfig(path: string) {
src/nl2mmd/catalog.ts:90                  return validateRuntimeConfig(
src/nl2mmd/catalog.ts:91                    {
src/nl2mmd/catalog.ts:92                      executor: "opencode",
src/nl2mmd/catalog.ts:93                      roleRepo: DEFAULT_ROLE_REPO,
src/nl2mmd/catalog.ts:94                      runsDir: ".ogs/runs",
src/nl2mmd/catalog.ts:95                      workspace: {
src/nl2mmd/catalog.ts:96                        rolesDir: "roles",
src/nl2mmd/catalog.ts:97                        privateDirName: "private",
src/nl2mmd/catalog.ts:98                        workspaceIsolation: "role"
src/nl2mmd/catalog.ts:99                      },
src/nl2mmd/catalog.ts:100                     redaction: {
src/nl2mmd/catalog.ts:101                       enabled: true
src/nl2mmd/catalog.ts:102                     },
src/nl2mmd/catalog.ts:103                     opencode: {
src/nl2mmd/catalog.ts:104                       baseArgs: ["run"]
src/nl2mmd/catalog.ts:105                     }
src/nl2mmd/catalog.ts:106                   },
src/nl2mmd/catalog.ts:107                   path
src/nl2mmd/catalog.ts:108                 );
src/nl2mmd/catalog.ts:109               }
src/nl2mmd/catalog.ts:110
src/nl2mmd/catalog.ts:111               function getOutputEvents(schema: unknown): string[] {
src/nl2mmd/catalog.ts:112                 if (typeof schema !== "object" || schema === null || Array.isArray(schema)) {
src/nl2mmd/catalog.ts:113                   return [];
src/nl2mmd/catalog.ts:114                 }
src/nl2mmd/catalog.ts:115                 const record = schema as {
src/nl2mmd/catalog.ts:116                   properties?: {
src/nl2mmd/catalog.ts:117                     event?: {
src/nl2mmd/catalog.ts:118                       enum?: unknown;
src/nl2mmd/catalog.ts:119                     };
src/nl2mmd/catalog.ts:120                   };
src/nl2mmd/catalog.ts:121                 };
src/nl2mmd/catalog.ts:122                 return Array.isArray(record.properties?.event?.enum)
src/nl2mmd/catalog.ts:123                   ? record.properties.event.enum.filter((item): item is string => typeof item === "string")
src/nl2mmd/catalog.ts:124                   : [];
src/nl2mmd/catalog.ts:125               }
src/nl2mmd/catalog.ts:126
src/nl2mmd/catalog.ts:127               export function getSupportedNl2MmdDictionary(): Nl2MmdSupportedDictionary {
src/nl2mmd/catalog.ts:128                 return {
src/nl2mmd/catalog.ts:129                   flowcharts: ["flowchart TD", "flowchart LR"],
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src/nl2mmd/catalog.ts:130                    boundaryTokens: ["input", "output"],
src/nl2mmd/catalog.ts:131                    exactMetadataKeys: [
src/nl2mmd/catalog.ts:132                      "engine",
src/nl2mmd/catalog.ts:133                      "system.id",
src/nl2mmd/catalog.ts:134                      "system.version",
src/nl2mmd/catalog.ts:135                      "law.global",
src/nl2mmd/catalog.ts:136                      "entry.role",
src/nl2mmd/catalog.ts:137                      "handoff.mode",
src/nl2mmd/catalog.ts:138                      "handoff.contracts"
src/nl2mmd/catalog.ts:139                    ],
src/nl2mmd/catalog.ts:140                    metadataPrefixes: [
src/nl2mmd/catalog.ts:141                      "exec.bind.",
src/nl2mmd/catalog.ts:142                      "role.mode.",
src/nl2mmd/catalog.ts:143                      "join.mode.",
src/nl2mmd/catalog.ts:144                      "join.min.",
src/nl2mmd/catalog.ts:145                      "join.sources.",
src/nl2mmd/catalog.ts:146                      "context.map.",
src/nl2mmd/catalog.ts:147                      "loop.max.",
src/nl2mmd/catalog.ts:148                      "route.order.",
src/nl2mmd/catalog.ts:149                      "review.mode.",
src/nl2mmd/catalog.ts:150                      "review.timeout.",
src/nl2mmd/catalog.ts:151                      "review.timeout.action.",
src/nl2mmd/catalog.ts:152                      "review.rework.target.",
src/nl2mmd/catalog.ts:153                      "review.rework.max.",
src/nl2mmd/catalog.ts:154                      "review.terminate.scope."
src/nl2mmd/catalog.ts:155                    ],
src/nl2mmd/catalog.ts:156                    roleModes: ["parallel_split"],
src/nl2mmd/catalog.ts:157                    joinModes: ["all_of", "quorum_of"],
src/nl2mmd/catalog.ts:158                    mentionPrefix: "@",
src/nl2mmd/catalog.ts:159                    nodeTokenPattern: "nodeId[Role:roleId]",
src/nl2mmd/catalog.ts:160                    edgePattern: "fromToken -->|EVENT| toToken (tokens: input/output or nodeId[Role:roleId])"
src/nl2mmd/catalog.ts:161                  };
src/nl2mmd/catalog.ts:162                }
src/nl2mmd/catalog.ts:163
src/nl2mmd/catalog.ts:164                function splitModelRef(modelRef: string): { provider: string; model: string } {
src/nl2mmd/catalog.ts:165                  const separator = modelRef.indexOf("/");
src/nl2mmd/catalog.ts:166                  if (separator <= 0 || separator === modelRef.length - 1) {
src/nl2mmd/catalog.ts:167                    return {
src/nl2mmd/catalog.ts:168                      provider: "unknown",
src/nl2mmd/catalog.ts:169                      model: modelRef
src/nl2mmd/catalog.ts:170                    };
src/nl2mmd/catalog.ts:171                  }
src/nl2mmd/catalog.ts:172                  return {
src/nl2mmd/catalog.ts:173                    provider: modelRef.slice(0, separator),
src/nl2mmd/catalog.ts:174                    model: modelRef.slice(separator + 1)
src/nl2mmd/catalog.ts:175                  };
src/nl2mmd/catalog.ts:176                }
src/nl2mmd/catalog.ts:177
src/nl2mmd/catalog.ts:178                function gatherSelectionModelRefs(context: Nl2MmdContext["modelSelection"]): string[] {
src/nl2mmd/catalog.ts:179                  if (!context) {
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src/nl2mmd/catalog.ts:180                    return [];
src/nl2mmd/catalog.ts:181                  }
src/nl2mmd/catalog.ts:182                  return [
src/nl2mmd/catalog.ts:183                    context.defaults?.backend && context.defaults.modelId ? `${context.defaults.backend}/${context.defaults.modelId}` : undefined,
src/nl2mmd/catalog.ts:184                    ...Object.values(context.roles ?? {}).map((entry) => entry.backend && entry.modelId ? `${entry.backend}/${entry.modelId}` :
                           undefined)
src/nl2mmd/catalog.ts:185                  ].filter((value): value is string => isDirectModelRef(value));
src/nl2mmd/catalog.ts:186                }
src/nl2mmd/catalog.ts:187
src/nl2mmd/catalog.ts:188                export async function loadNl2MmdContext(args: {
src/nl2mmd/catalog.ts:189                  workdir: string;
src/nl2mmd/catalog.ts:190                  runtimeConfigPath?: string;
src/nl2mmd/catalog.ts:191                  lawsPath?: string;
src/nl2mmd/catalog.ts:192                }): Promise<Nl2MmdContext> {
src/nl2mmd/catalog.ts:193                  const startedAt = Date.now();
src/nl2mmd/catalog.ts:194                  const runtimePath = args.runtimeConfigPath ?? resolve(args.workdir, ".ogs", "runtime.json");
src/nl2mmd/catalog.ts:195                  const runtimeConfigSource = await readJsonFileIfExists(runtimePath);
src/nl2mmd/catalog.ts:196                  const runtimeConfig =
src/nl2mmd/catalog.ts:197                    runtimeConfigSource !== undefined
src/nl2mmd/catalog.ts:198                      ? validateRuntimeConfig(runtimeConfigSource, runtimePath)
src/nl2mmd/catalog.ts:199                      : getDefaultRuntimeConfig(runtimePath);
src/nl2mmd/catalog.ts:200
src/nl2mmd/catalog.ts:201                  const roleRootDir = resolveProjectRoleRootDir(args.workdir, runtimeConfig.roleRepo);
src/nl2mmd/catalog.ts:202                  const templateRoleRootDir = resolveTemplateRoleRootDir();
src/nl2mmd/catalog.ts:203                  const modelCatalogPath = resolve(args.workdir, ".ogs", "model-catalog.json");
src/nl2mmd/catalog.ts:204                  const modelSelectionPath = resolve(args.workdir, ".ogs", "model-selection.json");
src/nl2mmd/catalog.ts:205                  const rawModelCatalog = await loadModelCatalog(modelCatalogPath);
src/nl2mmd/catalog.ts:206                  const modelSelection = await loadModelSelection(modelSelectionPath);
src/nl2mmd/catalog.ts:207
src/nl2mmd/catalog.ts:208                  const roleEntries = dedupeByKey(
src/nl2mmd/catalog.ts:209                    [
src/nl2mmd/catalog.ts:210                      ...(await listDirectoryNamesIfExists(roleRootDir)),
src/nl2mmd/catalog.ts:211                      ...(await listDirectoryNamesIfExists(templateRoleRootDir))
src/nl2mmd/catalog.ts:212                    ].filter((name) => !name.startsWith("_")),
src/nl2mmd/catalog.ts:213                    (name) => name
src/nl2mmd/catalog.ts:214                  );
src/nl2mmd/catalog.ts:215
src/nl2mmd/catalog.ts:216                  const roleCatalog = compact(
src/nl2mmd/catalog.ts:217                    await Promise.all(
src/nl2mmd/catalog.ts:218                      roleEntries.map(async (roleId) => {
src/nl2mmd/catalog.ts:219                        const rootsToTry = [roleRootDir, templateRoleRootDir];
src/nl2mmd/catalog.ts:220                        try {
src/nl2mmd/catalog.ts:221                          for (const candidateRoot of rootsToTry) {
src/nl2mmd/catalog.ts:222                            const manifestPath = resolve(candidateRoot, roleId, "role.json");
src/nl2mmd/catalog.ts:223                            try {
src/nl2mmd/catalog.ts:224                              const rolePackage = await loadRolePackage({ roleId, roleRootDir: candidateRoot });
src/nl2mmd/catalog.ts:225                              return {
src/nl2mmd/catalog.ts:226                                roleId,
src/nl2mmd/catalog.ts:227                                name: rolePackage.manifest.name,
src/nl2mmd/catalog.ts:228                                description: rolePackage.manifest.description,
src/nl2mmd/catalog.ts:229                                tags: rolePackage.manifest.tags ?? [],
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src/nl2mmd/catalog.ts:230                                preferredModelTags: rolePackage.manifest.preferredModelTags ?? [],
src/nl2mmd/catalog.ts:231                                outputEvents: getOutputEvents(rolePackage.outputSchema)
src/nl2mmd/catalog.ts:232                              };
src/nl2mmd/catalog.ts:233                            } catch (error) {
src/nl2mmd/catalog.ts:234                              if (isMissingPathError(error, manifestPath)) {
src/nl2mmd/catalog.ts:235                                continue;
src/nl2mmd/catalog.ts:236                              }
src/nl2mmd/catalog.ts:237                              throw error;
src/nl2mmd/catalog.ts:238                            }
src/nl2mmd/catalog.ts:239                          }
src/nl2mmd/catalog.ts:240                          return undefined;
src/nl2mmd/catalog.ts:241                        } catch (error) {
src/nl2mmd/catalog.ts:242                          throw error;
src/nl2mmd/catalog.ts:243                        }
src/nl2mmd/catalog.ts:244                      })
src/nl2mmd/catalog.ts:245                  )
src/nl2mmd/catalog.ts:246                );
src/nl2mmd/catalog.ts:247
src/nl2mmd/catalog.ts:248                const modelCatalog = dedupeByKey(
src/nl2mmd/catalog.ts:249                  [
src/nl2mmd/catalog.ts:250                    ...(rawModelCatalog?.models.map((entry) => ({
src/nl2mmd/catalog.ts:251                      modelRef: entry.ref,
src/nl2mmd/catalog.ts:252                      provider: entry.provider,
src/nl2mmd/catalog.ts:253                      model: entry.model,
src/nl2mmd/catalog.ts:254                      name: entry.name,
src/nl2mmd/catalog.ts:255                      status: entry.status,
src/nl2mmd/catalog.ts:256                      reasoningEffort: undefined,
src/nl2mmd/catalog.ts:257                      variants: entry.variants,
src/nl2mmd/catalog.ts:258                      tags: [
src/nl2mmd/catalog.ts:259                        ...(entry.status ? [entry.status] : []),
src/nl2mmd/catalog.ts:260                        ...(entry.capabilities.toolcall ? ["toolcall"] : []),
src/nl2mmd/catalog.ts:261                        ...(entry.capabilities.textInput ? ["text-input"] : []),
src/nl2mmd/catalog.ts:262                        ...(entry.capabilities.textOutput ? ["text-output"] : [])
src/nl2mmd/catalog.ts:263                      ]
src/nl2mmd/catalog.ts:264                    })) ?? []),
src/nl2mmd/catalog.ts:265                    ...gatherSelectionModelRefs(modelSelection).map((modelRef) => {
src/nl2mmd/catalog.ts:266                      const { provider, model } = splitModelRef(modelRef);
src/nl2mmd/catalog.ts:267                      return {
src/nl2mmd/catalog.ts:268                        modelRef,
src/nl2mmd/catalog.ts:269                        provider,
src/nl2mmd/catalog.ts:270                        model,
src/nl2mmd/catalog.ts:271                        name: undefined,
src/nl2mmd/catalog.ts:272                        status: undefined,
src/nl2mmd/catalog.ts:273                        reasoningEffort: undefined,
src/nl2mmd/catalog.ts:274                        variants: [],
src/nl2mmd/catalog.ts:275                        tags: ["selection"]
src/nl2mmd/catalog.ts:276                      };
src/nl2mmd/catalog.ts:277                    })
src/nl2mmd/catalog.ts:278                  ],
src/nl2mmd/catalog.ts:279                  (item) => item.modelRef
```

```
src/nl2mmd/catalog.ts:280                ).sort((left, right) => left.modelRef.localeCompare(right.modelRef));
src/nl2mmd/catalog.ts:281
src/nl2mmd/catalog.ts:282                const defaultCatalogModel = rawModelCatalog ? chooseDefaultModelFromCatalog(rawModelCatalog) : undefined;
src/nl2mmd/catalog.ts:283                const defaultModelRef = modelSelection?.defaults?.backend && modelSelection.defaults.modelId
src/nl2mmd/catalog.ts:284                  ? `${modelSelection.defaults.backend}/${modelSelection.defaults.modelId}`
src/nl2mmd/catalog.ts:285                  : defaultCatalogModel?.ref ?? modelCatalog[0]?.modelRef;
src/nl2mmd/catalog.ts:286                const defaultModelVariant = modelSelection?.defaults?.variant;
src/nl2mmd/catalog.ts:287                const defaultTimeoutMs = modelSelection?.defaults?.timeoutMs ?? 120000;
src/nl2mmd/catalog.ts:288                const defaultMaxOutputBytes = modelSelection?.defaults?.maxOutputBytes ?? 65536;
src/nl2mmd/catalog.ts:289
src/nl2mmd/catalog.ts:290                let lawIds: string[] = [];
src/nl2mmd/catalog.ts:291                const lawsPath = args.lawsPath ?? resolve(args.workdir, ".ogs", "laws.json");
src/nl2mmd/catalog.ts:292                const lawsSource = await readJsonFileIfExists(lawsPath);
src/nl2mmd/catalog.ts:293                if (lawsSource !== undefined) {
src/nl2mmd/catalog.ts:294                  const laws = validateLawsConfig(lawsSource, lawsPath);
src/nl2mmd/catalog.ts:295                  lawIds = laws.laws.map((item) => item.lawId).sort();
src/nl2mmd/catalog.ts:296                }
src/nl2mmd/catalog.ts:297
src/nl2mmd/catalog.ts:298                const context = {
src/nl2mmd/catalog.ts:299                  workdir: args.workdir,
src/nl2mmd/catalog.ts:300                  roleRootDir,
src/nl2mmd/catalog.ts:301                  templateRoleRootDir,
src/nl2mmd/catalog.ts:302                  roleCatalog,
src/nl2mmd/catalog.ts:303                  modelCatalog,
src/nl2mmd/catalog.ts:304                  rawModelCatalog,
src/nl2mmd/catalog.ts:305                  modelSelection,
src/nl2mmd/catalog.ts:306                  defaultModelRef,
src/nl2mmd/catalog.ts:307                  defaultModelVariant,
src/nl2mmd/catalog.ts:308                  defaultTimeoutMs,
src/nl2mmd/catalog.ts:309                  defaultMaxOutputBytes,
src/nl2mmd/catalog.ts:310                  lawIds,
src/nl2mmd/catalog.ts:311                  supportedDictionary: getSupportedNl2MmdDictionary()
src/nl2mmd/catalog.ts:312                };
src/nl2mmd/catalog.ts:313                logNl2MmdDebug("context.loaded", {
src/nl2mmd/catalog.ts:314                  workdir: args.workdir,
src/nl2mmd/catalog.ts:315                  runtimePath,
src/nl2mmd/catalog.ts:316                  runtimeConfigSource: runtimeConfigSource !== undefined ? "file" : "default",
src/nl2mmd/catalog.ts:317                  modelCatalogPath,
src/nl2mmd/catalog.ts:318                  modelCatalogLoaded: rawModelCatalog !== undefined,
src/nl2mmd/catalog.ts:319                  modelSelectionPath,
src/nl2mmd/catalog.ts:320                  modelSelectionLoaded: modelSelection !== undefined,
src/nl2mmd/catalog.ts:321                  defaultModelRef: defaultModelRef ?? "(none)",
src/nl2mmd/catalog.ts:322                  lawsPath,
src/nl2mmd/catalog.ts:323                  lawsLoaded: lawsSource !== undefined,
src/nl2mmd/catalog.ts:324                  roleCount: roleCatalog.length,
src/nl2mmd/catalog.ts:325                  modelCount: modelCatalog.length,
src/nl2mmd/catalog.ts:326                  durationMs: Date.now() - startedAt
src/nl2mmd/catalog.ts:327                });
src/nl2mmd/catalog.ts:328                return context;
src/nl2mmd/catalog.ts:329              }
```

```
src/nl2mmd/catalog.ts:330
src/nl2mmd/catalog.ts:331               export function extractRoleMentions(text: string): string[] {
src/nl2mmd/catalog.ts:332                 const matches = text.match(/@[A-Za-z0-9._:-]+/g) ?? [];
src/nl2mmd/catalog.ts:333                 return Array.from(new Set(matches.map((item) => item.slice(1))));
src/nl2mmd/catalog.ts:334               }
src/nl2mmd/catalog.ts:335
src/nl2mmd/catalog.ts:336               export function resolveRoleMentions(text: string, context: Nl2MmdContext): Nl2MmdRoleMention[] {
src/nl2mmd/catalog.ts:337                 const known = new Set(context.roleCatalog.map((item) => item.roleId));
src/nl2mmd/catalog.ts:338                 return extractRoleMentions(text).map((roleId) => ({
src/nl2mmd/catalog.ts:339                   mention: `@${roleId}`,
src/nl2mmd/catalog.ts:340                   roleId,
src/nl2mmd/catalog.ts:341                   exists: known.has(roleId)
src/nl2mmd/catalog.ts:342                 }));
src/nl2mmd/catalog.ts:343               }
src/nl2mmd/catalog.ts:344
src/nl2mmd/cli.ts:1                     /**
src/nl2mmd/cli.ts:2                      * @fileoverview Interactive and one-shot CLI for NL2MMD drafting workflow.
src/nl2mmd/cli.ts:3                      * File Set: nl2mmd-entry
src/nl2mmd/cli.ts:4                      * Responsibilities:
src/nl2mmd/cli.ts:5                      * - Manage REPL/session lifecycle and user commands.
src/nl2mmd/cli.ts:6                      * - Run NL2MMD turns, show suggestions, and print validation feedback.
src/nl2mmd/cli.ts:7                      * Boundaries:
src/nl2mmd/cli.ts:8                      * - Delegates generation/validation details to NL2MMD service modules.
src/nl2mmd/cli.ts:9                      */
src/nl2mmd/cli.ts:10                    import { stdin as input, stdout as output } from "node:process";
src/nl2mmd/cli.ts:11                    import { resolve } from "node:path";
src/nl2mmd/cli.ts:12                    import { parseArgs } from "node:util";
src/nl2mmd/cli.ts:13                    import { fileURLToPath } from "node:url";
src/nl2mmd/cli.ts:14                    import { createInterface } from "node:readline/promises";
src/nl2mmd/cli.ts:15
src/nl2mmd/cli.ts:16                    import { isDirectModelRef } from "../runtime/model-selection.js";
src/nl2mmd/cli.ts:17                    import {
src/nl2mmd/cli.ts:18                      createNl2MmdConversation,
src/nl2mmd/cli.ts:19                      detectSemanticHints,
src/nl2mmd/cli.ts:20                      loadNl2MmdContext,
src/nl2mmd/cli.ts:21                      runNl2MmdPreflight,
src/nl2mmd/cli.ts:22                      runNl2MmdTurn,
src/nl2mmd/cli.ts:23                      searchModels,
src/nl2mmd/cli.ts:24                      searchRoles,
src/nl2mmd/cli.ts:25                      validateNl2MmdCandidate
src/nl2mmd/cli.ts:26                    } from "./index.js";
src/nl2mmd/cli.ts:27                    import { syncProjectDependencies } from "../runtime/project-lifecycle.js";
src/nl2mmd/cli.ts:28                    import type { Nl2MmdConversation, Nl2MmdContext, Nl2MmdTurnResult } from "./types.js";
src/nl2mmd/cli.ts:29
src/nl2mmd/cli.ts:30                    export function usage(): string {
src/nl2mmd/cli.ts:31                      return [
src/nl2mmd/cli.ts:32                        "Usage:",
src/nl2mmd/cli.ts:33                        "  ogs nl2mmd [--message <text>] [--model <provider/model>]",
src/nl2mmd/cli.ts:34                        "",
src/nl2mmd/cli.ts:35                        "Base command:",
```

```
src/nl2mmd/cli.ts:36                    "  direct entrypoint for prompt generation and validation against local roles plus .ogs model-selection/model-catalog",
src/nl2mmd/cli.ts:37                    "",
src/nl2mmd/cli.ts:38                    "Source repository equivalent:",
src/nl2mmd/cli.ts:39                    "  pnpm run run:nl2mmd -- [--message <text>] [--model <provider/model>]",
src/nl2mmd/cli.ts:40                    "",
src/nl2mmd/cli.ts:41                    "Options:",
src/nl2mmd/cli.ts:42                    "  --message <text>       One-shot NL2MMD request; omit for interactive mode",
src/nl2mmd/cli.ts:43                    "  --model <provider/model>  Direct default model ref (overrides .ogs/model-selection.json defaults)",
src/nl2mmd/cli.ts:44                    "  --runtime <file>       Runtime config JSON (optional)",
src/nl2mmd/cli.ts:45                    "  --laws <file>          Laws JSON (optional)",
src/nl2mmd/cli.ts:46                    "  --profiles <file>      Profiles JSON for exec.bind validation (optional)",
src/nl2mmd/cli.ts:47                    "  --user-profile <file>  User profile JSON for validation (optional)",
src/nl2mmd/cli.ts:48                    "  --no-preflight         Skip startup preflight (default is preflight enabled)",
src/nl2mmd/cli.ts:49                    "  --ogs-dir <path>       OGS control-plane root (default: cwd)",
src/nl2mmd/cli.ts:50                    "  --help                 Show help",
src/nl2mmd/cli.ts:51                    "",
src/nl2mmd/cli.ts:52                    "Defaults:",
src/nl2mmd/cli.ts:53                    "  workdir defaults to the current directory",
src/nl2mmd/cli.ts:54                    "  model defaults to .ogs/model-selection.json, then .ogs/model-catalog.json",
src/nl2mmd/cli.ts:55                    "  preflight runs before the first turn unless disabled",
src/nl2mmd/cli.ts:56                    "",
src/nl2mmd/cli.ts:57                    "Interactive commands:",
src/nl2mmd/cli.ts:58                    "  /help                  Show commands",
src/nl2mmd/cli.ts:59                    "  /roles <query>         Search role repo",
src/nl2mmd/cli.ts:60                    "  /models <query>        Search local model catalog",
src/nl2mmd/cli.ts:61                    "  /laws                  List discovered law ids",
src/nl2mmd/cli.ts:62                    "  /use-model <provider/model>   Switch the conversation model",
src/nl2mmd/cli.ts:63                    "  /status                Show current draft/model/session status",
src/nl2mmd/cli.ts:64                    "  /validate              Re-run local validation for the current Mermaid draft",
src/nl2mmd/cli.ts:65                    "  /clear                 Clear current draft/validation state",
src/nl2mmd/cli.ts:66                    "  /quit                  Exit"
src/nl2mmd/cli.ts:67                  ].join("\n");
src/nl2mmd/cli.ts:68                }
src/nl2mmd/cli.ts:69
src/nl2mmd/cli.ts:70                function printSection(title: string, body?: string): void {
src/nl2mmd/cli.ts:71                  console.log(`\n${title}`);
src/nl2mmd/cli.ts:72                  if (body) {
src/nl2mmd/cli.ts:73                    console.log(body);
src/nl2mmd/cli.ts:74                  }
src/nl2mmd/cli.ts:75                }
src/nl2mmd/cli.ts:76
src/nl2mmd/cli.ts:77                function formatRoleSearch(context: Nl2MmdContext, query: string): string {
src/nl2mmd/cli.ts:78                  const matches = searchRoles(context, query, 8);
src/nl2mmd/cli.ts:79                  if (matches.length === 0) {
src/nl2mmd/cli.ts:80                    return "(no role matches)";
src/nl2mmd/cli.ts:81                  }
src/nl2mmd/cli.ts:82                  return matches
src/nl2mmd/cli.ts:83                    .map(
src/nl2mmd/cli.ts:84                      (item) =>
src/nl2mmd/cli.ts:85                        `${item.item.roleId} | ${item.item.name} | events=${item.item.outputEvents.join(",") || "-"} | ${item.reason}`
```

```
src/nl2mmd/cli.ts:86                    )
src/nl2mmd/cli.ts:87                    .join("\n");
src/nl2mmd/cli.ts:88                }
src/nl2mmd/cli.ts:89
src/nl2mmd/cli.ts:90                function formatModelSearch(context: Nl2MmdContext, query: string): string {
src/nl2mmd/cli.ts:91                  const matches = searchModels(context, query, 8);
src/nl2mmd/cli.ts:92                  if (matches.length === 0) {
src/nl2mmd/cli.ts:93                    return "(no model matches)";
src/nl2mmd/cli.ts:94                  }
src/nl2mmd/cli.ts:95                  return matches
src/nl2mmd/cli.ts:96                    .map(
src/nl2mmd/cli.ts:97                      (item) =>
src/nl2mmd/cli.ts:98                        `${item.item.modelRef} | ${item.item.model} | variants=${item.item.variants.join(",") || "-"} | ${item.reason}`
src/nl2mmd/cli.ts:99                    )
src/nl2mmd/cli.ts:100                   .join("\n");
src/nl2mmd/cli.ts:101               }
src/nl2mmd/cli.ts:102
src/nl2mmd/cli.ts:103               function formatSemanticHints(message: string): string {
src/nl2mmd/cli.ts:104                 const hints = detectSemanticHints(message);
src/nl2mmd/cli.ts:105                 if (hints.length === 0) {
src/nl2mmd/cli.ts:106                   return "(no fixed semantic hints detected)";
src/nl2mmd/cli.ts:107                 }
src/nl2mmd/cli.ts:108                 return hints.map((item) => `${item.label} | ${item.detail}`).join("\n");
src/nl2mmd/cli.ts:109               }
src/nl2mmd/cli.ts:110
src/nl2mmd/cli.ts:111               function printSuggestions(context: Nl2MmdContext, message: string): void {
src/nl2mmd/cli.ts:112                 printSection("Detected Semantic Hints", formatSemanticHints(message));
src/nl2mmd/cli.ts:113                 printSection("Role Suggestions", formatRoleSearch(context, message));
src/nl2mmd/cli.ts:114                 printSection("Model Suggestions", formatModelSearch(context, message));
src/nl2mmd/cli.ts:115               }
src/nl2mmd/cli.ts:116
src/nl2mmd/cli.ts:117               function printValidation(turn: Nl2MmdTurnResult): void {
src/nl2mmd/cli.ts:118                 if (!turn.validation) {
src/nl2mmd/cli.ts:119                   return;
src/nl2mmd/cli.ts:120                 }
src/nl2mmd/cli.ts:121                 const lines = [
src/nl2mmd/cli.ts:122                   `status=${turn.validation.status}`,
src/nl2mmd/cli.ts:123                   `errors=${turn.validation.errors.length}`,
src/nl2mmd/cli.ts:124                   `warnings=${turn.validation.warnings.length}`
src/nl2mmd/cli.ts:125                 ];
src/nl2mmd/cli.ts:126                 printSection("Validation", lines.join("\n"));
src/nl2mmd/cli.ts:127                 if (turn.validation.errors.length > 0) {
src/nl2mmd/cli.ts:128                   printSection("Validation Errors", turn.validation.errors.join("\n"));
src/nl2mmd/cli.ts:129                 }
src/nl2mmd/cli.ts:130                 if (turn.validation.warnings.length > 0) {
src/nl2mmd/cli.ts:131                   printSection("Validation Warnings", turn.validation.warnings.join("\n"));
src/nl2mmd/cli.ts:132                 }
src/nl2mmd/cli.ts:133               }
src/nl2mmd/cli.ts:134
src/nl2mmd/cli.ts:135               function printTurn(turn: Nl2MmdTurnResult): void {
```

```
src/nl2mmd/cli.ts:136                 printSection("Mode", turn.mode);
src/nl2mmd/cli.ts:137                 printSection("Summary", turn.summary);
src/nl2mmd/cli.ts:138
src/nl2mmd/cli.ts:139                 if (turn.questions.length > 0) {
src/nl2mmd/cli.ts:140                   printSection("Questions", turn.questions.map((item, index) => `${index + 1}. ${item}`).join("\n"));
src/nl2mmd/cli.ts:141                 }
src/nl2mmd/cli.ts:142                 if (turn.assumptions.length > 0) {
src/nl2mmd/cli.ts:143                   printSection("Assumptions", turn.assumptions.join("\n"));
src/nl2mmd/cli.ts:144                 }
src/nl2mmd/cli.ts:145                 if (turn.unresolvedItems.length > 0) {
src/nl2mmd/cli.ts:146                   printSection("Unresolved", turn.unresolvedItems.join("\n"));
src/nl2mmd/cli.ts:147                 }
src/nl2mmd/cli.ts:148                 if (turn.txtGraph) {
src/nl2mmd/cli.ts:149                   printSection("Txt Graph", turn.txtGraph);
src/nl2mmd/cli.ts:150                 }
src/nl2mmd/cli.ts:151                 if (turn.mermaid.trim()) {
src/nl2mmd/cli.ts:152                   printSection("Mermaid", turn.mermaid);
src/nl2mmd/cli.ts:153                 }
src/nl2mmd/cli.ts:154                 printValidation(turn);
src/nl2mmd/cli.ts:155               }
src/nl2mmd/cli.ts:156
src/nl2mmd/cli.ts:157               export async function main(): Promise<void> {
src/nl2mmd/cli.ts:158                 const { values } = parseArgs({
src/nl2mmd/cli.ts:159                   options: {
src/nl2mmd/cli.ts:160                     message: { type: "string" },
src/nl2mmd/cli.ts:161                     model: { type: "string" },
src/nl2mmd/cli.ts:162                     runtime: { type: "string" },
src/nl2mmd/cli.ts:163                     laws: { type: "string" },
src/nl2mmd/cli.ts:164                     profiles: { type: "string" },
src/nl2mmd/cli.ts:165                     "user-profile": { type: "string" },
src/nl2mmd/cli.ts:166                     "no-preflight": { type: "boolean" },
src/nl2mmd/cli.ts:167                     "ogs-dir": { type: "string" },
src/nl2mmd/cli.ts:168                     help: { type: "boolean", short: "h" }
src/nl2mmd/cli.ts:169                   },
src/nl2mmd/cli.ts:170                   allowPositionals: false
src/nl2mmd/cli.ts:171                 });
src/nl2mmd/cli.ts:172
src/nl2mmd/cli.ts:173                 if (values.help) {
src/nl2mmd/cli.ts:174                   console.log(usage());
src/nl2mmd/cli.ts:175                   return;
src/nl2mmd/cli.ts:176                 }
src/nl2mmd/cli.ts:177
src/nl2mmd/cli.ts:178                 const workdir = values["ogs-dir"] ?? process.cwd();
src/nl2mmd/cli.ts:179                 const context = await loadNl2MmdContext({
src/nl2mmd/cli.ts:180                   workdir,
src/nl2mmd/cli.ts:181                   runtimeConfigPath: values.runtime,
src/nl2mmd/cli.ts:182                   lawsPath: values.laws
src/nl2mmd/cli.ts:183                 });
src/nl2mmd/cli.ts:184
src/nl2mmd/cli.ts:185                 const initialModelRef = values.model ?? context.defaultModelRef ?? context.modelCatalog[0]?.modelRef;
```

```
src/nl2mmd/cli.ts:186                    if (!initialModelRef || !isDirectModelRef(initialModelRef)) {
src/nl2mmd/cli.ts:187                      throw new Error(
src/nl2mmd/cli.ts:188                        "No usable NL2MMD model ref found. Set --model <provider/model> or configure .ogs/model-selection.json."
src/nl2mmd/cli.ts:189                      );
src/nl2mmd/cli.ts:190                    }
src/nl2mmd/cli.ts:191                    let modelRef = initialModelRef;
src/nl2mmd/cli.ts:192                    let draftMermaid = "";
src/nl2mmd/cli.ts:193                    let lastTurn: Nl2MmdTurnResult | undefined;
src/nl2mmd/cli.ts:194                    let conversation: Nl2MmdConversation | undefined;
src/nl2mmd/cli.ts:195
src/nl2mmd/cli.ts:196                    async function ensureConversation(): Promise<Nl2MmdConversation> {
src/nl2mmd/cli.ts:197                      if (conversation) {
src/nl2mmd/cli.ts:198                        return conversation;
src/nl2mmd/cli.ts:199                      }
src/nl2mmd/cli.ts:200                      conversation = await createNl2MmdConversation({
src/nl2mmd/cli.ts:201                        workdir,
src/nl2mmd/cli.ts:202                        modelRef,
src/nl2mmd/cli.ts:203                        runtimeConfigPath: values.runtime,
src/nl2mmd/cli.ts:204                        lawsPath: values.laws,
src/nl2mmd/cli.ts:205                        context
src/nl2mmd/cli.ts:206                      });
src/nl2mmd/cli.ts:207                      return conversation;
src/nl2mmd/cli.ts:208                    }
src/nl2mmd/cli.ts:209
src/nl2mmd/cli.ts:210                    async function switchModel(nextModelRef: string): Promise<void> {
src/nl2mmd/cli.ts:211                      if (!isDirectModelRef(nextModelRef)) {
src/nl2mmd/cli.ts:212                        throw new Error(`Invalid model ref "${nextModelRef}". Expected provider/model.`);
src/nl2mmd/cli.ts:213                      }
src/nl2mmd/cli.ts:214                      conversation?.close();
src/nl2mmd/cli.ts:215                      conversation = undefined;
src/nl2mmd/cli.ts:216                      modelRef = nextModelRef;
src/nl2mmd/cli.ts:217                    }
src/nl2mmd/cli.ts:218
src/nl2mmd/cli.ts:219                    async function submit(message: string): Promise<Nl2MmdTurnResult> {
src/nl2mmd/cli.ts:220                      printSuggestions(context, message);
src/nl2mmd/cli.ts:221                      const activeConversation = await ensureConversation();
src/nl2mmd/cli.ts:222                      let turn = await runNl2MmdTurn({
src/nl2mmd/cli.ts:223                        conversation: activeConversation,
src/nl2mmd/cli.ts:224                        input: {
src/nl2mmd/cli.ts:225                          message,
src/nl2mmd/cli.ts:226                          draftMermaid,
src/nl2mmd/cli.ts:227                          validationErrors: lastTurn?.validation?.errors,
src/nl2mmd/cli.ts:228                          validationWarnings: lastTurn?.validation?.warnings
src/nl2mmd/cli.ts:229                        },
src/nl2mmd/cli.ts:230                        lawsPath: values.laws,
src/nl2mmd/cli.ts:231                        profilesPath: values.profiles,
src/nl2mmd/cli.ts:232                        userProfilePath: values["user-profile"]
src/nl2mmd/cli.ts:233                      });
src/nl2mmd/cli.ts:234                      if (turn.mermaid.trim()) {
src/nl2mmd/cli.ts:235                        const syncResult = await syncProjectDependencies({
```

```
src/nl2mmd/cli.ts:236                        workdir,
src/nl2mmd/cli.ts:237                        systemSource: turn.mermaid
src/nl2mmd/cli.ts:238                      });
src/nl2mmd/cli.ts:239                      if (syncResult.importedRoleIds.length > 0 || syncResult.importedModelIds.length > 0) {
src/nl2mmd/cli.ts:240                        printSection(
src/nl2mmd/cli.ts:241                          "Imported Project Dependencies",
src/nl2mmd/cli.ts:242                          [
src/nl2mmd/cli.ts:243                            `roles=${syncResult.importedRoleIds.join(", ") || "-"}`,
src/nl2mmd/cli.ts:244                            `models=${syncResult.importedModelIds.join(", ") || "-"}`
src/nl2mmd/cli.ts:245                          ].join("\n")
src/nl2mmd/cli.ts:246                        );
src/nl2mmd/cli.ts:247                      }
src/nl2mmd/cli.ts:248                      const validation = await validateNl2MmdCandidate({
src/nl2mmd/cli.ts:249                        mermaid: turn.mermaid,
src/nl2mmd/cli.ts:250                        context,
src/nl2mmd/cli.ts:251                        lawsPath: values.laws,
src/nl2mmd/cli.ts:252                        profilesPath: values.profiles,
src/nl2mmd/cli.ts:253                        userProfilePath: values["user-profile"]
src/nl2mmd/cli.ts:254                      });
src/nl2mmd/cli.ts:255                      turn = {
src/nl2mmd/cli.ts:256                        ...turn,
src/nl2mmd/cli.ts:257                        validation,
src/nl2mmd/cli.ts:258                        txtGraph: validation.txtGraph
src/nl2mmd/cli.ts:259                      };
src/nl2mmd/cli.ts:260                      draftMermaid = turn.mermaid;
src/nl2mmd/cli.ts:261                    }
src/nl2mmd/cli.ts:262                    lastTurn = turn;
src/nl2mmd/cli.ts:263                    printTurn(turn);
src/nl2mmd/cli.ts:264                    return turn;
src/nl2mmd/cli.ts:265                  }
src/nl2mmd/cli.ts:266
src/nl2mmd/cli.ts:267                  async function runPreflightIfNeeded(): Promise<void> {
src/nl2mmd/cli.ts:268                    if (values["no-preflight"]) {
src/nl2mmd/cli.ts:269                      return;
src/nl2mmd/cli.ts:270                    }
src/nl2mmd/cli.ts:271                    printSection("Preflight", "checking opencode lifecycle and model subscription...");
src/nl2mmd/cli.ts:272                    const activeConversation = await ensureConversation();
src/nl2mmd/cli.ts:273                    try {
src/nl2mmd/cli.ts:274                      await runNl2MmdPreflight({
src/nl2mmd/cli.ts:275                        conversation: activeConversation
src/nl2mmd/cli.ts:276                      });
src/nl2mmd/cli.ts:277                    } catch (error) {
src/nl2mmd/cli.ts:278                      const detail = error instanceof Error ? error.message : String(error);
src/nl2mmd/cli.ts:279                      throw new Error(
src/nl2mmd/cli.ts:280                        `NL2MMD preflight failed: ${detail}\nHint: verify Opencode account/group subscription and model access permissions.`
src/nl2mmd/cli.ts:281                      );
src/nl2mmd/cli.ts:282                    }
src/nl2mmd/cli.ts:283                    printSection("Preflight", "ok");
src/nl2mmd/cli.ts:284                  }
src/nl2mmd/cli.ts:285
```

```
src/nl2mmd/cli.ts:286                    try {
src/nl2mmd/cli.ts:287                      await runPreflightIfNeeded();
src/nl2mmd/cli.ts:288
src/nl2mmd/cli.ts:289                      if (values.message) {
src/nl2mmd/cli.ts:290                        await submit(values.message);
src/nl2mmd/cli.ts:291                        return;
src/nl2mmd/cli.ts:292                      }
src/nl2mmd/cli.ts:293
src/nl2mmd/cli.ts:294                      console.log("OGSystem NL2MMD interactive CLI");
src/nl2mmd/cli.ts:295                      console.log(`workdir=${workdir}`);
src/nl2mmd/cli.ts:296                      console.log(`model=${modelRef}`);
src/nl2mmd/cli.ts:297                      console.log('Type natural language requirements, or "/help" for commands.');
src/nl2mmd/cli.ts:298
src/nl2mmd/cli.ts:299                      const rl = createInterface({ input, output });
src/nl2mmd/cli.ts:300                      try {
src/nl2mmd/cli.ts:301                        while (true) {
src/nl2mmd/cli.ts:302                          const line = (await rl.question("\nnl2mmd> ")).trim();
src/nl2mmd/cli.ts:303                          if (!line) {
src/nl2mmd/cli.ts:304                            continue;
src/nl2mmd/cli.ts:305                          }
src/nl2mmd/cli.ts:306
src/nl2mmd/cli.ts:307                          if (line === "/quit" || line === "/exit") {
src/nl2mmd/cli.ts:308                            break;
src/nl2mmd/cli.ts:309                          }
src/nl2mmd/cli.ts:310                          if (line === "/help") {
src/nl2mmd/cli.ts:311                            console.log(usage());
src/nl2mmd/cli.ts:312                            continue;
src/nl2mmd/cli.ts:313                          }
src/nl2mmd/cli.ts:314                          if (line === "/laws") {
src/nl2mmd/cli.ts:315                            printSection("Laws", context.lawIds.length > 0 ? context.lawIds.join("\n") : "(none)");
src/nl2mmd/cli.ts:316                            continue;
src/nl2mmd/cli.ts:317                          }
src/nl2mmd/cli.ts:318                          if (line === "/status") {
src/nl2mmd/cli.ts:319                            printSection(
src/nl2mmd/cli.ts:320                              "Status",
src/nl2mmd/cli.ts:321                              [
src/nl2mmd/cli.ts:322                                `model=${modelRef}`,
src/nl2mmd/cli.ts:323                                `session=${conversation?.sessionId ?? "(none)"}`,
src/nl2mmd/cli.ts:324                                `hasDraft=${draftMermaid ? "yes" : "no"}`,
src/nl2mmd/cli.ts:325                                `lastMode=${lastTurn?.mode ?? "(none)"}`
src/nl2mmd/cli.ts:326                              ].join("\n")
src/nl2mmd/cli.ts:327                            );
src/nl2mmd/cli.ts:328                            continue;
src/nl2mmd/cli.ts:329                          }
src/nl2mmd/cli.ts:330                          if (line === "/clear") {
src/nl2mmd/cli.ts:331                            draftMermaid = "";
src/nl2mmd/cli.ts:332                            lastTurn = undefined;
src/nl2mmd/cli.ts:333                            printSection("State", "draft cleared");
src/nl2mmd/cli.ts:334                            continue;
src/nl2mmd/cli.ts:335                          }
```

```
src/nl2mmd/cli.ts:336                        if (line.startsWith("/roles ")) {
src/nl2mmd/cli.ts:337                          printSection("Role Search", formatRoleSearch(context, line.slice("/roles ".length).trim()));
src/nl2mmd/cli.ts:338                          continue;
src/nl2mmd/cli.ts:339                        }
src/nl2mmd/cli.ts:340                        if (line.startsWith("/models ")) {
src/nl2mmd/cli.ts:341                          printSection("Model Search", formatModelSearch(context, line.slice("/models ".length).trim()));
src/nl2mmd/cli.ts:342                          continue;
src/nl2mmd/cli.ts:343                        }
src/nl2mmd/cli.ts:344                        if (line.startsWith("/use-model ")) {
src/nl2mmd/cli.ts:345                          await switchModel(line.slice("/use-model ".length).trim());
src/nl2mmd/cli.ts:346                          printSection("Model", `switched to ${modelRef}`);
src/nl2mmd/cli.ts:347                          continue;
src/nl2mmd/cli.ts:348                        }
src/nl2mmd/cli.ts:349                        if (line === "/validate") {
src/nl2mmd/cli.ts:350                          if (!draftMermaid.trim()) {
src/nl2mmd/cli.ts:351                            printSection("Validation", "no draft Mermaid available");
src/nl2mmd/cli.ts:352                            continue;
src/nl2mmd/cli.ts:353                          }
src/nl2mmd/cli.ts:354                          const validation = await validateNl2MmdCandidate({
src/nl2mmd/cli.ts:355                            mermaid: draftMermaid,
src/nl2mmd/cli.ts:356                            context,
src/nl2mmd/cli.ts:357                            lawsPath: values.laws,
src/nl2mmd/cli.ts:358                            profilesPath: values.profiles,
src/nl2mmd/cli.ts:359                            userProfilePath: values["user-profile"]
src/nl2mmd/cli.ts:360                          });
src/nl2mmd/cli.ts:361                          printSection("Txt Graph", validation.txtGraph);
src/nl2mmd/cli.ts:362                          printSection(
src/nl2mmd/cli.ts:363                            "Validation",
src/nl2mmd/cli.ts:364                            [
src/nl2mmd/cli.ts:365                              `status=${validation.status}`,
src/nl2mmd/cli.ts:366                              `errors=${validation.errors.length}`,
src/nl2mmd/cli.ts:367                              `warnings=${validation.warnings.length}`
src/nl2mmd/cli.ts:368                            ].join("\n")
src/nl2mmd/cli.ts:369                          );
src/nl2mmd/cli.ts:370                          if (validation.errors.length > 0) {
src/nl2mmd/cli.ts:371                            printSection("Validation Errors", validation.errors.join("\n"));
src/nl2mmd/cli.ts:372                          }
src/nl2mmd/cli.ts:373                          if (validation.warnings.length > 0) {
src/nl2mmd/cli.ts:374                            printSection("Validation Warnings", validation.warnings.join("\n"));
src/nl2mmd/cli.ts:375                          }
src/nl2mmd/cli.ts:376                          continue;
src/nl2mmd/cli.ts:377                        }
src/nl2mmd/cli.ts:378
src/nl2mmd/cli.ts:379                        await submit(line);
src/nl2mmd/cli.ts:380                      }
src/nl2mmd/cli.ts:381                    } finally {
src/nl2mmd/cli.ts:382                      rl.close();
src/nl2mmd/cli.ts:383                    }
src/nl2mmd/cli.ts:384                  } finally {
src/nl2mmd/cli.ts:385                    conversation?.close();
```

```
src/nl2mmd/cli.ts:386                    }
src/nl2mmd/cli.ts:387                  }
src/nl2mmd/cli.ts:388
src/nl2mmd/cli.ts:389                  export function handleNl2MmdCliError(error: unknown): void {
src/nl2mmd/cli.ts:390                    console.error(error instanceof Error ? error.stack ?? error.message : String(error));
src/nl2mmd/cli.ts:391                    process.exitCode = 1;
src/nl2mmd/cli.ts:392                  }
src/nl2mmd/cli.ts:393
src/nl2mmd/cli.ts:394                  const isMainModule =
src/nl2mmd/cli.ts:395                    typeof process.argv[1] === "string" && fileURLToPath(import.meta.url) === resolve(process.argv[1]);
src/nl2mmd/cli.ts:396
src/nl2mmd/cli.ts:397                  export async function runNl2MmdCli(args: string[] = process.argv.slice(2)): Promise<void> {
src/nl2mmd/cli.ts:398                    const originalArgv = process.argv;
src/nl2mmd/cli.ts:399                    process.argv = [originalArgv[0] ?? "node", originalArgv[1] ?? "ogs", ...args];
src/nl2mmd/cli.ts:400                    try {
src/nl2mmd/cli.ts:401                      await main();
src/nl2mmd/cli.ts:402                    } catch (error) {
src/nl2mmd/cli.ts:403                      handleNl2MmdCliError(error);
src/nl2mmd/cli.ts:404                    } finally {
src/nl2mmd/cli.ts:405                      process.argv = originalArgv;
src/nl2mmd/cli.ts:406                    }
src/nl2mmd/cli.ts:407                  }
src/nl2mmd/cli.ts:408
src/nl2mmd/cli.ts:409                  if (isMainModule) {
src/nl2mmd/cli.ts:410                    void runNl2MmdCli();
src/nl2mmd/cli.ts:411                  }
src/nl2mmd/cli.ts:412
src/nl2mmd/index.ts:1                  /**
src/nl2mmd/index.ts:2                   * @fileoverview Barrel exports for NL2MMD public APIs and types.
src/nl2mmd/index.ts:3                   * File Set: nl2mmd-entry
src/nl2mmd/index.ts:4                   * Responsibilities:
src/nl2mmd/index.ts:5                   * - Re-export NL2MMD context/search/prompt/service/validation modules.
src/nl2mmd/index.ts:6                   * Boundaries:
src/nl2mmd/index.ts:7                   * - No runtime logic.
src/nl2mmd/index.ts:8                   */
src/nl2mmd/index.ts:9                  export {
src/nl2mmd/index.ts:10                   extractRoleMentions,
src/nl2mmd/index.ts:11                   getSupportedNl2MmdDictionary,
src/nl2mmd/index.ts:12                   loadNl2MmdContext,
src/nl2mmd/index.ts:13                   resolveRoleMentions
src/nl2mmd/index.ts:14                 } from "./catalog.js";
src/nl2mmd/index.ts:15                 export { detectSemanticHints, searchModels, searchRoles } from "./semantic-map.js";
src/nl2mmd/index.ts:16                 export {
src/nl2mmd/index.ts:17                   buildNl2MmdSystemPrompt,
src/nl2mmd/index.ts:18                   buildNl2MmdTurnPrompt,
src/nl2mmd/index.ts:19                   getNl2MmdTurnSchema
src/nl2mmd/index.ts:20                 } from "./prompt.js";
src/nl2mmd/index.ts:21                 export { normalizeNl2MmdMermaid, stabilizeNl2MmdMermaidForRuntime } from "./normalize-mermaid.js";
src/nl2mmd/index.ts:22                 export {
src/nl2mmd/index.ts:23                   getNl2MmdStructureTemplate,
```

```
src/nl2mmd/index.ts:24                    inferNl2MmdStructureTemplate,
src/nl2mmd/index.ts:25                    listNl2MmdStructureTemplates,
src/nl2mmd/index.ts:26                    suggestNl2MmdStructureTemplates
src/nl2mmd/index.ts:27                  } from "./structure-templates.js";
src/nl2mmd/index.ts:28                  export { createNl2MmdConversation, runNl2MmdPreflight, runNl2MmdTurn } from "./service.js";
src/nl2mmd/index.ts:29                  export { renderTxtGraphFromMermaidSource } from "./txt-graph.js";
src/nl2mmd/index.ts:30                  export { validateNl2MmdCandidate } from "./validate.js";
src/nl2mmd/index.ts:31                  export type {
src/nl2mmd/index.ts:32                    Nl2MmdContext,
src/nl2mmd/index.ts:33                    Nl2MmdCatalogSearchResult,
src/nl2mmd/index.ts:34                    Nl2MmdConversation,
src/nl2mmd/index.ts:35                    Nl2MmdModelSummary,
src/nl2mmd/index.ts:36                    Nl2MmdModelResponse,
src/nl2mmd/index.ts:37                    Nl2MmdRoleMention,
src/nl2mmd/index.ts:38                    Nl2MmdRoleSummary,
src/nl2mmd/index.ts:39                    Nl2MmdSemanticHint,
src/nl2mmd/index.ts:40                    Nl2MmdSupportedDictionary,
src/nl2mmd/index.ts:41                    Nl2MmdTurnInput,
src/nl2mmd/index.ts:42                    Nl2MmdTurnMode,
src/nl2mmd/index.ts:43                    Nl2MmdTurnResult,
src/nl2mmd/index.ts:44                    Nl2MmdValidationResult
src/nl2mmd/index.ts:45                  } from "./types.js";
src/nl2mmd/index.ts:46                  export type {
src/nl2mmd/index.ts:47                    Nl2MmdStructureTemplate,
src/nl2mmd/index.ts:48                    Nl2MmdStructureTemplateId,
src/nl2mmd/index.ts:49                    Nl2MmdStructureTemplateMatch,
src/nl2mmd/index.ts:50                    Nl2MmdStructureSlot
src/nl2mmd/index.ts:51                  } from "./structure-templates.js";
src/nl2mmd/index.ts:52
src/nl2mmd/logger.ts:1                  /**
src/nl2mmd/logger.ts:2                   * @fileoverview Lightweight debug logger for NL2MMD workflows.
src/nl2mmd/logger.ts:3                   * File Set: nl2mmd-observability
src/nl2mmd/logger.ts:4                   * Responsibilities:
src/nl2mmd/logger.ts:5                   * - Gate debug logs by environment flag.
src/nl2mmd/logger.ts:6                   * - Sanitize structured fields before printing.
src/nl2mmd/logger.ts:7                   * Boundaries:
src/nl2mmd/logger.ts:8                   * - No persistence or trace aggregation.
src/nl2mmd/logger.ts:9                   */
src/nl2mmd/logger.ts:10                 function isRecord(value: unknown): value is Record<string, unknown> {
src/nl2mmd/logger.ts:11                   return typeof value === "object" && value !== null && !Array.isArray(value);
src/nl2mmd/logger.ts:12                 }
src/nl2mmd/logger.ts:13
src/nl2mmd/logger.ts:14                 function sanitizeFields(fields: Record<string, unknown>): Record<string, unknown> {
src/nl2mmd/logger.ts:15                   const sanitized: Record<string, unknown> = {};
src/nl2mmd/logger.ts:16                   for (const [key, value] of Object.entries(fields)) {
src/nl2mmd/logger.ts:17                     if (value === undefined) {
src/nl2mmd/logger.ts:18                       continue;
src/nl2mmd/logger.ts:19                     }
src/nl2mmd/logger.ts:20                     if (
src/nl2mmd/logger.ts:21                       value === null ||
```

```
src/nl2mmd/logger.ts:22                       typeof value === "string" ||
src/nl2mmd/logger.ts:23                       typeof value === "number" ||
src/nl2mmd/logger.ts:24                       typeof value === "boolean"
src/nl2mmd/logger.ts:25                     ) {
src/nl2mmd/logger.ts:26                       sanitized[key] = value;
src/nl2mmd/logger.ts:27                       continue;
src/nl2mmd/logger.ts:28                     }
src/nl2mmd/logger.ts:29                     if (Array.isArray(value)) {
src/nl2mmd/logger.ts:30                       sanitized[key] = value.slice(0, 8);
src/nl2mmd/logger.ts:31                       continue;
src/nl2mmd/logger.ts:32                     }
src/nl2mmd/logger.ts:33                     if (isRecord(value)) {
src/nl2mmd/logger.ts:34                       sanitized[key] = "[object]";
src/nl2mmd/logger.ts:35                       continue;
src/nl2mmd/logger.ts:36                     }
src/nl2mmd/logger.ts:37                     sanitized[key] = String(value);
src/nl2mmd/logger.ts:38                   }
src/nl2mmd/logger.ts:39                   return sanitized;
src/nl2mmd/logger.ts:40                 }
src/nl2mmd/logger.ts:41
src/nl2mmd/logger.ts:42                 function isNl2MmdDebugEnabled(): boolean {
src/nl2mmd/logger.ts:43                   const value = process.env.OGSYSTEM_NL2MMD_DEBUG?.trim().toLowerCase();
src/nl2mmd/logger.ts:44                   return value === "1" || value === "true" || value === "yes" || value === "on";
src/nl2mmd/logger.ts:45                 }
src/nl2mmd/logger.ts:46
src/nl2mmd/logger.ts:47                 export function logNl2MmdDebug(event: string, fields?: Record<string, unknown>): void {
src/nl2mmd/logger.ts:48                   if (!isNl2MmdDebugEnabled()) {
src/nl2mmd/logger.ts:49                     return;
src/nl2mmd/logger.ts:50                   }
src/nl2mmd/logger.ts:51                   const payload = fields ? ` ${JSON.stringify(sanitizeFields(fields))}` : "";
src/nl2mmd/logger.ts:52                   console.error(`[nl2mmd] ${event}${payload}`);
src/nl2mmd/logger.ts:53                 }
src/nl2mmd/logger.ts:54
src/nl2mmd/normalize-mermaid.ts:1       /**
src/nl2mmd/normalize-mermaid.ts:2        * @fileoverview Mermaid normalizer/stabilizer for NL2MMD model outputs.
src/nl2mmd/normalize-mermaid.ts:3        * File Set: nl2mmd-normalization
src/nl2mmd/normalize-mermaid.ts:4        * Responsibilities:
src/nl2mmd/normalize-mermaid.ts:5        * - Canonicalize boundary/node tokens, metadata ordering, and join metadata placement.
src/nl2mmd/normalize-mermaid.ts:6        * - Repair common generation drift while preserving runtime-supported syntax.
src/nl2mmd/normalize-mermaid.ts:7        * Boundaries:
src/nl2mmd/normalize-mermaid.ts:8        * - Does not parse final runtime execution plan.
src/nl2mmd/normalize-mermaid.ts:9        */
src/nl2mmd/normalize-mermaid.ts:10      import type { Nl2MmdContext } from "./types.js";
src/nl2mmd/normalize-mermaid.ts:11
src/nl2mmd/normalize-mermaid.ts:12      const FLOWCHART_HEADER_REGEX = /^flowchart\s+(TD|LR)$/;
src/nl2mmd/normalize-mermaid.ts:13      const ROLE_TOKEN_REGEX = /^([A-Za-z0-9._:-]+)\[Role:([A-Za-z0-9._:-]+)\]$/;
src/nl2mmd/normalize-mermaid.ts:14      const BARE_NODE_TOKEN_REGEX = /^[A-Za-z0-9._:-]+$/;
src/nl2mmd/normalize-mermaid.ts:15      const EDGE_REGEX = /^(.+?)\s*-->\|(.+?)\|\s*(.+)$/;
src/nl2mmd/normalize-mermaid.ts:16      const METADATA_REGEX = /^%%\s*([A-Za-z0-9._:-]+)\s*=\s*(.+)$/;
src/nl2mmd/normalize-mermaid.ts:17      const ROLE_METADATA_PREFIXES = [
```

```
src/nl2mmd/normalize-mermaid.ts:18          "exec.bind.",
src/nl2mmd/normalize-mermaid.ts:19          "role.mode.",
src/nl2mmd/normalize-mermaid.ts:20          "join.mode.",
src/nl2mmd/normalize-mermaid.ts:21          "join.min.",
src/nl2mmd/normalize-mermaid.ts:22          "join.sources.",
src/nl2mmd/normalize-mermaid.ts:23          "loop.max.",
src/nl2mmd/normalize-mermaid.ts:24          "route.order."
src/nl2mmd/normalize-mermaid.ts:25        ];
src/nl2mmd/normalize-mermaid.ts:26        const SUPPORTED_EXACT_METADATA_KEYS = new Set([
src/nl2mmd/normalize-mermaid.ts:27          "engine",
src/nl2mmd/normalize-mermaid.ts:28          "system.id",
src/nl2mmd/normalize-mermaid.ts:29          "system.version",
src/nl2mmd/normalize-mermaid.ts:30          "law.global",
src/nl2mmd/normalize-mermaid.ts:31          "entry.role",
src/nl2mmd/normalize-mermaid.ts:32          "handoff.mode",
src/nl2mmd/normalize-mermaid.ts:33          "handoff.contracts"
src/nl2mmd/normalize-mermaid.ts:34        ]);
src/nl2mmd/normalize-mermaid.ts:35        const SUPPORTED_METADATA_PREFIXES = [
src/nl2mmd/normalize-mermaid.ts:36          "exec.bind.",
src/nl2mmd/normalize-mermaid.ts:37          "role.mode.",
src/nl2mmd/normalize-mermaid.ts:38          "join.mode.",
src/nl2mmd/normalize-mermaid.ts:39          "join.min.",
src/nl2mmd/normalize-mermaid.ts:40          "join.sources.",
src/nl2mmd/normalize-mermaid.ts:41          "context.map.",
src/nl2mmd/normalize-mermaid.ts:42          "loop.max.",
src/nl2mmd/normalize-mermaid.ts:43          "route.order."
src/nl2mmd/normalize-mermaid.ts:44        ];
src/nl2mmd/normalize-mermaid.ts:45
src/nl2mmd/normalize-mermaid.ts:46        function parseRoleToken(token: string): { nodeId: string; roleId: string } | null {
src/nl2mmd/normalize-mermaid.ts:47          const match = token.trim().match(ROLE_TOKEN_REGEX);
src/nl2mmd/normalize-mermaid.ts:48          if (!match) {
src/nl2mmd/normalize-mermaid.ts:49            return null;
src/nl2mmd/normalize-mermaid.ts:50          }
src/nl2mmd/normalize-mermaid.ts:51          return {
src/nl2mmd/normalize-mermaid.ts:52            nodeId: match[1],
src/nl2mmd/normalize-mermaid.ts:53            roleId: match[2]
src/nl2mmd/normalize-mermaid.ts:54          };
src/nl2mmd/normalize-mermaid.ts:55        }
src/nl2mmd/normalize-mermaid.ts:56
src/nl2mmd/normalize-mermaid.ts:57        function parseMetadata(line: string): { key: string; value: string } | null {
src/nl2mmd/normalize-mermaid.ts:58          const match = line.trim().match(METADATA_REGEX);
src/nl2mmd/normalize-mermaid.ts:59          if (!match) {
src/nl2mmd/normalize-mermaid.ts:60            return null;
src/nl2mmd/normalize-mermaid.ts:61          }
src/nl2mmd/normalize-mermaid.ts:62          return {
src/nl2mmd/normalize-mermaid.ts:63            key: match[1],
src/nl2mmd/normalize-mermaid.ts:64            value: match[2].trim()
src/nl2mmd/normalize-mermaid.ts:65          };
src/nl2mmd/normalize-mermaid.ts:66        }
src/nl2mmd/normalize-mermaid.ts:67
```

```
src/nl2mmd/normalize-mermaid.ts:68        function normalizeBoundaryToken(token: string): string {
src/nl2mmd/normalize-mermaid.ts:69          const trimmed = token.trim();
src/nl2mmd/normalize-mermaid.ts:70          const normalized = trimmed.toLowerCase();
src/nl2mmd/normalize-mermaid.ts:71          if (normalized === "start") {
src/nl2mmd/normalize-mermaid.ts:72            return "input";
src/nl2mmd/normalize-mermaid.ts:73          }
src/nl2mmd/normalize-mermaid.ts:74          if (normalized === "end" || normalized === "done") {
src/nl2mmd/normalize-mermaid.ts:75            return "output";
src/nl2mmd/normalize-mermaid.ts:76          }
src/nl2mmd/normalize-mermaid.ts:77          if (normalized === "input") {
src/nl2mmd/normalize-mermaid.ts:78            return "input";
src/nl2mmd/normalize-mermaid.ts:79          }
src/nl2mmd/normalize-mermaid.ts:80          if (normalized === "output") {
src/nl2mmd/normalize-mermaid.ts:81            return "output";
src/nl2mmd/normalize-mermaid.ts:82          }
src/nl2mmd/normalize-mermaid.ts:83          return trimmed;
src/nl2mmd/normalize-mermaid.ts:84        }
src/nl2mmd/normalize-mermaid.ts:85
src/nl2mmd/normalize-mermaid.ts:86        function collectKnownRoleIds(lines: string[]): Set<string> {
src/nl2mmd/normalize-mermaid.ts:87          const roleIds = new Set<string>();
src/nl2mmd/normalize-mermaid.ts:88          for (const line of lines) {
src/nl2mmd/normalize-mermaid.ts:89            const metadata = parseMetadata(line);
src/nl2mmd/normalize-mermaid.ts:90            if (!metadata) {
src/nl2mmd/normalize-mermaid.ts:91              continue;
src/nl2mmd/normalize-mermaid.ts:92            }
src/nl2mmd/normalize-mermaid.ts:93
src/nl2mmd/normalize-mermaid.ts:94            if (metadata.key === "entry.role" && BARE_NODE_TOKEN_REGEX.test(metadata.value)) {
src/nl2mmd/normalize-mermaid.ts:95              roleIds.add(metadata.value);
src/nl2mmd/normalize-mermaid.ts:96              continue;
src/nl2mmd/normalize-mermaid.ts:97            }
src/nl2mmd/normalize-mermaid.ts:98
src/nl2mmd/normalize-mermaid.ts:99            if (metadata.key.startsWith("context.map.")) {
src/nl2mmd/normalize-mermaid.ts:100             const rest = metadata.key.slice("context.map.".length);
src/nl2mmd/normalize-mermaid.ts:101             const dotIndex = rest.indexOf(".");
src/nl2mmd/normalize-mermaid.ts:102             if (dotIndex > 0) {
src/nl2mmd/normalize-mermaid.ts:103               roleIds.add(rest.slice(0, dotIndex));
src/nl2mmd/normalize-mermaid.ts:104             }
src/nl2mmd/normalize-mermaid.ts:105             continue;
src/nl2mmd/normalize-mermaid.ts:106           }
src/nl2mmd/normalize-mermaid.ts:107
src/nl2mmd/normalize-mermaid.ts:108           for (const prefix of ROLE_METADATA_PREFIXES) {
src/nl2mmd/normalize-mermaid.ts:109             if (!metadata.key.startsWith(prefix)) {
src/nl2mmd/normalize-mermaid.ts:110               continue;
src/nl2mmd/normalize-mermaid.ts:111             }
src/nl2mmd/normalize-mermaid.ts:112             const suffix = metadata.key.slice(prefix.length);
src/nl2mmd/normalize-mermaid.ts:113             if (BARE_NODE_TOKEN_REGEX.test(suffix)) {
src/nl2mmd/normalize-mermaid.ts:114               roleIds.add(suffix);
src/nl2mmd/normalize-mermaid.ts:115             }
src/nl2mmd/normalize-mermaid.ts:116             if (prefix === "join.sources.") {
src/nl2mmd/normalize-mermaid.ts:117               for (const sourceRoleId of metadata.value
```

```
src/nl2mmd/normalize-mermaid.ts:118                .split(",")
src/nl2mmd/normalize-mermaid.ts:119                .map((item) => item.trim())
src/nl2mmd/normalize-mermaid.ts:120                .filter(Boolean)) {
src/nl2mmd/normalize-mermaid.ts:121                if (BARE_NODE_TOKEN_REGEX.test(sourceRoleId)) {
src/nl2mmd/normalize-mermaid.ts:122                  roleIds.add(sourceRoleId);
src/nl2mmd/normalize-mermaid.ts:123                }
src/nl2mmd/normalize-mermaid.ts:124              }
src/nl2mmd/normalize-mermaid.ts:125            }
src/nl2mmd/normalize-mermaid.ts:126            break;
src/nl2mmd/normalize-mermaid.ts:127          }
src/nl2mmd/normalize-mermaid.ts:128        }
src/nl2mmd/normalize-mermaid.ts:129        return roleIds;
src/nl2mmd/normalize-mermaid.ts:130      }
src/nl2mmd/normalize-mermaid.ts:131
src/nl2mmd/normalize-mermaid.ts:132      function collectNodeRoleMap(lines: string[]): Map<string, string> {
src/nl2mmd/normalize-mermaid.ts:133        const nodeRoleMap = new Map<string, string>();
src/nl2mmd/normalize-mermaid.ts:134        const knownRoleIds = collectKnownRoleIds(lines);
src/nl2mmd/normalize-mermaid.ts:135
src/nl2mmd/normalize-mermaid.ts:136        for (const line of lines) {
src/nl2mmd/normalize-mermaid.ts:137          const trimmed = line.trim();
src/nl2mmd/normalize-mermaid.ts:138          const standaloneRoleToken = parseRoleToken(trimmed);
src/nl2mmd/normalize-mermaid.ts:139          if (standaloneRoleToken) {
src/nl2mmd/normalize-mermaid.ts:140            nodeRoleMap.set(standaloneRoleToken.nodeId, standaloneRoleToken.roleId);
src/nl2mmd/normalize-mermaid.ts:141            continue;
src/nl2mmd/normalize-mermaid.ts:142          }
src/nl2mmd/normalize-mermaid.ts:143
src/nl2mmd/normalize-mermaid.ts:144          const edgeMatch = trimmed.match(EDGE_REGEX);
src/nl2mmd/normalize-mermaid.ts:145          if (!edgeMatch) {
src/nl2mmd/normalize-mermaid.ts:146            continue;
src/nl2mmd/normalize-mermaid.ts:147          }
src/nl2mmd/normalize-mermaid.ts:148
src/nl2mmd/normalize-mermaid.ts:149          for (const endpoint of [edgeMatch[1], edgeMatch[3]]) {
src/nl2mmd/normalize-mermaid.ts:150            const roleToken = parseRoleToken(endpoint);
src/nl2mmd/normalize-mermaid.ts:151            if (!roleToken) {
src/nl2mmd/normalize-mermaid.ts:152              continue;
src/nl2mmd/normalize-mermaid.ts:153            }
src/nl2mmd/normalize-mermaid.ts:154            nodeRoleMap.set(roleToken.nodeId, roleToken.roleId);
src/nl2mmd/normalize-mermaid.ts:155          }
src/nl2mmd/normalize-mermaid.ts:156        }
src/nl2mmd/normalize-mermaid.ts:157
src/nl2mmd/normalize-mermaid.ts:158        for (const roleId of knownRoleIds) {
src/nl2mmd/normalize-mermaid.ts:159          if (!nodeRoleMap.has(roleId)) {
src/nl2mmd/normalize-mermaid.ts:160            nodeRoleMap.set(roleId, roleId);
src/nl2mmd/normalize-mermaid.ts:161          }
src/nl2mmd/normalize-mermaid.ts:162        }
src/nl2mmd/normalize-mermaid.ts:163
src/nl2mmd/normalize-mermaid.ts:164        return nodeRoleMap;
src/nl2mmd/normalize-mermaid.ts:165      }
src/nl2mmd/normalize-mermaid.ts:166
src/nl2mmd/normalize-mermaid.ts:167      function canonicalizeEndpointToken(token: string, nodeRoleMap: Map<string, string>): string {
```

```
src/nl2mmd/normalize-mermaid.ts:168         const normalizedBoundary = normalizeBoundaryToken(token);
src/nl2mmd/normalize-mermaid.ts:169         if (normalizedBoundary === "input" || normalizedBoundary === "output") {
src/nl2mmd/normalize-mermaid.ts:170           return normalizedBoundary;
src/nl2mmd/normalize-mermaid.ts:171         }
src/nl2mmd/normalize-mermaid.ts:172
src/nl2mmd/normalize-mermaid.ts:173         const roleToken = parseRoleToken(normalizedBoundary);
src/nl2mmd/normalize-mermaid.ts:174         if (roleToken) {
src/nl2mmd/normalize-mermaid.ts:175           return `${roleToken.nodeId}[Role:${roleToken.roleId}]`;
src/nl2mmd/normalize-mermaid.ts:176         }
src/nl2mmd/normalize-mermaid.ts:177
src/nl2mmd/normalize-mermaid.ts:178         if (BARE_NODE_TOKEN_REGEX.test(normalizedBoundary)) {
src/nl2mmd/normalize-mermaid.ts:179           const roleId = nodeRoleMap.get(normalizedBoundary);
src/nl2mmd/normalize-mermaid.ts:180           if (roleId) {
src/nl2mmd/normalize-mermaid.ts:181             return `${normalizedBoundary}[Role:${roleId}]`;
src/nl2mmd/normalize-mermaid.ts:182           }
src/nl2mmd/normalize-mermaid.ts:183         }
src/nl2mmd/normalize-mermaid.ts:184
src/nl2mmd/normalize-mermaid.ts:185         return normalizedBoundary;
src/nl2mmd/normalize-mermaid.ts:186       }
src/nl2mmd/normalize-mermaid.ts:187
src/nl2mmd/normalize-mermaid.ts:188       function isMarkdownFence(line: string): boolean {
src/nl2mmd/normalize-mermaid.ts:189         return /^```/.test(line.trim());
src/nl2mmd/normalize-mermaid.ts:190       }
src/nl2mmd/normalize-mermaid.ts:191
src/nl2mmd/normalize-mermaid.ts:192       function isSupportedMetadataKey(key: string): boolean {
src/nl2mmd/normalize-mermaid.ts:193         if (SUPPORTED_EXACT_METADATA_KEYS.has(key)) {
src/nl2mmd/normalize-mermaid.ts:194           return true;
src/nl2mmd/normalize-mermaid.ts:195         }
src/nl2mmd/normalize-mermaid.ts:196         return SUPPORTED_METADATA_PREFIXES.some((prefix) => key.startsWith(prefix));
src/nl2mmd/normalize-mermaid.ts:197       }
src/nl2mmd/normalize-mermaid.ts:198
src/nl2mmd/normalize-mermaid.ts:199       type CanonicalizedEdge = {
src/nl2mmd/normalize-mermaid.ts:200         fromToken: string;
src/nl2mmd/normalize-mermaid.ts:201         toToken: string;
src/nl2mmd/normalize-mermaid.ts:202         eventType: string;
src/nl2mmd/normalize-mermaid.ts:203       };
src/nl2mmd/normalize-mermaid.ts:204
src/nl2mmd/normalize-mermaid.ts:205       function buildIncomingSourcesByRole(edges: CanonicalizedEdge[]): Map<string, string[]> {
src/nl2mmd/normalize-mermaid.ts:206         const incoming = new Map<string, Set<string>>();
src/nl2mmd/normalize-mermaid.ts:207         for (const edge of edges) {
src/nl2mmd/normalize-mermaid.ts:208           const fromRole = parseRoleToken(edge.fromToken);
src/nl2mmd/normalize-mermaid.ts:209           const toRole = parseRoleToken(edge.toToken);
src/nl2mmd/normalize-mermaid.ts:210           if (!fromRole || !toRole) {
src/nl2mmd/normalize-mermaid.ts:211             continue;
src/nl2mmd/normalize-mermaid.ts:212           }
src/nl2mmd/normalize-mermaid.ts:213           const bucket = incoming.get(toRole.roleId) ?? new Set<string>();
src/nl2mmd/normalize-mermaid.ts:214           bucket.add(fromRole.roleId);
src/nl2mmd/normalize-mermaid.ts:215           incoming.set(toRole.roleId, bucket);
src/nl2mmd/normalize-mermaid.ts:216         }
src/nl2mmd/normalize-mermaid.ts:217         return new Map(
```

```
src/nl2mmd/normalize-mermaid.ts:218           Array.from(incoming.entries()).map(([roleId, sources]) => [
src/nl2mmd/normalize-mermaid.ts:219             roleId,
src/nl2mmd/normalize-mermaid.ts:220             Array.from(sources).sort((left, right) => left.localeCompare(right))
src/nl2mmd/normalize-mermaid.ts:221           ])
src/nl2mmd/normalize-mermaid.ts:222         );
src/nl2mmd/normalize-mermaid.ts:223       }
src/nl2mmd/normalize-mermaid.ts:224
src/nl2mmd/normalize-mermaid.ts:225       function sameRoleSet(left: string[], right: string[]): boolean {
src/nl2mmd/normalize-mermaid.ts:226         if (left.length !== right.length) {
src/nl2mmd/normalize-mermaid.ts:227           return false;
src/nl2mmd/normalize-mermaid.ts:228         }
src/nl2mmd/normalize-mermaid.ts:229         return left.every((roleId, index) => roleId === right[index]);
src/nl2mmd/normalize-mermaid.ts:230       }
src/nl2mmd/normalize-mermaid.ts:231
src/nl2mmd/normalize-mermaid.ts:232       function rewriteMetadataKeyTarget(key: string, fromRoleId: string, toRoleId: string): string {
src/nl2mmd/normalize-mermaid.ts:233         if (key.startsWith(`context.map.${fromRoleId}.`)) {
src/nl2mmd/normalize-mermaid.ts:234           return `context.map.${toRoleId}.${key.slice(`context.map.${fromRoleId}.`.length)}`;
src/nl2mmd/normalize-mermaid.ts:235         }
src/nl2mmd/normalize-mermaid.ts:236         for (const prefix of ["join.mode.", "join.sources.", "join.min."]) {
src/nl2mmd/normalize-mermaid.ts:237           if (key === `${prefix}${fromRoleId}`) {
src/nl2mmd/normalize-mermaid.ts:238             return `${prefix}${toRoleId}`;
src/nl2mmd/normalize-mermaid.ts:239           }
src/nl2mmd/normalize-mermaid.ts:240         }
src/nl2mmd/normalize-mermaid.ts:241         return key;
src/nl2mmd/normalize-mermaid.ts:242       }
src/nl2mmd/normalize-mermaid.ts:243
src/nl2mmd/normalize-mermaid.ts:244       function repairMisplacedJoinMetadata(args: {
src/nl2mmd/normalize-mermaid.ts:245         metadata: Map<string, string>;
src/nl2mmd/normalize-mermaid.ts:246         metadataOrder: string[];
src/nl2mmd/normalize-mermaid.ts:247         edges: CanonicalizedEdge[];
src/nl2mmd/normalize-mermaid.ts:248       }): void {
src/nl2mmd/normalize-mermaid.ts:249         // Recovery semantics for generation drift: if join metadata was attached to the wrong role
src/nl2mmd/normalize-mermaid.ts:250         // but uniquely matches another role's incoming sources, move metadata to that role.
src/nl2mmd/normalize-mermaid.ts:251         const incomingSourcesByRole = buildIncomingSourcesByRole(args.edges);
src/nl2mmd/normalize-mermaid.ts:252         const joinRoleIds = Array.from(args.metadata.keys())
src/nl2mmd/normalize-mermaid.ts:253           .filter((key) => key.startsWith("join.mode."))
src/nl2mmd/normalize-mermaid.ts:254           .map((key) => key.slice("join.mode.".length));
src/nl2mmd/normalize-mermaid.ts:255
src/nl2mmd/normalize-mermaid.ts:256         for (const roleId of joinRoleIds) {
src/nl2mmd/normalize-mermaid.ts:257           const joinSourcesKey = `join.sources.${roleId}`;
src/nl2mmd/normalize-mermaid.ts:258           const joinSourcesValue = args.metadata.get(joinSourcesKey);
src/nl2mmd/normalize-mermaid.ts:259           if (!joinSourcesValue) {
src/nl2mmd/normalize-mermaid.ts:260             continue;
src/nl2mmd/normalize-mermaid.ts:261           }
src/nl2mmd/normalize-mermaid.ts:262
src/nl2mmd/normalize-mermaid.ts:263           const declaredSources = Array.from(
src/nl2mmd/normalize-mermaid.ts:264             new Set(
src/nl2mmd/normalize-mermaid.ts:265               joinSourcesValue
src/nl2mmd/normalize-mermaid.ts:266                 .split(",")
src/nl2mmd/normalize-mermaid.ts:267                 .map((item) => item.trim())
```

```
src/nl2mmd/normalize-mermaid.ts:268                .filter(Boolean)
src/nl2mmd/normalize-mermaid.ts:269              )
src/nl2mmd/normalize-mermaid.ts:270            ).sort((left, right) => left.localeCompare(right));
src/nl2mmd/normalize-mermaid.ts:271            const currentIncomingSources = incomingSourcesByRole.get(roleId) ?? [];
src/nl2mmd/normalize-mermaid.ts:272            if (sameRoleSet(declaredSources, currentIncomingSources)) {
src/nl2mmd/normalize-mermaid.ts:273              continue;
src/nl2mmd/normalize-mermaid.ts:274            }
src/nl2mmd/normalize-mermaid.ts:275
src/nl2mmd/normalize-mermaid.ts:276            const candidates = Array.from(incomingSourcesByRole.entries())
src/nl2mmd/normalize-mermaid.ts:277              .filter(([candidateRoleId, sources]) => candidateRoleId !== roleId && sameRoleSet(declaredSources, sources))
src/nl2mmd/normalize-mermaid.ts:278              .map(([candidateRoleId]) => candidateRoleId);
src/nl2mmd/normalize-mermaid.ts:279            if (candidates.length !== 1) {
src/nl2mmd/normalize-mermaid.ts:280              continue;
src/nl2mmd/normalize-mermaid.ts:281            }
src/nl2mmd/normalize-mermaid.ts:282
src/nl2mmd/normalize-mermaid.ts:283            const targetRoleId = candidates[0];
src/nl2mmd/normalize-mermaid.ts:284            const keysToMove = Array.from(args.metadata.keys()).filter(
src/nl2mmd/normalize-mermaid.ts:285              (key) =>
src/nl2mmd/normalize-mermaid.ts:286                key === `join.mode.${roleId}` ||
src/nl2mmd/normalize-mermaid.ts:287                key === `join.sources.${roleId}` ||
src/nl2mmd/normalize-mermaid.ts:288                key === `join.min.${roleId}` ||
src/nl2mmd/normalize-mermaid.ts:289                key.startsWith(`context.map.${roleId}.`)
src/nl2mmd/normalize-mermaid.ts:290            );
src/nl2mmd/normalize-mermaid.ts:291
src/nl2mmd/normalize-mermaid.ts:292            for (const key of keysToMove) {
src/nl2mmd/normalize-mermaid.ts:293              const nextKey = rewriteMetadataKeyTarget(key, roleId, targetRoleId);
src/nl2mmd/normalize-mermaid.ts:294              if (!args.metadata.has(nextKey)) {
src/nl2mmd/normalize-mermaid.ts:295                args.metadata.set(nextKey, args.metadata.get(key) as string);
src/nl2mmd/normalize-mermaid.ts:296                if (!args.metadataOrder.includes(nextKey)) {
src/nl2mmd/normalize-mermaid.ts:297                  args.metadataOrder.push(nextKey);
src/nl2mmd/normalize-mermaid.ts:298                }
src/nl2mmd/normalize-mermaid.ts:299              }
src/nl2mmd/normalize-mermaid.ts:300              args.metadata.delete(key);
src/nl2mmd/normalize-mermaid.ts:301            }
src/nl2mmd/normalize-mermaid.ts:302
src/nl2mmd/normalize-mermaid.ts:303            for (let index = args.metadataOrder.length - 1; index >= 0; index -= 1) {
src/nl2mmd/normalize-mermaid.ts:304              const key = args.metadataOrder[index];
src/nl2mmd/normalize-mermaid.ts:305              if (
src/nl2mmd/normalize-mermaid.ts:306                key === `join.mode.${roleId}` ||
src/nl2mmd/normalize-mermaid.ts:307                key === `join.sources.${roleId}` ||
src/nl2mmd/normalize-mermaid.ts:308                key === `join.min.${roleId}` ||
src/nl2mmd/normalize-mermaid.ts:309                key.startsWith(`context.map.${roleId}.`)
src/nl2mmd/normalize-mermaid.ts:310              ) {
src/nl2mmd/normalize-mermaid.ts:311                args.metadataOrder.splice(index, 1);
src/nl2mmd/normalize-mermaid.ts:312              }
src/nl2mmd/normalize-mermaid.ts:313            }
src/nl2mmd/normalize-mermaid.ts:314          }
src/nl2mmd/normalize-mermaid.ts:315        }
src/nl2mmd/normalize-mermaid.ts:316
src/nl2mmd/normalize-mermaid.ts:317        function collectCyclicRoleComponents(edges: CanonicalizedEdge[]): string[][] {
```

```
src/nl2mmd/normalize-mermaid.ts:318         const adjacency = new Map<string, Set<string>>();
src/nl2mmd/normalize-mermaid.ts:319         const roleIds = new Set<string>();
src/nl2mmd/normalize-mermaid.ts:320         const selfLoops = new Set<string>();
src/nl2mmd/normalize-mermaid.ts:321
src/nl2mmd/normalize-mermaid.ts:322         for (const edge of edges) {
src/nl2mmd/normalize-mermaid.ts:323           const fromRole = parseRoleToken(edge.fromToken);
src/nl2mmd/normalize-mermaid.ts:324           const toRole = parseRoleToken(edge.toToken);
src/nl2mmd/normalize-mermaid.ts:325           if (!fromRole || !toRole) {
src/nl2mmd/normalize-mermaid.ts:326             continue;
src/nl2mmd/normalize-mermaid.ts:327           }
src/nl2mmd/normalize-mermaid.ts:328           roleIds.add(fromRole.roleId);
src/nl2mmd/normalize-mermaid.ts:329           roleIds.add(toRole.roleId);
src/nl2mmd/normalize-mermaid.ts:330           const bucket = adjacency.get(fromRole.roleId) ?? new Set<string>();
src/nl2mmd/normalize-mermaid.ts:331           bucket.add(toRole.roleId);
src/nl2mmd/normalize-mermaid.ts:332           adjacency.set(fromRole.roleId, bucket);
src/nl2mmd/normalize-mermaid.ts:333           if (fromRole.roleId === toRole.roleId) {
src/nl2mmd/normalize-mermaid.ts:334             selfLoops.add(fromRole.roleId);
src/nl2mmd/normalize-mermaid.ts:335           }
src/nl2mmd/normalize-mermaid.ts:336         }
src/nl2mmd/normalize-mermaid.ts:337
src/nl2mmd/normalize-mermaid.ts:338         let index = 0;
src/nl2mmd/normalize-mermaid.ts:339         const indices = new Map<string, number>();
src/nl2mmd/normalize-mermaid.ts:340         const lowlink = new Map<string, number>();
src/nl2mmd/normalize-mermaid.ts:341         const stack: string[] = [];
src/nl2mmd/normalize-mermaid.ts:342         const onStack = new Set<string>();
src/nl2mmd/normalize-mermaid.ts:343         const components: string[][] = [];
src/nl2mmd/normalize-mermaid.ts:344
src/nl2mmd/normalize-mermaid.ts:345         const strongConnect = (roleId: string): void => {
src/nl2mmd/normalize-mermaid.ts:346           indices.set(roleId, index);
src/nl2mmd/normalize-mermaid.ts:347           lowlink.set(roleId, index);
src/nl2mmd/normalize-mermaid.ts:348           index += 1;
src/nl2mmd/normalize-mermaid.ts:349           stack.push(roleId);
src/nl2mmd/normalize-mermaid.ts:350           onStack.add(roleId);
src/nl2mmd/normalize-mermaid.ts:351
src/nl2mmd/normalize-mermaid.ts:352           const nextRoles = adjacency.get(roleId) ?? new Set<string>();
src/nl2mmd/normalize-mermaid.ts:353           for (const nextRoleId of nextRoles) {
src/nl2mmd/normalize-mermaid.ts:354             if (!indices.has(nextRoleId)) {
src/nl2mmd/normalize-mermaid.ts:355               strongConnect(nextRoleId);
src/nl2mmd/normalize-mermaid.ts:356               const nextLowlink = lowlink.get(nextRoleId);
src/nl2mmd/normalize-mermaid.ts:357               const currentLowlink = lowlink.get(roleId);
src/nl2mmd/normalize-mermaid.ts:358               if (nextLowlink !== undefined && currentLowlink !== undefined && nextLowlink < currentLowlink) {
src/nl2mmd/normalize-mermaid.ts:359                 lowlink.set(roleId, nextLowlink);
src/nl2mmd/normalize-mermaid.ts:360               }
src/nl2mmd/normalize-mermaid.ts:361               continue;
src/nl2mmd/normalize-mermaid.ts:362             }
src/nl2mmd/normalize-mermaid.ts:363             if (!onStack.has(nextRoleId)) {
src/nl2mmd/normalize-mermaid.ts:364               continue;
src/nl2mmd/normalize-mermaid.ts:365             }
src/nl2mmd/normalize-mermaid.ts:366             const nextIndex = indices.get(nextRoleId);
src/nl2mmd/normalize-mermaid.ts:367             const currentLowlink = lowlink.get(roleId);
```

```
src/nl2mmd/normalize-mermaid.ts:368             if (nextIndex !== undefined && currentLowlink !== undefined && nextIndex < currentLowlink) {
src/nl2mmd/normalize-mermaid.ts:369               lowlink.set(roleId, nextIndex);
src/nl2mmd/normalize-mermaid.ts:370             }
src/nl2mmd/normalize-mermaid.ts:371           }
src/nl2mmd/normalize-mermaid.ts:372
src/nl2mmd/normalize-mermaid.ts:373           if (lowlink.get(roleId) !== indices.get(roleId)) {
src/nl2mmd/normalize-mermaid.ts:374             return;
src/nl2mmd/normalize-mermaid.ts:375           }
src/nl2mmd/normalize-mermaid.ts:376           const component: string[] = [];
src/nl2mmd/normalize-mermaid.ts:377           while (stack.length > 0) {
src/nl2mmd/normalize-mermaid.ts:378             const node = stack.pop() as string;
src/nl2mmd/normalize-mermaid.ts:379             onStack.delete(node);
src/nl2mmd/normalize-mermaid.ts:380             component.push(node);
src/nl2mmd/normalize-mermaid.ts:381             if (node === roleId) {
src/nl2mmd/normalize-mermaid.ts:382               break;
src/nl2mmd/normalize-mermaid.ts:383             }
src/nl2mmd/normalize-mermaid.ts:384           }
src/nl2mmd/normalize-mermaid.ts:385           if (component.length > 1) {
src/nl2mmd/normalize-mermaid.ts:386             components.push(component);
src/nl2mmd/normalize-mermaid.ts:387             return;
src/nl2mmd/normalize-mermaid.ts:388           }
src/nl2mmd/normalize-mermaid.ts:389           if (component.length === 1 && selfLoops.has(component[0])) {
src/nl2mmd/normalize-mermaid.ts:390             components.push(component);
src/nl2mmd/normalize-mermaid.ts:391           }
src/nl2mmd/normalize-mermaid.ts:392         };
src/nl2mmd/normalize-mermaid.ts:393
src/nl2mmd/normalize-mermaid.ts:394         for (const roleId of roleIds) {
src/nl2mmd/normalize-mermaid.ts:395           if (!indices.has(roleId)) {
src/nl2mmd/normalize-mermaid.ts:396             strongConnect(roleId);
src/nl2mmd/normalize-mermaid.ts:397           }
src/nl2mmd/normalize-mermaid.ts:398         }
src/nl2mmd/normalize-mermaid.ts:399
src/nl2mmd/normalize-mermaid.ts:400         return components;
src/nl2mmd/normalize-mermaid.ts:401       }
src/nl2mmd/normalize-mermaid.ts:402
src/nl2mmd/normalize-mermaid.ts:403       function pickLoopBudgetRole(component: string[]): string {
src/nl2mmd/normalize-mermaid.ts:404         const priorityRegex = /(moderator|coordinator|dispatch|controller|judge)/i;
src/nl2mmd/normalize-mermaid.ts:405         const preferred = component.find((roleId) => priorityRegex.test(roleId));
src/nl2mmd/normalize-mermaid.ts:406         if (preferred) {
src/nl2mmd/normalize-mermaid.ts:407           return preferred;
src/nl2mmd/normalize-mermaid.ts:408         }
src/nl2mmd/normalize-mermaid.ts:409         return component.slice().sort((left, right) => left.localeCompare(right))[0];
src/nl2mmd/normalize-mermaid.ts:410       }
src/nl2mmd/normalize-mermaid.ts:411
src/nl2mmd/normalize-mermaid.ts:412       function canonicalizeWithRuntimeDefaults(args: {
src/nl2mmd/normalize-mermaid.ts:413         normalizedMermaid: string;
src/nl2mmd/normalize-mermaid.ts:414         context: Nl2MmdContext;
src/nl2mmd/normalize-mermaid.ts:415       }): string {
src/nl2mmd/normalize-mermaid.ts:416         const lines = args.normalizedMermaid.split(/\r?\n/);
src/nl2mmd/normalize-mermaid.ts:417         const metadata = new Map<string, string>();
```

```
src/nl2mmd/normalize-mermaid.ts:418         const metadataOrder: string[] = [];
src/nl2mmd/normalize-mermaid.ts:419         const edges: CanonicalizedEdge[] = [];
src/nl2mmd/normalize-mermaid.ts:420         const roleIds = new Set<string>();
src/nl2mmd/normalize-mermaid.ts:421         const inputTargets: string[] = [];
src/nl2mmd/normalize-mermaid.ts:422         let header: string | undefined;
src/nl2mmd/normalize-mermaid.ts:423
src/nl2mmd/normalize-mermaid.ts:424         for (const line of lines) {
src/nl2mmd/normalize-mermaid.ts:425           const trimmed = line.trim();
src/nl2mmd/normalize-mermaid.ts:426           if (!trimmed) {
src/nl2mmd/normalize-mermaid.ts:427             continue;
src/nl2mmd/normalize-mermaid.ts:428           }
src/nl2mmd/normalize-mermaid.ts:429
src/nl2mmd/normalize-mermaid.ts:430           if (FLOWCHART_HEADER_REGEX.test(trimmed)) {
src/nl2mmd/normalize-mermaid.ts:431             header ??= trimmed;
src/nl2mmd/normalize-mermaid.ts:432             continue;
src/nl2mmd/normalize-mermaid.ts:433           }
src/nl2mmd/normalize-mermaid.ts:434
src/nl2mmd/normalize-mermaid.ts:435           const metadataItem = parseMetadata(trimmed);
src/nl2mmd/normalize-mermaid.ts:436           if (metadataItem) {
src/nl2mmd/normalize-mermaid.ts:437             if (!isSupportedMetadataKey(metadataItem.key)) {
src/nl2mmd/normalize-mermaid.ts:438               continue;
src/nl2mmd/normalize-mermaid.ts:439             }
src/nl2mmd/normalize-mermaid.ts:440             if (!metadata.has(metadataItem.key)) {
src/nl2mmd/normalize-mermaid.ts:441               metadataOrder.push(metadataItem.key);
src/nl2mmd/normalize-mermaid.ts:442             }
src/nl2mmd/normalize-mermaid.ts:443             metadata.set(metadataItem.key, metadataItem.value);
src/nl2mmd/normalize-mermaid.ts:444             continue;
src/nl2mmd/normalize-mermaid.ts:445           }
src/nl2mmd/normalize-mermaid.ts:446
src/nl2mmd/normalize-mermaid.ts:447           const edgeMatch = trimmed.match(EDGE_REGEX);
src/nl2mmd/normalize-mermaid.ts:448           if (!edgeMatch) {
src/nl2mmd/normalize-mermaid.ts:449             continue;
src/nl2mmd/normalize-mermaid.ts:450           }
src/nl2mmd/normalize-mermaid.ts:451           const fromToken = edgeMatch[1].trim();
src/nl2mmd/normalize-mermaid.ts:452           const toToken = edgeMatch[3].trim();
src/nl2mmd/normalize-mermaid.ts:453           const eventType = edgeMatch[2].trim();
src/nl2mmd/normalize-mermaid.ts:454           edges.push({ fromToken, toToken, eventType });
src/nl2mmd/normalize-mermaid.ts:455
src/nl2mmd/normalize-mermaid.ts:456           const fromRole = parseRoleToken(fromToken);
src/nl2mmd/normalize-mermaid.ts:457           const toRole = parseRoleToken(toToken);
src/nl2mmd/normalize-mermaid.ts:458           if (fromRole) {
src/nl2mmd/normalize-mermaid.ts:459             roleIds.add(fromRole.roleId);
src/nl2mmd/normalize-mermaid.ts:460           }
src/nl2mmd/normalize-mermaid.ts:461           if (toRole) {
src/nl2mmd/normalize-mermaid.ts:462             roleIds.add(toRole.roleId);
src/nl2mmd/normalize-mermaid.ts:463           }
src/nl2mmd/normalize-mermaid.ts:464           if (fromToken === "input" && toRole) {
src/nl2mmd/normalize-mermaid.ts:465             inputTargets.push(toRole.roleId);
src/nl2mmd/normalize-mermaid.ts:466           }
src/nl2mmd/normalize-mermaid.ts:467         }
```

```
src/nl2mmd/normalize-mermaid.ts:468
src/nl2mmd/normalize-mermaid.ts:469         if (!header) {
src/nl2mmd/normalize-mermaid.ts:470           header = "flowchart TD";
src/nl2mmd/normalize-mermaid.ts:471         }
src/nl2mmd/normalize-mermaid.ts:472         if (!metadata.get("system.id")) {
src/nl2mmd/normalize-mermaid.ts:473           metadata.set("system.id", "ogsystem.nl2mmd.autofix");
src/nl2mmd/normalize-mermaid.ts:474           metadataOrder.push("system.id");
src/nl2mmd/normalize-mermaid.ts:475         }
src/nl2mmd/normalize-mermaid.ts:476         if (!metadata.get("system.version")) {
src/nl2mmd/normalize-mermaid.ts:477           metadata.set("system.version", "1");
src/nl2mmd/normalize-mermaid.ts:478           metadataOrder.push("system.version");
src/nl2mmd/normalize-mermaid.ts:479         }
src/nl2mmd/normalize-mermaid.ts:480         if (!metadata.get("law.global")) {
src/nl2mmd/normalize-mermaid.ts:481           metadata.set("law.global", args.context.lawIds[0] ?? "law.console.base");
src/nl2mmd/normalize-mermaid.ts:482           metadataOrder.push("law.global");
src/nl2mmd/normalize-mermaid.ts:483         }
src/nl2mmd/normalize-mermaid.ts:484
src/nl2mmd/normalize-mermaid.ts:485         const entryRole = metadata.get("entry.role");
src/nl2mmd/normalize-mermaid.ts:486         const inferredEntry = inputTargets[0] ?? Array.from(roleIds)[0];
src/nl2mmd/normalize-mermaid.ts:487         if (!entryRole && inferredEntry) {
src/nl2mmd/normalize-mermaid.ts:488           metadata.set("entry.role", inferredEntry);
src/nl2mmd/normalize-mermaid.ts:489           metadataOrder.push("entry.role");
src/nl2mmd/normalize-mermaid.ts:490         } else if (entryRole && !roleIds.has(entryRole) && inferredEntry) {
src/nl2mmd/normalize-mermaid.ts:491           metadata.set("entry.role", inferredEntry);
src/nl2mmd/normalize-mermaid.ts:492         }
src/nl2mmd/normalize-mermaid.ts:493
src/nl2mmd/normalize-mermaid.ts:494         repairMisplacedJoinMetadata({
src/nl2mmd/normalize-mermaid.ts:495           metadata,
src/nl2mmd/normalize-mermaid.ts:496           metadataOrder,
src/nl2mmd/normalize-mermaid.ts:497           edges
src/nl2mmd/normalize-mermaid.ts:498         });
src/nl2mmd/normalize-mermaid.ts:499
src/nl2mmd/normalize-mermaid.ts:500         const cycleComponents = collectCyclicRoleComponents(edges);
src/nl2mmd/normalize-mermaid.ts:501         for (const component of cycleComponents) {
src/nl2mmd/normalize-mermaid.ts:502           const hasLoopBudget = component.some((roleId) => metadata.has(`loop.max.${roleId}`));
src/nl2mmd/normalize-mermaid.ts:503           if (hasLoopBudget) {
src/nl2mmd/normalize-mermaid.ts:504             continue;
src/nl2mmd/normalize-mermaid.ts:505           }
src/nl2mmd/normalize-mermaid.ts:506           // Fail-closed default: inject a conservative loop budget so generated cyclic graphs
src/nl2mmd/normalize-mermaid.ts:507           // remain parseable by runtime validators without silently creating unbounded loops.
src/nl2mmd/normalize-mermaid.ts:508           const loopBudgetRole = pickLoopBudgetRole(component);
src/nl2mmd/normalize-mermaid.ts:509           const loopKey = `loop.max.${loopBudgetRole}`;
src/nl2mmd/normalize-mermaid.ts:510           metadata.set(loopKey, "3");
src/nl2mmd/normalize-mermaid.ts:511           metadataOrder.push(loopKey);
src/nl2mmd/normalize-mermaid.ts:512         }
src/nl2mmd/normalize-mermaid.ts:513
src/nl2mmd/normalize-mermaid.ts:514         const requiredOrder = ["engine", "system.id", "system.version", "law.global", "entry.role"];
src/nl2mmd/normalize-mermaid.ts:515         const emittedMetadataLines: string[] = [];
src/nl2mmd/normalize-mermaid.ts:516         const emittedKeys = new Set<string>();
src/nl2mmd/normalize-mermaid.ts:517         for (const key of requiredOrder) {
```

```
src/nl2mmd/normalize-mermaid.ts:518           const value = metadata.get(key);
src/nl2mmd/normalize-mermaid.ts:519           if (!value) {
src/nl2mmd/normalize-mermaid.ts:520             continue;
src/nl2mmd/normalize-mermaid.ts:521           }
src/nl2mmd/normalize-mermaid.ts:522           emittedMetadataLines.push(`%% ${key}=${value}`);
src/nl2mmd/normalize-mermaid.ts:523           emittedKeys.add(key);
src/nl2mmd/normalize-mermaid.ts:524         }
src/nl2mmd/normalize-mermaid.ts:525         for (const key of metadataOrder) {
src/nl2mmd/normalize-mermaid.ts:526           if (emittedKeys.has(key)) {
src/nl2mmd/normalize-mermaid.ts:527             continue;
src/nl2mmd/normalize-mermaid.ts:528           }
src/nl2mmd/normalize-mermaid.ts:529           const value = metadata.get(key);
src/nl2mmd/normalize-mermaid.ts:530           if (!value) {
src/nl2mmd/normalize-mermaid.ts:531             continue;
src/nl2mmd/normalize-mermaid.ts:532           }
src/nl2mmd/normalize-mermaid.ts:533           emittedMetadataLines.push(`%% ${key}=${value}`);
src/nl2mmd/normalize-mermaid.ts:534           emittedKeys.add(key);
src/nl2mmd/normalize-mermaid.ts:535         }
src/nl2mmd/normalize-mermaid.ts:536
src/nl2mmd/normalize-mermaid.ts:537         const edgeLines = edges.map((edge) => `${edge.fromToken} -->|${edge.eventType}| ${edge.toToken}`);
src/nl2mmd/normalize-mermaid.ts:538         return [header, ...emittedMetadataLines, ...edgeLines].join("\n").trim();
src/nl2mmd/normalize-mermaid.ts:539       }
src/nl2mmd/normalize-mermaid.ts:540
src/nl2mmd/normalize-mermaid.ts:541       /**
src/nl2mmd/normalize-mermaid.ts:542        * Normalize NL2MMD Mermaid output into the strict runtime subset.
src/nl2mmd/normalize-mermaid.ts:543        * This closes common model gaps (standalone node declarations and bare edge node ids).
src/nl2mmd/normalize-mermaid.ts:544        */
src/nl2mmd/normalize-mermaid.ts:545       export function normalizeNl2MmdMermaid(mermaid: string): string {
src/nl2mmd/normalize-mermaid.ts:546         const lines = mermaid.split(/\r?\n/).filter((line) => !isMarkdownFence(line));
src/nl2mmd/normalize-mermaid.ts:547         const nodeRoleMap = collectNodeRoleMap(lines);
src/nl2mmd/normalize-mermaid.ts:548         const normalizedLines: string[] = [];
src/nl2mmd/normalize-mermaid.ts:549
src/nl2mmd/normalize-mermaid.ts:550         for (const line of lines) {
src/nl2mmd/normalize-mermaid.ts:551           const trimmed = line.trim();
src/nl2mmd/normalize-mermaid.ts:552           if (!trimmed) {
src/nl2mmd/normalize-mermaid.ts:553             normalizedLines.push("");
src/nl2mmd/normalize-mermaid.ts:554             continue;
src/nl2mmd/normalize-mermaid.ts:555           }
src/nl2mmd/normalize-mermaid.ts:556
src/nl2mmd/normalize-mermaid.ts:557           if (parseRoleToken(trimmed)) {
src/nl2mmd/normalize-mermaid.ts:558             continue;
src/nl2mmd/normalize-mermaid.ts:559           }
src/nl2mmd/normalize-mermaid.ts:560
src/nl2mmd/normalize-mermaid.ts:561           const edgeMatch = trimmed.match(EDGE_REGEX);
src/nl2mmd/normalize-mermaid.ts:562           if (!edgeMatch) {
src/nl2mmd/normalize-mermaid.ts:563             normalizedLines.push(trimmed);
src/nl2mmd/normalize-mermaid.ts:564             continue;
src/nl2mmd/normalize-mermaid.ts:565           }
src/nl2mmd/normalize-mermaid.ts:566
src/nl2mmd/normalize-mermaid.ts:567           const fromToken = canonicalizeEndpointToken(edgeMatch[1], nodeRoleMap);
```

src/visualizer/studio-client/studio-graph.ts:2904
```
      const view = this.graph.findViewByCell(cell);
```
src/visualizer/studio-client/studio-graph.ts:2905
```
      const container = view?.container as Element | undefined;
```
src/visualizer/studio-client/studio-graph.ts:2906
```
      if (!container) {
```
src/visualizer/studio-client/studio-graph.ts:2907
```
        continue;
```
src/visualizer/studio-client/studio-graph.ts:2908
```
      }
```
src/visualizer/studio-client/studio-graph.ts:2909
```
      container.classList.toggle("is-selection-active", selectedId === cell.id);
```
src/visualizer/studio-client/studio-graph.ts:2910
```
      if (selectedId !== cell.id) {
```
src/visualizer/studio-client/studio-graph.ts:2911
```
        container.classList.remove("is-selection-focus-pulse");
```
src/visualizer/studio-client/studio-graph.ts:2912
```
      }
```
src/visualizer/studio-client/studio-graph.ts:2913
```
    }
```
src/visualizer/studio-client/studio-graph.ts:2914
```
    if (!selectedId || this.reducedMotion || this.focusMotionCellId === selectedId) {
```
src/visualizer/studio-client/studio-graph.ts:2915
```
      return;
```
src/visualizer/studio-client/studio-graph.ts:2916
```
    }
```
src/visualizer/studio-client/studio-graph.ts:2917
```
    this.focusMotionCellId = selectedId;
```
src/visualizer/studio-client/studio-graph.ts:2918
```
    const selectedCell = this.graph.getCellById(selectedId);
```
src/visualizer/studio-client/studio-graph.ts:2919
```
    if (!selectedCell) {
```
src/visualizer/studio-client/studio-graph.ts:2920
```
      return;
```
src/visualizer/studio-client/studio-graph.ts:2921
```
    }
```
src/visualizer/studio-client/studio-graph.ts:2922
```
    this.focusCell(selectedCell, { preserveViewport: options.preserveViewport === true });
```
src/visualizer/studio-client/studio-graph.ts:2923
```
    this.renderMinimap();
```
src/visualizer/studio-client/studio-graph.ts:2924
```
  }
```
src/visualizer/studio-client/studio-graph.ts:2925
```

```
src/visualizer/studio-client/studio-graph.ts:2926
```
  private focusCell(
```
src/visualizer/studio-client/studio-graph.ts:2927
```
    cell: ReturnType<Graph["getSelectedCells"]>[number],
```
src/visualizer/studio-client/studio-graph.ts:2928
```
    options: { preserveViewport?: boolean } = {}
```
src/visualizer/studio-client/studio-graph.ts:2929
```
  ): void {
```
src/visualizer/studio-client/studio-graph.ts:2930
```
    if (!cell) {
```
src/visualizer/studio-client/studio-graph.ts:2931
```
      return;
```
src/visualizer/studio-client/studio-graph.ts:2932
```
    }
```
src/visualizer/studio-client/studio-graph.ts:2933
```
    this.focusMotionCellId = cell.id;
```
src/visualizer/studio-client/studio-graph.ts:2934
```
    const selectedView = this.graph.findViewByCell(cell);
```
src/visualizer/studio-client/studio-graph.ts:2935
```
    const container = selectedView?.container as Element | undefined;
```
src/visualizer/studio-client/studio-graph.ts:2936
```
    if (!options.preserveViewport && this.shouldCenterSelectionInViewport(container)) {
```
src/visualizer/studio-client/studio-graph.ts:2937
```
      this.graph.centerCell(cell);
```
src/visualizer/studio-client/studio-graph.ts:2938
```
    }
```
src/visualizer/studio-client/studio-graph.ts:2939
```
    if (!container) {
```
src/visualizer/studio-client/studio-graph.ts:2940
```
      return;
```
src/visualizer/studio-client/studio-graph.ts:2941
```
    }
```
src/visualizer/studio-client/studio-graph.ts:2942
```
    container.classList.add("is-selection-focus-pulse");
```
src/visualizer/studio-client/studio-graph.ts:2943
```
    if (this.focusMotionTimer) {
```
src/visualizer/studio-client/studio-graph.ts:2944
```
      clearTimeout(this.focusMotionTimer);
```
src/visualizer/studio-client/studio-graph.ts:2945
```
    }
```
src/visualizer/studio-client/studio-graph.ts:2946
```
    this.focusMotionTimer = setTimeout(() => {
```
src/visualizer/studio-client/studio-graph.ts:2947
```
      container.classList.remove("is-selection-focus-pulse");
```
src/visualizer/studio-client/studio-graph.ts:2948
```
      if (this.focusMotionCellId === cell.id) {
```
src/visualizer/studio-client/studio-graph.ts:2949
```
        this.focusMotionCellId = "";
```
src/visualizer/studio-client/studio-graph.ts:2950
```
      }
```
src/visualizer/studio-client/studio-graph.ts:2951
```
      this.focusMotionTimer = null;
```
src/visualizer/studio-client/studio-graph.ts:2952
```
    }, 900);
```
src/visualizer/studio-client/studio-graph.ts:2953
```
  }
```

src/visualizer/studio-client/studio-graph.ts:2954
src/visualizer/studio-client/studio-graph.ts:2955 private renderMinimap(): void {
src/visualizer/studio-client/studio-graph.ts:2956 const allNodes = this.graph.getNodes().filter((node) => !isStudioLayoutJunction(node));
src/visualizer/studio-client/studio-graph.ts:2957 const roleNodes = allNodes.filter((node) => {
src/visualizer/studio-client/studio-graph.ts:2958 const data = node.getData() as { studioNode?: { kind?: string } } | undefined;
src/visualizer/studio-client/studio-graph.ts:2959 return data?.studioNode?.roleSeat === true;
src/visualizer/studio-client/studio-graph.ts:2960 });
src/visualizer/studio-client/studio-graph.ts:2961 if (!roleNodes.length) {
src/visualizer/studio-client/studio-graph.ts:2962 this.minimapEl.hidden = true;
src/visualizer/studio-client/studio-graph.ts:2963 this.minimapContentEl.innerHTML = "";
src/visualizer/studio-client/studio-graph.ts:2964 return;
src/visualizer/studio-client/studio-graph.ts:2965 }
src/visualizer/studio-client/studio-graph.ts:2966 this.minimapEl.hidden = false;
src/visualizer/studio-client/studio-graph.ts:2967 const metrics = allNodes.map((node) => {
src/visualizer/studio-client/studio-graph.ts:2968 const data = node.getData() as { studioNode?: { kind?: string } } | undefined;
src/visualizer/studio-client/studio-graph.ts:2969 const position = node.getPosition();
src/visualizer/studio-client/studio-graph.ts:2970 const size = node.getSize();
src/visualizer/studio-client/studio-graph.ts:2971 return {
src/visualizer/studio-client/studio-graph.ts:2972 id: node.id,
src/visualizer/studio-client/studio-graph.ts:2973 kind: data?.studioNode?.kind || "role",
src/visualizer/studio-client/studio-graph.ts:2974 x: position.x,
src/visualizer/studio-client/studio-graph.ts:2975 y: position.y,
src/visualizer/studio-client/studio-graph.ts:2976 width: size.width,
src/visualizer/studio-client/studio-graph.ts:2977 height: size.height
src/visualizer/studio-client/studio-graph.ts:2978 };
src/visualizer/studio-client/studio-graph.ts:2979 });
src/visualizer/studio-client/studio-graph.ts:2980 const minX = Math.min(...metrics.map((item) => item.x));
src/visualizer/studio-client/studio-graph.ts:2981 const minY = Math.min(...metrics.map((item) => item.y));
src/visualizer/studio-client/studio-graph.ts:2982 const maxX = Math.max(...metrics.map((item) => item.x + item.width));
src/visualizer/studio-client/studio-graph.ts:2983 const maxY = Math.max(...metrics.map((item) => item.y + item.height));
src/visualizer/studio-client/studio-graph.ts:2984 const totalWidth = Math.max(maxX - minX, 1);
src/visualizer/studio-client/studio-graph.ts:2985 const totalHeight = Math.max(maxY - minY, 1);
src/visualizer/studio-client/studio-graph.ts:2986 const edgeLines = this.graph.getEdges().filter((edge) => !isStudioLayoutBundleEdge(edge)).map((edge) => {
src/visualizer/studio-client/studio-graph.ts:2987 const sourceNode = metrics.find((m) => m.id === edge.getSourceCellId());
src/visualizer/studio-client/studio-graph.ts:2988 const targetNode = metrics.find((m) => m.id === edge.getTargetCellId());
src/visualizer/studio-client/studio-graph.ts:2989 if (!sourceNode || !targetNode) return "";
src/visualizer/studio-client/studio-graph.ts:2990 const x1 = ((sourceNode.x + sourceNode.width - minX) / totalWidth) * 100;
src/visualizer/studio-client/studio-graph.ts:2991 const y1 = ((sourceNode.y + sourceNode.height / 2 - minY) / totalHeight) * 100;
src/visualizer/studio-client/studio-graph.ts:2992 const x2 = ((targetNode.x - minX) / totalWidth) * 100;
src/visualizer/studio-client/studio-graph.ts:2993 const y2 = ((targetNode.y + targetNode.height / 2 - minY) / totalHeight) * 100;
src/visualizer/studio-client/studio-graph.ts:2994 return '<line x1="' + x1.toFixed(2) + '" y1="' + y1.toFixed(2) + '" x2="' + x2.toFixed(2) + '" y2="' + y2.toFixed(2) + '" />';
src/visualizer/studio-client/studio-graph.ts:2995 }).filter(Boolean);
src/visualizer/studio-client/studio-graph.ts:2996 const nodeHtml = metrics.map((item) => {
src/visualizer/studio-client/studio-graph.ts:2997 const left = ((item.x - minX) / totalWidth) * 100;
src/visualizer/studio-client/studio-graph.ts:2998 const top = ((item.y - minY) / totalHeight) * 100;
src/visualizer/studio-client/studio-graph.ts:2999 const width = Math.max((item.width / totalWidth) * 100, 5);
src/visualizer/studio-client/studio-graph.ts:3000 const height = Math.max((item.height / totalHeight) * 100, 8);
src/visualizer/studio-client/studio-graph.ts:3001 const selected = this.options.selectedRoleId === item.id;
src/visualizer/studio-client/studio-graph.ts:3002 const kindClass = item.kind === "boundary" ? " is-boundary" : "";
src/visualizer/studio-client/studio-graph.ts:3003 return '<div class="studio-graph-minimap-node' + kindClass + (selected ? " is-selected" : "") +

src/visualizer/studio-client/studio-graph.ts:3004
            '" data-minimap-role-id="' + this.escapeHtml(item.id) +
src/visualizer/studio-client/studio-graph.ts:3005
            '" style="left:' + left.toFixed(3) + "%;top:" + top.toFixed(3) + "%;width:" + width.toFixed(3) + "%;height:" + height.toFixed(3) + '%"> ';
src/visualizer/studio-client/studio-graph.ts:3006
        }).join("");
src/visualizer/studio-client/studio-graph.ts:3007
        const svgHtml = edgeLines.length
src/visualizer/studio-client/studio-graph.ts:3008
          ? '<svg class="studio-graph-minimap-edges" viewBox="0 0 100 100" preserveAspectRatio="none">' + edgeLines.join("") + '</svg>'
src/visualizer/studio-client/studio-graph.ts:3009
          : "";
src/visualizer/studio-client/studio-graph.ts:3010
        this.minimapContentEl.innerHTML = svgHtml + nodeHtml;
src/visualizer/studio-client/studio-graph.ts:3011
        const translate = this.graph.translate();
src/visualizer/studio-client/studio-graph.ts:3012
        const scale = this.graph.zoom();
src/visualizer/studio-client/studio-graph.ts:3013
        const viewportWidth = Math.max(this.canvasEl.clientWidth, 1);
src/visualizer/studio-client/studio-graph.ts:3014
        const viewportHeight = Math.max(this.canvasEl.clientHeight, 1);
src/visualizer/studio-client/studio-graph.ts:3015
        const viewportBoxWidth = Math.max(10, Math.min((viewportWidth / scale / totalWidth) * 100, 100));
src/visualizer/studio-client/studio-graph.ts:3016
        const viewportBoxHeight = Math.max(10, Math.min((viewportHeight / scale / totalHeight) * 100, 100));
src/visualizer/studio-client/studio-graph.ts:3017
        const viewportLeft = Math.max(0, Math.min(((-translate.tx - minX) / totalWidth) * 100, 100 - viewportBoxWidth));
src/visualizer/studio-client/studio-graph.ts:3018
        const viewportTop = Math.max(0, Math.min(((-translate.ty - minY) / totalHeight) * 100, 100 - viewportBoxHeight));
src/visualizer/studio-client/studio-graph.ts:3019
        this.minimapViewportEl.style.left = `${viewportLeft}%`;
src/visualizer/studio-client/studio-graph.ts:3020
        this.minimapViewportEl.style.top = `${viewportTop}%`;
src/visualizer/studio-client/studio-graph.ts:3021
        this.minimapViewportEl.style.width = `${viewportBoxWidth}%`;
src/visualizer/studio-client/studio-graph.ts:3022
        this.minimapViewportEl.style.height = `${viewportBoxHeight}%`;
src/visualizer/studio-client/studio-graph.ts:3023
      }
src/visualizer/studio-client/studio-graph.ts:3024
src/visualizer/studio-client/studio-graph.ts:3025
      private shouldCenterSelectionInViewport(container: Element | undefined): boolean {
src/visualizer/studio-client/studio-graph.ts:3026
        if (!container || typeof (container as HTMLElement).getBoundingClientRect !== "function") {
src/visualizer/studio-client/studio-graph.ts:3027
          return false;
src/visualizer/studio-client/studio-graph.ts:3028
        }
src/visualizer/studio-client/studio-graph.ts:3029
        const stageRect = this.stageEl.getBoundingClientRect();
src/visualizer/studio-client/studio-graph.ts:3030
        const cellRect = (container as HTMLElement).getBoundingClientRect();
src/visualizer/studio-client/studio-graph.ts:3031
        if (
src/visualizer/studio-client/studio-graph.ts:3032
          stageRect.width <= 0 ||
src/visualizer/studio-client/studio-graph.ts:3033
          stageRect.height <= 0 ||
src/visualizer/studio-client/studio-graph.ts:3034
          cellRect.width <= 0 ||
src/visualizer/studio-client/studio-graph.ts:3035
          cellRect.height <= 0
src/visualizer/studio-client/studio-graph.ts:3036
) {
src/visualizer/studio-client/studio-graph.ts:3037
          return false;
src/visualizer/studio-client/studio-graph.ts:3038
        }
src/visualizer/studio-client/studio-graph.ts:3039
        const margin = 24;
src/visualizer/studio-client/studio-graph.ts:3040
        const withinHorizontalViewport =
src/visualizer/studio-client/studio-graph.ts:3041
          cellRect.left >= stageRect.left + margin &&
src/visualizer/studio-client/studio-graph.ts:3042
          cellRect.right <= stageRect.right - margin;
src/visualizer/studio-client/studio-graph.ts:3043
        const withinVerticalViewport =
src/visualizer/studio-client/studio-graph.ts:3044
          cellRect.top >= stageRect.top + margin &&
src/visualizer/studio-client/studio-graph.ts:3045
          cellRect.bottom <= stageRect.bottom - margin;
src/visualizer/studio-client/studio-graph.ts:3046
        return !(withinHorizontalViewport && withinVerticalViewport);
src/visualizer/studio-client/studio-graph.ts:3047
      }
src/visualizer/studio-client/studio-graph.ts:3048
src/visualizer/studio-client/studio-graph.ts:3049
      private detectReducedMotion(): boolean {
src/visualizer/studio-client/studio-graph.ts:3050
        try {
src/visualizer/studio-client/studio-graph.ts:3051
          return Boolean(typeof window !== "undefined"
src/visualizer/studio-client/studio-graph.ts:3052
            && typeof window.matchMedia === "function"
src/visualizer/studio-client/studio-graph.ts:3053
            && window.matchMedia("(prefers-reduced-motion: reduce)").matches);

```
src/visualizer/studio-client/studio-graph.ts:3054          } catch {
src/visualizer/studio-client/studio-graph.ts:3055            return false;
src/visualizer/studio-client/studio-graph.ts:3056          }
src/visualizer/studio-client/studio-graph.ts:3057        }
src/visualizer/studio-client/studio-graph.ts:3058      }
src/visualizer/studio-client/studio-graph.ts:3059
src/visualizer/studio-client/styles.ts:1  const STYLE_ID = "ogs-studio-graph-island-styles";
src/visualizer/studio-client/styles.ts:2
src/visualizer/studio-client/styles.ts:3  export function injectStudioGraphStyles(): void {
src/visualizer/studio-client/styles.ts:4    if (document.getElementById(STYLE_ID)) {
src/visualizer/studio-client/styles.ts:5      return;
src/visualizer/studio-client/styles.ts:6    }
src/visualizer/studio-client/styles.ts:7    const style = document.createElement("style");
src/visualizer/studio-client/styles.ts:8    style.id = STYLE_ID;
src/visualizer/studio-client/styles.ts:9    style.textContent = `
src/visualizer/studio-client/styles.ts:10     .studio-graph-island {
src/visualizer/studio-client/styles.ts:11       display: grid;
src/visualizer/studio-client/styles.ts:12       grid-template-rows: auto minmax(0, 1fr);
src/visualizer/studio-client/styles.ts:13       min-height: 390px;
src/visualizer/studio-client/styles.ts:14       height: 100%;
src/visualizer/studio-client/styles.ts:15       border: 1px solid rgba(148, 163, 184, 0.18);
src/visualizer/studio-client/styles.ts:16       background: rgba(4, 8, 16, 0.54);
src/visualizer/studio-client/styles.ts:17       min-width: 0;
src/visualizer/studio-client/styles.ts:18       overflow: hidden;
src/visualizer/studio-client/styles.ts:19       container-type: inline-size;
src/visualizer/studio-client/styles.ts:20     }
src/visualizer/studio-client/styles.ts:21     .studio-canvas-shell.is-fullscreen .studio-graph-island {
src/visualizer/studio-client/styles.ts:22       height: calc(100vh - 108px);
src/visualizer/studio-client/styles.ts:23     }
src/visualizer/studio-client/styles.ts:24     .studio-graph-toolbar {
src/visualizer/studio-client/styles.ts:25       display: flex;
src/visualizer/studio-client/styles.ts:26       justify-content: space-between;
src/visualizer/studio-client/styles.ts:27       align-items: center;
src/visualizer/studio-client/styles.ts:28       gap: 8px;
src/visualizer/studio-client/styles.ts:29       padding: 5px 6px;
src/visualizer/studio-client/styles.ts:30       border-bottom: 1px solid rgba(148, 163, 184, 0.14);
src/visualizer/studio-client/styles.ts:31       background:
src/visualizer/studio-client/styles.ts:32         linear-gradient(180deg, rgba(15, 23, 42, 0.92), rgba(8, 13, 26, 0.76));
src/visualizer/studio-client/styles.ts:33       backdrop-filter: blur(14px);
src/visualizer/studio-client/styles.ts:34       min-width: 0;
src/visualizer/studio-client/styles.ts:35     }
src/visualizer/studio-client/styles.ts:36     .studio-graph-toolbar-main {
src/visualizer/studio-client/styles.ts:37       display: flex;
src/visualizer/studio-client/styles.ts:38       flex-wrap: nowrap;
src/visualizer/studio-client/styles.ts:39       align-items: center;
src/visualizer/studio-client/styles.ts:40       gap: 5px;
src/visualizer/studio-client/styles.ts:41       min-width: 0;
src/visualizer/studio-client/styles.ts:42       overflow-x: auto;
src/visualizer/studio-client/styles.ts:43       scrollbar-width: thin;
src/visualizer/studio-client/styles.ts:44     }
```

```
src/visualizer/studio-client/styles.ts:45     .studio-graph-toolbar-group {
src/visualizer/studio-client/styles.ts:46       display: flex;
src/visualizer/studio-client/styles.ts:47       flex-wrap: nowrap;
src/visualizer/studio-client/styles.ts:48       align-items: center;
src/visualizer/studio-client/styles.ts:49       gap: 3px;
src/visualizer/studio-client/styles.ts:50       min-width: 0;
src/visualizer/studio-client/styles.ts:51       flex-shrink: 0;
src/visualizer/studio-client/styles.ts:52       padding: 2px;
src/visualizer/studio-client/styles.ts:53       border: 1px solid rgba(148, 163, 184, 0.14);
src/visualizer/studio-client/styles.ts:54       background: rgba(255, 255, 255, 0.035);
src/visualizer/studio-client/styles.ts:55     }
src/visualizer/studio-client/styles.ts:56     .studio-graph-toolbar-group[hidden],
src/visualizer/studio-client/styles.ts:57     .studio-graph-toolbar button[hidden] {
src/visualizer/studio-client/styles.ts:58       display: none;
src/visualizer/studio-client/styles.ts:59     }
src/visualizer/studio-client/styles.ts:60     .studio-graph-toolbar button {
src/visualizer/studio-client/styles.ts:61       display: inline-flex;
src/visualizer/studio-client/styles.ts:62       align-items: center;
src/visualizer/studio-client/styles.ts:63       justify-content: center;
src/visualizer/studio-client/styles.ts:64       gap: 5px;
src/visualizer/studio-client/styles.ts:65       border: 1px solid transparent;
src/visualizer/studio-client/styles.ts:66       background: transparent;
src/visualizer/studio-client/styles.ts:67       color: #e5eefb;
src/visualizer/studio-client/styles.ts:68       padding: 4px 6px;
src/visualizer/studio-client/styles.ts:69       min-width: 28px;
src/visualizer/studio-client/styles.ts:70       min-height: 28px;
src/visualizer/studio-client/styles.ts:71       cursor: pointer;
src/visualizer/studio-client/styles.ts:72       font-size: 11px;
src/visualizer/studio-client/styles.ts:73       line-height: 1;
src/visualizer/studio-client/styles.ts:74       white-space: nowrap;
src/visualizer/studio-client/styles.ts:75       flex: 0 0 auto;
src/visualizer/studio-client/styles.ts:76       transition: background 120ms ease, border-color 120ms ease, color 120ms ease;
src/visualizer/studio-client/styles.ts:77     }
src/visualizer/studio-client/styles.ts:78     .studio-graph-toolbar button:hover {
src/visualizer/studio-client/styles.ts:79       border-color: rgba(56, 189, 248, 0.32);
src/visualizer/studio-client/styles.ts:80       background: rgba(56, 189, 248, 0.13);
src/visualizer/studio-client/styles.ts:81       color: #bae6fd;
src/visualizer/studio-client/styles.ts:82     }
src/visualizer/studio-client/styles.ts:83     .studio-graph-toolbar button:disabled {
src/visualizer/studio-client/styles.ts:84       cursor: not-allowed;
src/visualizer/studio-client/styles.ts:85       opacity: 0.55;
src/visualizer/studio-client/styles.ts:86     }
src/visualizer/studio-client/styles.ts:87     .studio-graph-status {
src/visualizer/studio-client/styles.ts:88       color: #8fa1c3;
src/visualizer/studio-client/styles.ts:89       font-size: 12px;
src/visualizer/studio-client/styles.ts:90       flex: 0 1 auto;
src/visualizer/studio-client/styles.ts:91       overflow-wrap: anywhere;
src/visualizer/studio-client/styles.ts:92     }
src/visualizer/studio-client/styles.ts:93     .studio-graph-toolbar-icon {
src/visualizer/studio-client/styles.ts:94       display: inline-flex;
```

```
src/visualizer/studio-client/styles.ts:95       width: 16px;
src/visualizer/studio-client/styles.ts:96       height: 16px;
src/visualizer/studio-client/styles.ts:97       align-items: center;
src/visualizer/studio-client/styles.ts:98       justify-content: center;
src/visualizer/studio-client/styles.ts:99       flex: 0 0 16px;
src/visualizer/studio-client/styles.ts:100    }
src/visualizer/studio-client/styles.ts:101    .studio-graph-toolbar-icon svg {
src/visualizer/studio-client/styles.ts:102      display: block;
src/visualizer/studio-client/styles.ts:103      width: 16px;
src/visualizer/studio-client/styles.ts:104      height: 16px;
src/visualizer/studio-client/styles.ts:105      fill: none;
src/visualizer/studio-client/styles.ts:106      stroke: currentColor;
src/visualizer/studio-client/styles.ts:107      stroke-width: 1.8;
src/visualizer/studio-client/styles.ts:108      stroke-linecap: round;
src/visualizer/studio-client/styles.ts:109      stroke-linejoin: round;
src/visualizer/studio-client/styles.ts:110    }
src/visualizer/studio-client/styles.ts:111    .studio-graph-toolbar-text {
src/visualizer/studio-client/styles.ts:112      font-size: 10px;
src/visualizer/studio-client/styles.ts:113    }
src/visualizer/studio-client/styles.ts:114    @media (max-width: 720px) {
src/visualizer/studio-client/styles.ts:115      .studio-graph-toolbar {
src/visualizer/studio-client/styles.ts:116        align-items: stretch;
src/visualizer/studio-client/styles.ts:117        flex-direction: column;
src/visualizer/studio-client/styles.ts:118      }
src/visualizer/studio-client/styles.ts:119      .studio-graph-toolbar-main {
src/visualizer/studio-client/styles.ts:120        overflow-x: auto;
src/visualizer/studio-client/styles.ts:121        width: 100%;
src/visualizer/studio-client/styles.ts:122      }
src/visualizer/studio-client/styles.ts:123      .studio-graph-toolbar-text {
src/visualizer/studio-client/styles.ts:124        display: none;
src/visualizer/studio-client/styles.ts:125      }
src/visualizer/studio-client/styles.ts:126      .studio-graph-quick-debug {
src/visualizer/studio-client/styles.ts:127        inset: auto 12px 12px 12px;
src/visualizer/studio-client/styles.ts:128        width: auto;
src/visualizer/studio-client/styles.ts:129      }
src/visualizer/studio-client/styles.ts:130    }
src/visualizer/studio-client/styles.ts:131    @container (max-width: 920px) {
src/visualizer/studio-client/styles.ts:132      .studio-graph-toolbar-text {
src/visualizer/studio-client/styles.ts:133        display: none;
src/visualizer/studio-client/styles.ts:134      }
src/visualizer/studio-client/styles.ts:135    }
src/visualizer/studio-client/styles.ts:136    .studio-graph-stage {
src/visualizer/studio-client/styles.ts:137      min-height: 320px;
src/visualizer/studio-client/styles.ts:138      height: 100%;
src/visualizer/studio-client/styles.ts:139      min-width: 0;
src/visualizer/studio-client/styles.ts:140      position: relative;
src/visualizer/studio-client/styles.ts:141      overflow: hidden;
src/visualizer/studio-client/styles.ts:142    }
src/visualizer/studio-client/styles.ts:143    .studio-graph-canvas {
src/visualizer/studio-client/styles.ts:144      min-height: 320px;
```

```
src/visualizer/studio-client/styles.ts:145      width: 100%;
src/visualizer/studio-client/styles.ts:146      height: 100%;
src/visualizer/studio-client/styles.ts:147      min-width: 0;
src/visualizer/studio-client/styles.ts:148      position: relative;
src/visualizer/studio-client/styles.ts:149    }
src/visualizer/studio-client/styles.ts:150    .studio-graph-canvas .x6-graph-svg,
src/visualizer/studio-client/styles.ts:151    .studio-graph-canvas svg {
src/visualizer/studio-client/styles.ts:152      width: 100% !important;
src/visualizer/studio-client/styles.ts:153      height: 100% !important;
src/visualizer/studio-client/styles.ts:154      inset: 0;
src/visualizer/studio-client/styles.ts:155    }
src/visualizer/studio-client/styles.ts:156    .studio-graph-canvas .x6-node .x6-port-body {
src/visualizer/studio-client/styles.ts:157      transition: stroke 140ms ease, fill 140ms ease, opacity 140ms ease;
src/visualizer/studio-client/styles.ts:158    }
src/visualizer/studio-client/styles.ts:159    .studio-graph-canvas .x6-node .x6-port-body,
src/visualizer/studio-client/styles.ts:160    .studio-graph-canvas .x6-node [data-studio-port] {
src/visualizer/studio-client/styles.ts:161      vector-effect: non-scaling-stroke;
src/visualizer/studio-client/styles.ts:162    }
src/visualizer/studio-client/styles.ts:163    .studio-graph-canvas .x6-node:hover rect,
src/visualizer/studio-client/styles.ts:164    .studio-graph-canvas .x6-node:hover path:first-of-type {
src/visualizer/studio-client/styles.ts:165      stroke: #7dd3fc;
src/visualizer/studio-client/styles.ts:166    }
src/visualizer/studio-client/styles.ts:167    .studio-graph-canvas .x6-node.is-selection-active rect,
src/visualizer/studio-client/styles.ts:168    .studio-graph-canvas .x6-node.is-selection-active path:first-of-type {
src/visualizer/studio-client/styles.ts:169      filter: drop-shadow(0 0 0.35rem rgba(56, 189, 248, 0.32));
src/visualizer/studio-client/styles.ts:170      stroke: #38bdf8;
src/visualizer/studio-client/styles.ts:171    }
src/visualizer/studio-client/styles.ts:172    .studio-graph-canvas .x6-node.is-selection-focus-pulse rect,
src/visualizer/studio-client/styles.ts:173    .studio-graph-canvas .x6-node.is-selection-focus-pulse path:first-of-type {
src/visualizer/studio-client/styles.ts:174      animation: studio-node-focus-pulse 820ms ease-out 1;
src/visualizer/studio-client/styles.ts:175    }
src/visualizer/studio-client/styles.ts:176    .studio-graph-canvas .x6-node rect,
src/visualizer/studio-client/styles.ts:177    .studio-graph-canvas .x6-node path:first-of-type {
src/visualizer/studio-client/styles.ts:178      transition: filter 220ms ease, stroke 220ms ease, opacity 220ms ease;
src/visualizer/studio-client/styles.ts:179    }
src/visualizer/studio-client/styles.ts:180    .studio-graph-canvas .x6-node.is-runtime-active rect,
src/visualizer/studio-client/styles.ts:181    .studio-graph-canvas .x6-node.is-runtime-active path:first-of-type {
src/visualizer/studio-client/styles.ts:182      filter: drop-shadow(0 0 0.55rem rgba(56, 189, 248, 0.22));
src/visualizer/studio-client/styles.ts:183      animation: studio-node-active-pulse 2400ms ease-in-out infinite;
src/visualizer/studio-client/styles.ts:184    }
src/visualizer/studio-client/styles.ts:185    .studio-graph-canvas .x6-node[data-runtime-status="failed"] rect,
src/visualizer/studio-client/styles.ts:186    .studio-graph-canvas .x6-node[data-runtime-status="failed"] path:first-of-type {
src/visualizer/studio-client/styles.ts:187      animation: studio-node-error-shake 400ms ease-out 1;
src/visualizer/studio-client/styles.ts:188    }
src/visualizer/studio-client/styles.ts:189    .studio-graph-canvas .x6-node.is-runtime-waiting-review rect,
src/visualizer/studio-client/styles.ts:190    .studio-graph-canvas .x6-node.is-runtime-waiting-review path:first-of-type {
src/visualizer/studio-client/styles.ts:191      animation: studio-human-gate-pulse 2200ms ease-in-out infinite;
src/visualizer/studio-client/styles.ts:192    }
src/visualizer/studio-client/styles.ts:193    .studio-graph-canvas .x6-node.has-human-gate:not(.is-runtime-waiting-review) rect,
src/visualizer/studio-client/styles.ts:194    .studio-graph-canvas .x6-node.has-human-gate:not(.is-runtime-waiting-review) path:first-of-type {
```

```
src/visualizer/studio-client/styles.ts:195      stroke-dasharray: 8 4;
src/visualizer/studio-client/styles.ts:196      filter: drop-shadow(0 0 0.2rem rgba(251, 191, 36, 0.22));
src/visualizer/studio-client/styles.ts:197    }
src/visualizer/studio-client/styles.ts:198    .studio-graph-canvas .x6-edge .connection {
src/visualizer/studio-client/styles.ts:199      transition: stroke 160ms ease, stroke-width 160ms ease, opacity 160ms ease, stroke-dasharray 160ms ease;
src/visualizer/studio-client/styles.ts:200      stroke-linecap: round;
src/visualizer/studio-client/styles.ts:201      stroke-linejoin: round;
src/visualizer/studio-client/styles.ts:202    }
src/visualizer/studio-client/styles.ts:203    .studio-graph-canvas .x6-edge-label {
src/visualizer/studio-client/styles.ts:204      pointer-events: none;
src/visualizer/studio-client/styles.ts:205    }
src/visualizer/studio-client/styles.ts:206    .studio-graph-canvas .x6-edge:hover .connection {
src/visualizer/studio-client/styles.ts:207      stroke-width: 2.1px !important;
src/visualizer/studio-client/styles.ts:208    }
src/visualizer/studio-client/styles.ts:209    .studio-graph-canvas .x6-edge.is-selection-active .connection {
src/visualizer/studio-client/styles.ts:210      stroke-width: 2.6px !important;
src/visualizer/studio-client/styles.ts:211      filter: drop-shadow(0 0 0.3rem rgba(56, 189, 248, 0.25));
src/visualizer/studio-client/styles.ts:212    }
src/visualizer/studio-client/styles.ts:213    .studio-graph-canvas .x6-edge.is-pending-preview .connection {
src/visualizer/studio-client/styles.ts:214      stroke: #38bdf8 !important;
src/visualizer/studio-client/styles.ts:215      stroke-width: 2px !important;
src/visualizer/studio-client/styles.ts:216      stroke-dasharray: 7 5;
src/visualizer/studio-client/styles.ts:217      filter: drop-shadow(0 0 0.22rem rgba(56, 189, 248, 0.2));
src/visualizer/studio-client/styles.ts:218      animation: studio-edge-flow 760ms linear infinite;
src/visualizer/studio-client/styles.ts:219      opacity: 0.95;
src/visualizer/studio-client/styles.ts:220    }
src/visualizer/studio-client/styles.ts:221    .studio-graph-canvas .x6-edge.is-pending-preview .x6-edge-label rect {
src/visualizer/studio-client/styles.ts:222      stroke: #38bdf8;
src/visualizer/studio-client/styles.ts:223      stroke-dasharray: 6 4;
src/visualizer/studio-client/styles.ts:224    }
src/visualizer/studio-client/styles.ts:225    .studio-graph-canvas .x6-edge.is-runtime-active .connection {
src/visualizer/studio-client/styles.ts:226      stroke-width: 2.5px !important;
src/visualizer/studio-client/styles.ts:227      stroke-dasharray: 10 10;
src/visualizer/studio-client/styles.ts:228      filter: drop-shadow(0 0 0.32rem rgba(56, 189, 248, 0.24));
src/visualizer/studio-client/styles.ts:229      animation: studio-edge-flow 680ms linear infinite;
src/visualizer/studio-client/styles.ts:230    }
src/visualizer/studio-client/styles.ts:231    .studio-graph-canvas .x6-edge.is-runtime-error .connection {
src/visualizer/studio-client/styles.ts:232      stroke-dasharray: 7 5;
src/visualizer/studio-client/styles.ts:233    }
src/visualizer/studio-client/styles.ts:234    .studio-graph-canvas .x6-edge.is-loop-flow .connection,
src/visualizer/studio-client/styles.ts:235    .studio-graph-canvas .x6-edge.is-loop-flow path[marker-end],
src/visualizer/studio-client/styles.ts:236    .studio-graph-canvas .x6-edge.is-loop-back .connection,
src/visualizer/studio-client/styles.ts:237    .studio-graph-canvas .x6-edge.is-loop-back path[marker-end] {
src/visualizer/studio-client/styles.ts:238      stroke-dasharray: 7 5;
src/visualizer/studio-client/styles.ts:239      stroke-width: 2.6px;
src/visualizer/studio-client/styles.ts:240      stroke-linecap: round;
src/visualizer/studio-client/styles.ts:241      filter: drop-shadow(0 0 0.3rem rgba(45, 212, 191, 0.52));
src/visualizer/studio-client/styles.ts:242    }
src/visualizer/studio-client/styles.ts:243    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node.is-selection-focus-pulse rect,
src/visualizer/studio-client/styles.ts:244    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node.is-selection-focus-pulse path:first-of-type,
```

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src/visualizer/studio-client/styles.ts:245    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node.is-runtime-active rect,
src/visualizer/studio-client/styles.ts:246    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node.is-runtime-active path:first-of-type,
src/visualizer/studio-client/styles.ts:247    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node[data-runtime-status="failed"] rect,
src/visualizer/studio-client/styles.ts:248    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node[data-runtime-status="failed"] path:first-of-type,
src/visualizer/studio-client/styles.ts:249    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node.is-runtime-waiting-review rect,
src/visualizer/studio-client/styles.ts:250    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-node.is-runtime-waiting-review path:first-of-type,
src/visualizer/studio-client/styles.ts:251    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-edge.is-pending-preview .connection,
src/visualizer/studio-client/styles.ts:252    .studio-graph-island[data-reduced-motion="on"] .studio-graph-canvas .x6-edge.is-runtime-active .connection {
src/visualizer/studio-client/styles.ts:253      animation: none !important;
src/visualizer/studio-client/styles.ts:254    }
src/visualizer/studio-client/styles.ts:255    @keyframes studio-node-focus-pulse {
src/visualizer/studio-client/styles.ts:256      0% {
src/visualizer/studio-client/styles.ts:257        transform: scale(1);
src/visualizer/studio-client/styles.ts:258      }
src/visualizer/studio-client/styles.ts:259      28% {
src/visualizer/studio-client/styles.ts:260        transform: scale(1.025);
src/visualizer/studio-client/styles.ts:261      }
src/visualizer/studio-client/styles.ts:262      100% {
src/visualizer/studio-client/styles.ts:263        transform: scale(1);
src/visualizer/studio-client/styles.ts:264      }
src/visualizer/studio-client/styles.ts:265    }
src/visualizer/studio-client/styles.ts:266    @keyframes studio-edge-flow {
src/visualizer/studio-client/styles.ts:267      from {
src/visualizer/studio-client/styles.ts:268        stroke-dashoffset: 18;
src/visualizer/studio-client/styles.ts:269      }
src/visualizer/studio-client/styles.ts:270      to {
src/visualizer/studio-client/styles.ts:271        stroke-dashoffset: 0;
src/visualizer/studio-client/styles.ts:272      }
src/visualizer/studio-client/styles.ts:273    }
src/visualizer/studio-client/styles.ts:274    @keyframes studio-human-gate-pulse {
src/visualizer/studio-client/styles.ts:275      0%,
src/visualizer/studio-client/styles.ts:276      100% {
src/visualizer/studio-client/styles.ts:277        filter: drop-shadow(0 0 0 rgba(251, 191, 36, 0));
src/visualizer/studio-client/styles.ts:278      }
src/visualizer/studio-client/styles.ts:279      50% {
src/visualizer/studio-client/styles.ts:280        filter: drop-shadow(0 0 0.45rem rgba(251, 191, 36, 0.32));
src/visualizer/studio-client/styles.ts:281      }
src/visualizer/studio-client/styles.ts:282    }
src/visualizer/studio-client/styles.ts:283    @keyframes studio-node-active-pulse {
src/visualizer/studio-client/styles.ts:284      0%, 100% {
src/visualizer/studio-client/styles.ts:285        filter: drop-shadow(0 0 0.35rem rgba(56, 189, 248, 0.15));
src/visualizer/studio-client/styles.ts:286      }
src/visualizer/studio-client/styles.ts:287      50% {
src/visualizer/studio-client/styles.ts:288        filter: drop-shadow(0 0 0.7rem rgba(56, 189, 248, 0.35));
src/visualizer/studio-client/styles.ts:289      }
src/visualizer/studio-client/styles.ts:290    }
src/visualizer/studio-client/styles.ts:291    @keyframes studio-node-error-shake {
src/visualizer/studio-client/styles.ts:292      0%, 100% { transform: translateX(0); }
src/visualizer/studio-client/styles.ts:293      20% { transform: translateX(-2px); }
src/visualizer/studio-client/styles.ts:294      40% { transform: translateX(2px); }
```

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src/visualizer/studio-client/styles.ts:295      60% { transform: translateX(-1.5px); }
src/visualizer/studio-client/styles.ts:296      80% { transform: translateX(1px); }
src/visualizer/studio-client/styles.ts:297    }
src/visualizer/studio-client/styles.ts:298    .studio-graph-empty {
src/visualizer/studio-client/styles.ts:299      display: grid;
src/visualizer/studio-client/styles.ts:300      place-items: center;
src/visualizer/studio-client/styles.ts:301      position: absolute;
src/visualizer/studio-client/styles.ts:302      inset: 0;
src/visualizer/studio-client/styles.ts:303      z-index: 2;
src/visualizer/studio-client/styles.ts:304      padding: 24px;
src/visualizer/studio-client/styles.ts:305      text-align: center;
src/visualizer/studio-client/styles.ts:306      color: #8fa1c3;
src/visualizer/studio-client/styles.ts:307      font-size: 13px;
src/visualizer/studio-client/styles.ts:308      background: rgba(4, 8, 16, 0.76);
src/visualizer/studio-client/styles.ts:309    }
src/visualizer/studio-client/styles.ts:310    .studio-graph-empty[hidden] {
src/visualizer/studio-client/styles.ts:311      display: none;
src/visualizer/studio-client/styles.ts:312    }
src/visualizer/studio-client/styles.ts:313    .studio-graph-diagnostic-card {
src/visualizer/studio-client/styles.ts:314      position: absolute;
src/visualizer/studio-client/styles.ts:315      z-index: 6;
src/visualizer/studio-client/styles.ts:316      max-width: 320px;
src/visualizer/studio-client/styles.ts:317      padding: 10px 12px;
src/visualizer/studio-client/styles.ts:318      border: 1px solid rgba(148, 163, 184, 0.28);
src/visualizer/studio-client/styles.ts:319      border-radius: 10px;
src/visualizer/studio-client/styles.ts:320      background: rgba(8, 13, 26, 0.96);
src/visualizer/studio-client/styles.ts:321      color: #dbeafe;
src/visualizer/studio-client/styles.ts:322      box-shadow: 0 18px 40px rgba(0, 0, 0, 0.36);
src/visualizer/studio-client/styles.ts:323      pointer-events: none;
src/visualizer/studio-client/styles.ts:324      display: grid;
src/visualizer/studio-client/styles.ts:325      gap: 6px;
src/visualizer/studio-client/styles.ts:326      font-size: 12px;
src/visualizer/studio-client/styles.ts:327      line-height: 1.45;
src/visualizer/studio-client/styles.ts:328    }
src/visualizer/studio-client/styles.ts:329    .studio-graph-diagnostic-card[hidden] {
src/visualizer/studio-client/styles.ts:330      display: none;
src/visualizer/studio-client/styles.ts:331    }
src/visualizer/studio-client/styles.ts:332    .studio-graph-minimap {
src/visualizer/studio-client/styles.ts:333      position: absolute;
src/visualizer/studio-client/styles.ts:334      right: 14px;
src/visualizer/studio-client/styles.ts:335      bottom: 14px;
src/visualizer/studio-client/styles.ts:336      width: 180px;
src/visualizer/studio-client/styles.ts:337      height: 120px;
src/visualizer/studio-client/styles.ts:338      border: 1px solid rgba(148, 163, 184, 0.28);
src/visualizer/studio-client/styles.ts:339      border-radius: 10px;
src/visualizer/studio-client/styles.ts:340      background: rgba(8, 13, 26, 0.92);
src/visualizer/studio-client/styles.ts:341      box-shadow: 0 16px 32px rgba(0, 0, 0, 0.24);
src/visualizer/studio-client/styles.ts:342      overflow: hidden;
src/visualizer/studio-client/styles.ts:343      z-index: 4;
src/visualizer/studio-client/styles.ts:344      pointer-events: auto;
```

```
src/visualizer/studio-client/styles.ts:345      cursor: pointer;
src/visualizer/studio-client/styles.ts:346    }
src/visualizer/studio-client/styles.ts:347    .studio-graph-minimap[hidden] {
src/visualizer/studio-client/styles.ts:348      display: none;
src/visualizer/studio-client/styles.ts:349    }
src/visualizer/studio-client/styles.ts:350    .studio-graph-minimap-content {
src/visualizer/studio-client/styles.ts:351      position: absolute;
src/visualizer/studio-client/styles.ts:352      inset: 8px;
src/visualizer/studio-client/styles.ts:353      overflow: hidden;
src/visualizer/studio-client/styles.ts:354    }
src/visualizer/studio-client/styles.ts:355    .studio-graph-minimap-edges {
src/visualizer/studio-client/styles.ts:356      position: absolute;
src/visualizer/studio-client/styles.ts:357      inset: 0;
src/visualizer/studio-client/styles.ts:358      width: 100%;
src/visualizer/studio-client/styles.ts:359      height: 100%;
src/visualizer/studio-client/styles.ts:360      pointer-events: none;
src/visualizer/studio-client/styles.ts:361    }
src/visualizer/studio-client/styles.ts:362    .studio-graph-minimap-edges line {
src/visualizer/studio-client/styles.ts:363      stroke: rgba(148, 163, 184, 0.35);
src/visualizer/studio-client/styles.ts:364      stroke-width: 0.6;
src/visualizer/studio-client/styles.ts:365    }
src/visualizer/studio-client/styles.ts:366    .studio-graph-minimap-node {
src/visualizer/studio-client/styles.ts:367      position: absolute;
src/visualizer/studio-client/styles.ts:368      min-width: 4px;
src/visualizer/studio-client/styles.ts:369      min-height: 6px;
src/visualizer/studio-client/styles.ts:370      border-radius: 3px;
src/visualizer/studio-client/styles.ts:371      background: rgba(56, 189, 248, 0.55);
src/visualizer/studio-client/styles.ts:372      border: 1px solid rgba(186, 230, 253, 0.4);
src/visualizer/studio-client/styles.ts:373    }
src/visualizer/studio-client/styles.ts:374    .studio-graph-minimap-node.is-boundary {
src/visualizer/studio-client/styles.ts:375      background: rgba(100, 116, 139, 0.45);
src/visualizer/studio-client/styles.ts:376      border-color: rgba(148, 163, 184, 0.4);
src/visualizer/studio-client/styles.ts:377      border-style: dashed;
src/visualizer/studio-client/styles.ts:378      border-radius: 2px;
src/visualizer/studio-client/styles.ts:379    }
src/visualizer/studio-client/styles.ts:380    .studio-graph-minimap-node.is-selected {
src/visualizer/studio-client/styles.ts:381      background: rgba(56, 189, 248, 0.85);
src/visualizer/studio-client/styles.ts:382      border-color: rgba(186, 230, 253, 0.8);
src/visualizer/studio-client/styles.ts:383      box-shadow: 0 0 4px rgba(56, 189, 248, 0.5);
src/visualizer/studio-client/styles.ts:384    }
src/visualizer/studio-client/styles.ts:385    .studio-graph-minimap-viewport {
src/visualizer/studio-client/styles.ts:386      position: absolute;
src/visualizer/studio-client/styles.ts:387      border: 1px solid rgba(56, 189, 248, 0.9);
src/visualizer/studio-client/styles.ts:388      background: rgba(56, 189, 248, 0.08);
src/visualizer/studio-client/styles.ts:389      border-radius: 6px;
src/visualizer/studio-client/styles.ts:390      box-shadow: inset 0 0 0 1px rgba(186, 230, 253, 0.22);
src/visualizer/studio-client/styles.ts:391      pointer-events: none;
src/visualizer/studio-client/styles.ts:392    }
src/visualizer/studio-client/styles.ts:393    .studio-graph-context-menu {
src/visualizer/studio-client/styles.ts:394      position: absolute;
```

```
src/visualizer/studio-client/styles.ts:395      z-index: 8;
src/visualizer/studio-client/styles.ts:396      display: grid;
src/visualizer/studio-client/styles.ts:397      gap: 4px;
src/visualizer/studio-client/styles.ts:398      min-width: 152px;
src/visualizer/studio-client/styles.ts:399      padding: 6px;
src/visualizer/studio-client/styles.ts:400      border: 1px solid rgba(148, 163, 184, 0.24);
src/visualizer/studio-client/styles.ts:401      border-radius: 12px;
src/visualizer/studio-client/styles.ts:402      background: rgba(8, 13, 26, 0.97);
src/visualizer/studio-client/styles.ts:403      box-shadow: 0 18px 36px rgba(0, 0, 0, 0.34);
src/visualizer/studio-client/styles.ts:404      backdrop-filter: blur(12px);
src/visualizer/studio-client/styles.ts:405    }
src/visualizer/studio-client/styles.ts:406    .studio-graph-context-menu[hidden] {
src/visualizer/studio-client/styles.ts:407      display: none;
src/visualizer/studio-client/styles.ts:408    }
src/visualizer/studio-client/styles.ts:409    .studio-graph-context-menu-item {
src/visualizer/studio-client/styles.ts:410      display: flex;
src/visualizer/studio-client/styles.ts:411      align-items: center;
src/visualizer/studio-client/styles.ts:412      justify-content: flex-start;
src/visualizer/studio-client/styles.ts:413      width: 100%;
src/visualizer/studio-client/styles.ts:414      border: 1px solid transparent;
src/visualizer/studio-client/styles.ts:415      border-radius: 8px;
src/visualizer/studio-client/styles.ts:416      background: transparent;
src/visualizer/studio-client/styles.ts:417      color: #e5eefb;
src/visualizer/studio-client/styles.ts:418      padding: 8px 10px;
src/visualizer/studio-client/styles.ts:419      font-size: 12px;
src/visualizer/studio-client/styles.ts:420      text-align: left;
src/visualizer/studio-client/styles.ts:421      cursor: pointer;
src/visualizer/studio-client/styles.ts:422      transition: background 120ms ease, border-color 120ms ease, color 120ms ease, opacity 120ms ease;
src/visualizer/studio-client/styles.ts:423    }
src/visualizer/studio-client/styles.ts:424    .studio-graph-context-menu-item:hover {
src/visualizer/studio-client/styles.ts:425      border-color: rgba(56, 189, 248, 0.34);
src/visualizer/studio-client/styles.ts:426      background: rgba(56, 189, 248, 0.12);
src/visualizer/studio-client/styles.ts:427      color: #bae6fd;
src/visualizer/studio-client/styles.ts:428    }
src/visualizer/studio-client/styles.ts:429    .studio-graph-context-menu-item.is-destructive:hover {
src/visualizer/studio-client/styles.ts:430      border-color: rgba(248, 113, 113, 0.36);
src/visualizer/studio-client/styles.ts:431      background: rgba(248, 113, 113, 0.12);
src/visualizer/studio-client/styles.ts:432      color: #fecaca;
src/visualizer/studio-client/styles.ts:433    }
src/visualizer/studio-client/styles.ts:434    .studio-graph-context-menu-item:disabled {
src/visualizer/studio-client/styles.ts:435      opacity: 0.42;
src/visualizer/studio-client/styles.ts:436      cursor: not-allowed;
src/visualizer/studio-client/styles.ts:437    }
src/visualizer/studio-client/styles.ts:438    .studio-graph-quick-open {
src/visualizer/studio-client/styles.ts:439      position: absolute;
src/visualizer/studio-client/styles.ts:440      inset: 16px auto auto 16px;
src/visualizer/studio-client/styles.ts:441      z-index: 7;
src/visualizer/studio-client/styles.ts:442      width: min(420px, calc(100% - 32px));
src/visualizer/studio-client/styles.ts:443    }
src/visualizer/studio-client/styles.ts:444    .studio-graph-quick-open[hidden] {
```

```
src/visualizer/studio-client/styles.ts:445      display: none;
src/visualizer/studio-client/styles.ts:446    }
src/visualizer/studio-client/styles.ts:447    .studio-graph-quick-open-panel {
src/visualizer/studio-client/styles.ts:448      display: grid;
src/visualizer/studio-client/styles.ts:449      gap: 8px;
src/visualizer/studio-client/styles.ts:450      padding: 12px;
src/visualizer/studio-client/styles.ts:451      border: 1px solid rgba(148, 163, 184, 0.24);
src/visualizer/studio-client/styles.ts:452      border-radius: 12px;
src/visualizer/studio-client/styles.ts:453      background: rgba(8, 13, 26, 0.96);
src/visualizer/studio-client/styles.ts:454      box-shadow: 0 24px 48px rgba(0, 0, 0, 0.34);
src/visualizer/studio-client/styles.ts:455    }
src/visualizer/studio-client/styles.ts:456    .studio-graph-quick-open-label {
src/visualizer/studio-client/styles.ts:457      font-size: 12px;
src/visualizer/studio-client/styles.ts:458      color: #8fa1c3;
src/visualizer/studio-client/styles.ts:459    }
src/visualizer/studio-client/styles.ts:460    .studio-graph-quick-open-input {
src/visualizer/studio-client/styles.ts:461      width: 100%;
src/visualizer/studio-client/styles.ts:462      min-width: 0;
src/visualizer/studio-client/styles.ts:463      border: 1px solid rgba(148, 163, 184, 0.24);
src/visualizer/studio-client/styles.ts:464      background: rgba(4, 8, 16, 0.92);
src/visualizer/studio-client/styles.ts:465      color: #e5eefb;
src/visualizer/studio-client/styles.ts:466      padding: 9px 10px;
src/visualizer/studio-client/styles.ts:467      box-sizing: border-box;
src/visualizer/studio-client/styles.ts:468    }
src/visualizer/studio-client/styles.ts:469    .studio-graph-quick-open-results {
src/visualizer/studio-client/styles.ts:470      display: grid;
src/visualizer/studio-client/styles.ts:471      gap: 6px;
src/visualizer/studio-client/styles.ts:472      max-height: 240px;
src/visualizer/studio-client/styles.ts:473      overflow: auto;
src/visualizer/studio-client/styles.ts:474    }
src/visualizer/studio-client/styles.ts:475    .studio-graph-quick-open-item {
src/visualizer/studio-client/styles.ts:476      display: grid;
src/visualizer/studio-client/styles.ts:477      gap: 3px;
src/visualizer/studio-client/styles.ts:478      width: 100%;
src/visualizer/studio-client/styles.ts:479      text-align: left;
src/visualizer/studio-client/styles.ts:480      border: 1px solid rgba(148, 163, 184, 0.18);
src/visualizer/studio-client/styles.ts:481      background: rgba(255, 255, 255, 0.03);
src/visualizer/studio-client/styles.ts:482      color: #e5eefb;
src/visualizer/studio-client/styles.ts:483      padding: 9px 10px;
src/visualizer/studio-client/styles.ts:484      cursor: pointer;
src/visualizer/studio-client/styles.ts:485    }
src/visualizer/studio-client/styles.ts:486    .studio-graph-quick-open-item.is-active,
src/visualizer/studio-client/styles.ts:487    .studio-graph-quick-open-item:hover {
src/visualizer/studio-client/styles.ts:488      border-color: rgba(56, 189, 248, 0.4);
src/visualizer/studio-client/styles.ts:489      background: rgba(56, 189, 248, 0.12);
src/visualizer/studio-client/styles.ts:490      color: #bae6fd;
src/visualizer/studio-client/styles.ts:491    }
src/visualizer/studio-client/styles.ts:492    .studio-graph-quick-open-kind,
src/visualizer/studio-client/styles.ts:493    .studio-graph-quick-open-empty {
src/visualizer/studio-client/styles.ts:494      font-size: 11px;
```

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src/visualizer/studio-client/styles.ts:495      color: #8fa1c3;
src/visualizer/studio-client/styles.ts:496    }
src/visualizer/studio-client/styles.ts:497    .studio-graph-quick-debug {
src/visualizer/studio-client/styles.ts:498      position: absolute;
src/visualizer/studio-client/styles.ts:499      inset: 16px 16px auto auto;
src/visualizer/studio-client/styles.ts:500      z-index: 7;
src/visualizer/studio-client/styles.ts:501      width: min(380px, calc(100% - 32px));
src/visualizer/studio-client/styles.ts:502    }
src/visualizer/studio-client/styles.ts:503    .studio-graph-quick-debug[hidden] {
src/visualizer/studio-client/styles.ts:504      display: none;
src/visualizer/studio-client/styles.ts:505    }
src/visualizer/studio-client/styles.ts:506    .studio-graph-quick-debug-panel {
src/visualizer/studio-client/styles.ts:507      display: grid;
src/visualizer/studio-client/styles.ts:508      gap: 8px;
src/visualizer/studio-client/styles.ts:509      padding: 12px;
src/visualizer/studio-client/styles.ts:510      border: 1px solid rgba(148, 163, 184, 0.24);
src/visualizer/studio-client/styles.ts:511      border-radius: 12px;
src/visualizer/studio-client/styles.ts:512      background: rgba(8, 13, 26, 0.96);
src/visualizer/studio-client/styles.ts:513      box-shadow: 0 24px 48px rgba(0, 0, 0, 0.34);
src/visualizer/studio-client/styles.ts:514    }
src/visualizer/studio-client/styles.ts:515    .studio-graph-quick-debug-head {
src/visualizer/studio-client/styles.ts:516      display: flex;
src/visualizer/studio-client/styles.ts:517      align-items: center;
src/visualizer/studio-client/styles.ts:518      justify-content: space-between;
src/visualizer/studio-client/styles.ts:519      gap: 8px;
src/visualizer/studio-client/styles.ts:520    }
src/visualizer/studio-client/styles.ts:521    .studio-graph-quick-debug-label {
src/visualizer/studio-client/styles.ts:522      font-size: 12px;
src/visualizer/studio-client/styles.ts:523      color: #8fa1c3;
src/visualizer/studio-client/styles.ts:524    }
src/visualizer/studio-client/styles.ts:525    .studio-graph-quick-debug-input {
src/visualizer/studio-client/styles.ts:526      width: 100%;
src/visualizer/studio-client/styles.ts:527      min-width: 0;
src/visualizer/studio-client/styles.ts:528      min-height: 104px;
src/visualizer/studio-client/styles.ts:529      box-sizing: border-box;
src/visualizer/studio-client/styles.ts:530      resize: vertical;
src/visualizer/studio-client/styles.ts:531    }
src/visualizer/studio-client/styles.ts:532    .studio-graph-quick-debug-actions {
src/visualizer/studio-client/styles.ts:533      display: flex;
src/visualizer/studio-client/styles.ts:534      flex-wrap: wrap;
src/visualizer/studio-client/styles.ts:535      justify-content: flex-end;
src/visualizer/studio-client/styles.ts:536      gap: 8px;
src/visualizer/studio-client/styles.ts:537    }
src/visualizer/studio-client/styles.ts:538    .studio-graph-quick-debug-hint {
src/visualizer/studio-client/styles.ts:539      margin-top: -2px;
src/visualizer/studio-client/styles.ts:540    }
src/visualizer/studio-client/styles.ts:541    .studio-graph-stage.is-empty .studio-graph-canvas {
src/visualizer/studio-client/styles.ts:542      opacity: 0.34;
src/visualizer/studio-client/styles.ts:543    }
src/visualizer/studio-client/styles.ts:544    .studio-command-dialog {
```

```
src/visualizer/studio-client/styles.ts:545      width: 100%;
src/visualizer/studio-client/styles.ts:546      max-width: 100%;
src/visualizer/studio-client/styles.ts:547      overflow: auto;
src/visualizer/studio-client/styles.ts:548      border: 1px solid rgba(148, 163, 184, 0.28);
src/visualizer/studio-client/styles.ts:549      background: rgba(8, 13, 26, 0.96);
src/visualizer/studio-client/styles.ts:550      box-shadow: 0 18px 60px rgba(0, 0, 0, 0.36);
src/visualizer/studio-client/styles.ts:551    }
src/visualizer/studio-client/styles.ts:552    .studio-command-dialog[hidden] {
src/visualizer/studio-client/styles.ts:553      display: none;
src/visualizer/studio-client/styles.ts:554    }
src/visualizer/studio-client/styles.ts:555    .studio-command-form {
src/visualizer/studio-client/styles.ts:556      display: grid;
src/visualizer/studio-client/styles.ts:557      gap: 10px;
src/visualizer/studio-client/styles.ts:558      padding: 12px;
src/visualizer/studio-client/styles.ts:559      color: #dbeafe;
src/visualizer/studio-client/styles.ts:560      font-size: 13px;
src/visualizer/studio-client/styles.ts:561    }
src/visualizer/studio-client/styles.ts:562    .studio-command-form-header,
src/visualizer/studio-client/styles.ts:563    .studio-command-form-actions {
src/visualizer/studio-client/styles.ts:564      display: flex;
src/visualizer/studio-client/styles.ts:565      align-items: center;
src/visualizer/studio-client/styles.ts:566      justify-content: space-between;
src/visualizer/studio-client/styles.ts:567      gap: 8px;
src/visualizer/studio-client/styles.ts:568    }
src/visualizer/studio-client/styles.ts:569    .studio-command-form-header button,
src/visualizer/studio-client/styles.ts:570    .studio-command-form-actions button {
src/visualizer/studio-client/styles.ts:571      border: 1px solid rgba(148, 163, 184, 0.24);
src/visualizer/studio-client/styles.ts:572      background: rgba(255, 255, 255, 0.05);
src/visualizer/studio-client/styles.ts:573      color: #e5eefb;
src/visualizer/studio-client/styles.ts:574      padding: 7px 10px;
src/visualizer/studio-client/styles.ts:575      cursor: pointer;
src/visualizer/studio-client/styles.ts:576    }
src/visualizer/studio-client/styles.ts:577    .studio-command-form-actions button {
src/visualizer/studio-client/styles.ts:578      background: rgba(56, 189, 248, 0.16);
src/visualizer/studio-client/styles.ts:579      border-color: rgba(56, 189, 248, 0.36);
src/visualizer/studio-client/styles.ts:580    }
src/visualizer/studio-client/styles.ts:581    .studio-command-form-actions button:disabled {
src/visualizer/studio-client/styles.ts:582      cursor: not-allowed;
src/visualizer/studio-client/styles.ts:583      opacity: 0.55;
src/visualizer/studio-client/styles.ts:584    }
src/visualizer/studio-client/styles.ts:585    .studio-command-form-row {
src/visualizer/studio-client/styles.ts:586      display: grid;
src/visualizer/studio-client/styles.ts:587      gap: 5px;
src/visualizer/studio-client/styles.ts:588      min-width: 0;
src/visualizer/studio-client/styles.ts:589    }
src/visualizer/studio-client/styles.ts:590    .studio-command-form-row.segmented {
src/visualizer/studio-client/styles.ts:591      grid-template-columns: repeat(2, minmax(0, 1fr));
src/visualizer/studio-client/styles.ts:592      gap: 8px;
src/visualizer/studio-client/styles.ts:593    }
src/visualizer/studio-client/styles.ts:594    .studio-command-form-row.segmented label,
```

```
src/visualizer/studio-client/styles.ts:595    .studio-command-form-check {
src/visualizer/studio-client/styles.ts:596      display: flex;
src/visualizer/studio-client/styles.ts:597      align-items: center;
src/visualizer/studio-client/styles.ts:598      gap: 7px;
src/visualizer/studio-client/styles.ts:599      min-width: 0;
src/visualizer/studio-client/styles.ts:600    }
src/visualizer/studio-client/styles.ts:601    .studio-command-form input,
src/visualizer/studio-client/styles.ts:602    .studio-command-form select {
src/visualizer/studio-client/styles.ts:603      width: 100%;
src/visualizer/studio-client/styles.ts:604      min-width: 0;
src/visualizer/studio-client/styles.ts:605      min-height: 36px;
src/visualizer/studio-client/styles.ts:606      border: 1px solid var(--border);
src/visualizer/studio-client/styles.ts:607      border-radius: 7px;
src/visualizer/studio-client/styles.ts:608      background: var(--control-bg);
src/visualizer/studio-client/styles.ts:609      color: var(--text);
src/visualizer/studio-client/styles.ts:610      padding: 7px 10px;
src/visualizer/studio-client/styles.ts:611      box-sizing: border-box;
src/visualizer/studio-client/styles.ts:612    }
src/visualizer/studio-client/styles.ts:613    .studio-command-form input[type="radio"],
src/visualizer/studio-client/styles.ts:614    .studio-command-form input[type="checkbox"] {
src/visualizer/studio-client/styles.ts:615      width: auto;
src/visualizer/studio-client/styles.ts:616      flex: 0 0 auto;
src/visualizer/studio-client/styles.ts:617    }
src/visualizer/studio-client/styles.ts:618    .studio-command-form-error,
src/visualizer/studio-client/styles.ts:619    .studio-command-form-diagnostic {
src/visualizer/studio-client/styles.ts:620      color: #fecaca;
src/visualizer/studio-client/styles.ts:621      font-size: 12px;
src/visualizer/studio-client/styles.ts:622      line-height: 1.35;
src/visualizer/studio-client/styles.ts:623      overflow-wrap: anywhere;
src/visualizer/studio-client/styles.ts:624    }
src/visualizer/studio-client/styles.ts:625    .studio-command-form-diagnostics {
src/visualizer/studio-client/styles.ts:626      display: grid;
src/visualizer/studio-client/styles.ts:627      gap: 6px;
src/visualizer/studio-client/styles.ts:628    }
src/visualizer/studio-client/styles.ts:629    .studio-command-form-diagnostic.warning {
src/visualizer/studio-client/styles.ts:630      color: #fde68a;
src/visualizer/studio-client/styles.ts:631    }
src/visualizer/studio-client/styles.ts:632    .studio-command-form-diagnostic.info {
src/visualizer/studio-client/styles.ts:633      color: #bae6fd;
src/visualizer/studio-client/styles.ts:634    }
src/visualizer/studio-client/styles.ts:635    @media (max-width: 540px) {
src/visualizer/studio-client/styles.ts:636      .studio-graph-island {
src/visualizer/studio-client/styles.ts:637        min-height: 340px;
src/visualizer/studio-client/styles.ts:638      }
src/visualizer/studio-client/styles.ts:639      .studio-graph-stage,
src/visualizer/studio-client/styles.ts:640      .studio-graph-canvas {
src/visualizer/studio-client/styles.ts:641        min-height: 270px;
src/visualizer/studio-client/styles.ts:642      }
src/visualizer/studio-client/styles.ts:643      .studio-graph-minimap {
src/visualizer/studio-client/styles.ts:644        right: 8px;
```

```
src/visualizer/studio-client/styles.ts:645        bottom: 8px;
src/visualizer/studio-client/styles.ts:646        width: 132px;
src/visualizer/studio-client/styles.ts:647        height: 88px;
src/visualizer/studio-client/styles.ts:648      }
src/visualizer/studio-client/styles.ts:649      .studio-graph-context-menu {
src/visualizer/studio-client/styles.ts:650        max-width: calc(100% - 16px);
src/visualizer/studio-client/styles.ts:651      }
src/visualizer/studio-client/styles.ts:652    }
src/visualizer/studio-client/styles.ts:653  `;
src/visualizer/studio-client/styles.ts:654  document.head.appendChild(style);
src/visualizer/studio-client/styles.ts:655}
src/visualizer/studio-client/styles.ts:656
src/visualizer/studio-contracts.ts:1      export const STUDIO_SYSTEM_END_ROLE_ID = "__system_end__";
src/visualizer/studio-contracts.ts:2
src/visualizer/studio-contracts.ts:3      export type StudioHumanReviewSpec = {
src/visualizer/studio-contracts.ts:4        mode?: string;
src/visualizer/studio-contracts.ts:5        timeoutSeconds?: number;
src/visualizer/studio-contracts.ts:6        timeoutAction?: string;
src/visualizer/studio-contracts.ts:7        reworkTargetRoleId?: string;
src/visualizer/studio-contracts.ts:8        reworkMax?: number;
src/visualizer/studio-contracts.ts:9        terminateScope?: string;
src/visualizer/studio-contracts.ts:10     };
src/visualizer/studio-contracts.ts:11
src/visualizer/studio-contracts.ts:12     export type StudioAuthoringDocument = {
src/visualizer/studio-contracts.ts:13       version: 1;
src/visualizer/studio-contracts.ts:14       project: {
src/visualizer/studio-contracts.ts:15         workdir: string;
src/visualizer/studio-contracts.ts:16         systemPath: string;
src/visualizer/studio-contracts.ts:17       };
src/visualizer/studio-contracts.ts:18       system: {
src/visualizer/studio-contracts.ts:19         systemId: string;
src/visualizer/studio-contracts.ts:20         systemVersion: string;
src/visualizer/studio-contracts.ts:21         legacyEngine?: "langgraph";
src/visualizer/studio-contracts.ts:22         entryRoleId: string;
src/visualizer/studio-contracts.ts:23         entryEventType?: string;
src/visualizer/studio-contracts.ts:24         lawGlobalRef: string;
src/visualizer/studio-contracts.ts:25         handoffMode?: string;
src/visualizer/studio-contracts.ts:26         handoffContracts?: string;
src/visualizer/studio-contracts.ts:27       };
src/visualizer/studio-contracts.ts:28       roles: Record<string, StudioAuthoringRole>;
src/visualizer/studio-contracts.ts:29       flows: Record<string, StudioAuthoringFlow>;
src/visualizer/studio-contracts.ts:30       layout: {
src/visualizer/studio-contracts.ts:31         nodes: Record<string, { x: number; y: number; width?: number; height?: number }>;
src/visualizer/studio-contracts.ts:32         viewport?: { x: number; y: number; zoom: number };
src/visualizer/studio-contracts.ts:33       };
src/visualizer/studio-contracts.ts:34     };
src/visualizer/studio-contracts.ts:35
src/visualizer/studio-contracts.ts:36     export type StudioAuthoringRole = {
src/visualizer/studio-contracts.ts:37       roleId: string;
src/visualizer/studio-contracts.ts:38       title?: string;
```

```
src/visualizer/studio-contracts.ts:39       bindingKind: "model" | "exec" | "noop";
src/visualizer/studio-contracts.ts:40       backend?: string;
src/visualizer/studio-contracts.ts:41       modelId?: string;
src/visualizer/studio-contracts.ts:42       modelSelectionSource?: "default" | "role";
src/visualizer/studio-contracts.ts:43       modelRef?: string;
src/visualizer/studio-contracts.ts:44       profileId?: string;
src/visualizer/studio-contracts.ts:45       routingMode?: "parallel_split";
src/visualizer/studio-contracts.ts:46       routeOrder?: string[];
src/visualizer/studio-contracts.ts:47       joinMode?: "all_of" | "quorum_of";
src/visualizer/studio-contracts.ts:48       joinMin?: number;
src/visualizer/studio-contracts.ts:49       joinSources?: string[];
src/visualizer/studio-contracts.ts:50       loopMax?: number;
src/visualizer/studio-contracts.ts:51       review?: StudioHumanReviewSpec;
src/visualizer/studio-contracts.ts:52       contextMap?: Record<string, string>;
src/visualizer/studio-contracts.ts:53     };
src/visualizer/studio-contracts.ts:54
src/visualizer/studio-contracts.ts:55     export type StudioAuthoringFlow = {
src/visualizer/studio-contracts.ts:56       flowId: string;
src/visualizer/studio-contracts.ts:57       fromRoleId: string;
src/visualizer/studio-contracts.ts:58       toRoleId: string;
src/visualizer/studio-contracts.ts:59       eventType: string;
src/visualizer/studio-contracts.ts:60       label?: string;
src/visualizer/studio-contracts.ts:61       runtimeOnlyErrorFlow?: boolean;
src/visualizer/studio-contracts.ts:62     };
src/visualizer/studio-contracts.ts:63
src/visualizer/studio-contracts.ts:64     /** Internal graph snapshot consumed by X6 and the authoring adapter. */
src/visualizer/studio-contracts.ts:65     export type StudioGraphSnapshot = {
src/visualizer/studio-contracts.ts:66       version: 1;
src/visualizer/studio-contracts.ts:67       nodes: Array<{
src/visualizer/studio-contracts.ts:68         id: string;
src/visualizer/studio-contracts.ts:69         roleId: string;
src/visualizer/studio-contracts.ts:70         x: number;
src/visualizer/studio-contracts.ts:71         y: number;
src/visualizer/studio-contracts.ts:72         width: number;
src/visualizer/studio-contracts.ts:73         height: number;
src/visualizer/studio-contracts.ts:74         label: string;
src/visualizer/studio-contracts.ts:75         badges: string[];
src/visualizer/studio-contracts.ts:76         bindingKind: StudioAuthoringRole["bindingKind"];
src/visualizer/studio-contracts.ts:77       }>;
src/visualizer/studio-contracts.ts:78       edges: Array<{
src/visualizer/studio-contracts.ts:79         id?: string;
src/visualizer/studio-contracts.ts:80         source: string;
src/visualizer/studio-contracts.ts:81         target: string;
src/visualizer/studio-contracts.ts:82         label: string;
src/visualizer/studio-contracts.ts:83         eventType: string;
src/visualizer/studio-contracts.ts:84         runtimeOnlyErrorFlow: boolean;
src/visualizer/studio-contracts.ts:85         participatesInJoin: boolean;
src/visualizer/studio-contracts.ts:86       }>;
src/visualizer/studio-contracts.ts:87       viewport?: { x: number; y: number; zoom: number };
src/visualizer/studio-contracts.ts:88     };
```

```
src/visualizer/studio-contracts.ts:89
src/visualizer/studio-contracts.ts:90     export type StudioDiagnosticDto = {
src/visualizer/studio-contracts.ts:91       source: "client-preflight" | "server-validation" | "parser" | "compiler" | "readiness" | "capability";
src/visualizer/studio-contracts.ts:92       severity: "info" | "warning" | "error";
src/visualizer/studio-contracts.ts:93       fieldPath?: string;
src/visualizer/studio-contracts.ts:94       roleId?: string;
src/visualizer/studio-contracts.ts:95       flowKey?: string;
src/visualizer/studio-contracts.ts:96       code: string;
src/visualizer/studio-contracts.ts:97       messageKey: string;
src/visualizer/studio-contracts.ts:98       message?: string;
src/visualizer/studio-contracts.ts:99       vars?: Record<string, unknown>;
src/visualizer/studio-contracts.ts:100      selector?: string;
src/visualizer/studio-contracts.ts:101    };
src/visualizer/studio-contracts.ts:102
src/visualizer/studio-contracts.ts:103    export type GraphViewModelMode = "edit" | "run";
src/visualizer/studio-contracts.ts:104
src/visualizer/studio-contracts.ts:105    export type GraphViewModelLayout = {
src/visualizer/studio-contracts.ts:106      x: number;
src/visualizer/studio-contracts.ts:107      y: number;
src/visualizer/studio-contracts.ts:108      width: number;
src/visualizer/studio-contracts.ts:109      height: number;
src/visualizer/studio-contracts.ts:110    };
src/visualizer/studio-contracts.ts:111
src/visualizer/studio-contracts.ts:112    export type GraphViewModelNodeStructure = {
src/visualizer/studio-contracts.ts:113      routingMode?: StudioAuthoringRole["routingMode"];
src/visualizer/studio-contracts.ts:114      routeOrder?: string[];
src/visualizer/studio-contracts.ts:115      joinMode?: StudioAuthoringRole["joinMode"];
src/visualizer/studio-contracts.ts:116      joinMin?: number;
src/visualizer/studio-contracts.ts:117      joinSources?: string[];
src/visualizer/studio-contracts.ts:118      loopMax?: number;
src/visualizer/studio-contracts.ts:119      review?: StudioHumanReviewSpec;
src/visualizer/studio-contracts.ts:120      contextFields?: string[];
src/visualizer/studio-contracts.ts:121      modes?: string[];
src/visualizer/studio-contracts.ts:122      loopScope?: {
src/visualizer/studio-contracts.ts:123        loopId: string;
src/visualizer/studio-contracts.ts:124        boundaryRoleId: string;
src/visualizer/studio-contracts.ts:125        maxRounds: number;
src/visualizer/studio-contracts.ts:126        onExhausted: string;
src/visualizer/studio-contracts.ts:127      };
src/visualizer/studio-contracts.ts:128    };
src/visualizer/studio-contracts.ts:129
src/visualizer/studio-contracts.ts:130    export type GraphViewModelJoinWaiting = {
src/visualizer/studio-contracts.ts:131      expectedCount: number;
src/visualizer/studio-contracts.ts:132      readyCount: number;
src/visualizer/studio-contracts.ts:133      missingCount: number;
src/visualizer/studio-contracts.ts:134    };
src/visualizer/studio-contracts.ts:135
src/visualizer/studio-contracts.ts:136    export type GraphViewModelNodeRuntime = {
src/visualizer/studio-contracts.ts:137      status: string;
src/visualizer/studio-contracts.ts:138      activeBranchCount: number;
```

```
src/visualizer/studio-contracts.ts:139      completedBranchCount: number;
src/visualizer/studio-contracts.ts:140      waitingReviewCount: number;
src/visualizer/studio-contracts.ts:141      pendingReviewCount: number;
src/visualizer/studio-contracts.ts:142      loopIteration: number;
src/visualizer/studio-contracts.ts:143      lastErrorCode?: string;
src/visualizer/studio-contracts.ts:144      lastSelectedEvent?: string;
src/visualizer/studio-contracts.ts:145      expectedSources: string[];
src/visualizer/studio-contracts.ts:146      readySources: string[];
src/visualizer/studio-contracts.ts:147      missingSources: string[];
src/visualizer/studio-contracts.ts:148      joinWaitingSummary: GraphViewModelJoinWaiting | null;
src/visualizer/studio-contracts.ts:149      lastFailure?: Record<string, unknown>;
src/visualizer/studio-contracts.ts:150    };
src/visualizer/studio-contracts.ts:151
src/visualizer/studio-contracts.ts:152    export type GraphViewModelDiagnostic = {
src/visualizer/studio-contracts.ts:153      severity: "warning" | "error";
src/visualizer/studio-contracts.ts:154      code?: string;
src/visualizer/studio-contracts.ts:155      message?: string;
src/visualizer/studio-contracts.ts:156    };
src/visualizer/studio-contracts.ts:157
src/visualizer/studio-contracts.ts:158    type GraphViewModelNodeBase = {
src/visualizer/studio-contracts.ts:159      id: string;
src/visualizer/studio-contracts.ts:160      roleId: string;
src/visualizer/studio-contracts.ts:161      label: string;
src/visualizer/studio-contracts.ts:162      badges: string[];
src/visualizer/studio-contracts.ts:163      structure: GraphViewModelNodeStructure;
src/visualizer/studio-contracts.ts:164      layout: GraphViewModelLayout;
src/visualizer/studio-contracts.ts:165      runtime?: GraphViewModelNodeRuntime;
src/visualizer/studio-contracts.ts:166      diagnostic?: GraphViewModelDiagnostic;
src/visualizer/studio-contracts.ts:167      editable: boolean;
src/visualizer/studio-contracts.ts:168      topologyComponentId?: string;
src/visualizer/studio-contracts.ts:169    };
src/visualizer/studio-contracts.ts:170
src/visualizer/studio-contracts.ts:171    /** A rendered node that represents one accountable role seat and its aggregate runtime data. */
src/visualizer/studio-contracts.ts:172    export type ResponsibilitySeatNode = GraphViewModelNodeBase & {
src/visualizer/studio-contracts.ts:173      kind: "roleSeat";
src/visualizer/studio-contracts.ts:174      entityKind: "responsibility_seat";
src/visualizer/studio-contracts.ts:175      roleSeat: true;
src/visualizer/studio-contracts.ts:176      executionScope: "roleAggregate";
src/visualizer/studio-contracts.ts:177      bindingKind: StudioAuthoringRole["bindingKind"];
src/visualizer/studio-contracts.ts:178    };
src/visualizer/studio-contracts.ts:179
src/visualizer/studio-contracts.ts:180    /** A synthetic graph boundary; it can never be edited or treated as a role execution seat. */
src/visualizer/studio-contracts.ts:181    export type BoundaryNode = GraphViewModelNodeBase & {
src/visualizer/studio-contracts.ts:182      kind: "boundary";
src/visualizer/studio-contracts.ts:183      entityKind: "boundary";
src/visualizer/studio-contracts.ts:184      roleSeat: false;
src/visualizer/studio-contracts.ts:185      executionScope: "boundary";
src/visualizer/studio-contracts.ts:186      bindingKind: "boundary";
src/visualizer/studio-contracts.ts:187    };
src/visualizer/studio-contracts.ts:188
```

```
src/visualizer/studio-contracts.ts:189  export type GraphViewModelNode = ResponsibilitySeatNode | BoundaryNode;
src/visualizer/studio-contracts.ts:190
src/visualizer/studio-contracts.ts:191  export type GraphViewModelEdgeRuntime = {
src/visualizer/studio-contracts.ts:192    recentlyActivated: boolean;
src/visualizer/studio-contracts.ts:193  };
src/visualizer/studio-contracts.ts:194
src/visualizer/studio-contracts.ts:195  export type GraphViewModelEdge = {
src/visualizer/studio-contracts.ts:196    id: string;
src/visualizer/studio-contracts.ts:197    source: string;
src/visualizer/studio-contracts.ts:198    target: string;
src/visualizer/studio-contracts.ts:199    eventType: string;
src/visualizer/studio-contracts.ts:200    label: string;
src/visualizer/studio-contracts.ts:201    runtimeOnlyErrorFlow: boolean;
src/visualizer/studio-contracts.ts:202    participatesInJoin: boolean;
src/visualizer/studio-contracts.ts:203    conditionSummary?: string;
src/visualizer/studio-contracts.ts:204    priority?: number;
src/visualizer/studio-contracts.ts:205    channel?: "normal" | "error" | "loop" | "join" | "feedback";
src/visualizer/studio-contracts.ts:206    runtime?: GraphViewModelEdgeRuntime;
src/visualizer/studio-contracts.ts:207    diagnostic?: GraphViewModelDiagnostic;
src/visualizer/studio-contracts.ts:208    editable: boolean;
src/visualizer/studio-contracts.ts:209    topologyOrder?: string;
src/visualizer/studio-contracts.ts:210  };
src/visualizer/studio-contracts.ts:211
src/visualizer/studio-contracts.ts:212  export type GraphViewModel = {
src/visualizer/studio-contracts.ts:213    version: 1;
src/visualizer/studio-contracts.ts:214    mode: GraphViewModelMode;
src/visualizer/studio-contracts.ts:215    nodes: GraphViewModelNode[];
src/visualizer/studio-contracts.ts:216    edges: GraphViewModelEdge[];
src/visualizer/studio-contracts.ts:217    viewport?: { x: number; y: number; zoom: number };
src/visualizer/studio-contracts.ts:218    capabilities: {
src/visualizer/studio-contracts.ts:219      editable: boolean;
src/visualizer/studio-contracts.ts:220      canAddRole: boolean;
src/visualizer/studio-contracts.ts:221      canAddEdge: boolean;
src/visualizer/studio-contracts.ts:222      canDelete: boolean;
src/visualizer/studio-contracts.ts:223    };
src/visualizer/studio-contracts.ts:224    validation: {
src/visualizer/studio-contracts.ts:225      ok: boolean;
src/visualizer/studio-contracts.ts:226      diagnostics: StudioDiagnosticDto[];
src/visualizer/studio-contracts.ts:227    };
src/visualizer/studio-contracts.ts:228  };
src/visualizer/studio-contracts.ts:229
src/visualizer/studio-contracts.ts:230  export type GraphReadingMode = "all" | "upstream" | "downstream" | "route";
src/visualizer/studio-contracts.ts:231
src/visualizer/studio-contracts.ts:232  /** Stable graph reading state shared by URL serialization and pure view projections. */
src/visualizer/studio-contracts.ts:233  export type GraphReadingState = {
src/visualizer/studio-contracts.ts:234    mode: GraphReadingMode;
src/visualizer/studio-contracts.ts:235    roleId?: string;
src/visualizer/studio-contracts.ts:236    flowKey?: string;
src/visualizer/studio-contracts.ts:237    channel?: "normal" | "error" | "loop" | "join" | "feedback";
src/visualizer/studio-contracts.ts:238  };
```

```
src/visualizer/studio-contracts.ts:239
src/visualizer/studio-contracts.ts:240  export function normalizeStudioGraphTargetRoleId(roleId: unknown): string {
src/visualizer/studio-contracts.ts:241    const value = String(roleId ?? "");
src/visualizer/studio-contracts.ts:242    return value === STUDIO_SYSTEM_END_ROLE_ID ? "output" : value;
src/visualizer/studio-contracts.ts:243  }
src/visualizer/studio-contracts.ts:244
src/visualizer/studio-contracts.ts:245  export function normalizeStudioGraphStoredRoleId(roleId: unknown): string {
src/visualizer/studio-contracts.ts:246    const value = String(roleId ?? "");
src/visualizer/studio-contracts.ts:247    return value === "output" ? STUDIO_SYSTEM_END_ROLE_ID : value;
src/visualizer/studio-contracts.ts:248  }
src/visualizer/studio-contracts.ts:249
src/visualizer/studio-contracts.ts:250  export function studioFlowKey(flow: Pick<StudioAuthoringFlow, "fromRoleId" | "eventType" | "toRoleId">): string {
src/visualizer/studio-contracts.ts:251    return `${flow.fromRoleId}:${flow.eventType}:${normalizeStudioGraphTargetRoleId(flow.toRoleId)}`;
src/visualizer/studio-contracts.ts:252  }
src/visualizer/studio-contracts.ts:253
src/visualizer/studio-edge-semantics.ts:1 import type { GraphViewModelEdge } from "./studio-contracts.js";
src/visualizer/studio-edge-semantics.ts:2
src/visualizer/studio-edge-semantics.ts:3 function formatStudioEdgeDetails(edge: GraphViewModelEdge, includeTopology: boolean): string[] {
src/visualizer/studio-edge-semantics.ts:4   return [
src/visualizer/studio-edge-semantics.ts:5     includeTopology && edge.topologyOrder ? `#${edge.topologyOrder}` : "",
src/visualizer/studio-edge-semantics.ts:6     edge.channel,
src/visualizer/studio-edge-semantics.ts:7     edge.priority === undefined ? "" : `p${edge.priority}`,
src/visualizer/studio-edge-semantics.ts:8     edge.conditionSummary ? `when:${edge.conditionSummary}` : ""
src/visualizer/studio-edge-semantics.ts:9   ].filter((value): value is string => Boolean(value));
src/visualizer/studio-edge-semantics.ts:10 }
src/visualizer/studio-edge-semantics.ts:11
src/visualizer/studio-edge-semantics.ts:12 /** Produces the semantic portion of the label rendered by the X6 graph edge. */
src/visualizer/studio-edge-semantics.ts:13 export function formatStudioEdgeLabel(edge: GraphViewModelEdge): string {
src/visualizer/studio-edge-semantics.ts:14   const details = formatStudioEdgeDetails(edge, true);
src/visualizer/studio-edge-semantics.ts:15   return details.length ? `${edge.label}  [${details.join(" ")}]` : edge.label;
src/visualizer/studio-edge-semantics.ts:16 }
src/visualizer/studio-edge-semantics.ts:17
src/visualizer/studio-edge-semantics.ts:18 /** Keeps the primary flow label compact when the topology overlay is enabled. */
src/visualizer/studio-edge-semantics.ts:19 export function formatStudioEdgeSummaryLabel(edge: GraphViewModelEdge): string {
src/visualizer/studio-edge-semantics.ts:20   const details = formatStudioEdgeDetails(edge, false);
src/visualizer/studio-edge-semantics.ts:21   return details.length ? `${edge.label}  [${details.join(" ")}]` : edge.label;
src/visualizer/studio-edge-semantics.ts:22 }
src/visualizer/studio-edge-semantics.ts:23
src/visualizer/studio-edge-semantics.ts:24 export function formatStudioEdgeTopologyLabel(edge: GraphViewModelEdge): string {
src/visualizer/studio-edge-semantics.ts:25   return edge.topologyOrder ? `#${edge.topologyOrder}` : "";
src/visualizer/studio-edge-semantics.ts:26 }
src/visualizer/studio-edge-semantics.ts:27
src/visualizer/studio-graph-command-      export {
forms.ts:1
src/visualizer/studio-graph-command-        commandFromStudioCommandFormState,
forms.ts:2
src/visualizer/studio-graph-command-        createDefaultStudioCommandFormState,
forms.ts:3
src/visualizer/studio-graph-command-        renderStudioCommandForm,
forms.ts:4
src/visualizer/studio-graph-command-        type StudioCommandFormKind,
forms.ts:5
src/visualizer/studio-graph-command-        type StudioCommandFormLabels,
forms.ts:6
src/visualizer/studio-graph-command-        type StudioCommandFormState
forms.ts:7
src/visualizer/studio-graph-command-      } from "./studio-client/studio-graph-command-forms.js";
forms.ts:8
```

src/visualizer/studio-graph-command-forms.ts:9

```
src/visualizer/studio-graph-commands.ts:1 export {
src/visualizer/studio-graph-commands.ts:2   applyStudioAuthoringCommand,
src/visualizer/studio-graph-commands.ts:3   deriveInverseCommand,
src/visualizer/studio-graph-commands.ts:4   type StudioAuthoringCommand,
src/visualizer/studio-graph-commands.ts:5   type StudioAuthoringCommandResult
src/visualizer/studio-graph-commands.ts:6 } from "./studio-client/studio-graph-commands.js";
src/visualizer/studio-graph-commands.ts:7
src/visualizer/studio-templates.ts:1      import {
src/visualizer/studio-templates.ts:2        importMermaidToAuthoring,
src/visualizer/studio-templates.ts:3        type StudioAuthoringDocument
src/visualizer/studio-templates.ts:4      } from "./studio-authoring.js";
src/visualizer/studio-templates.ts:5
src/visualizer/studio-templates.ts:6      export type StudioTemplateId =
src/visualizer/studio-templates.ts:7        | "debate"
src/visualizer/studio-templates.ts:8        | "consultation"
src/visualizer/studio-templates.ts:9        | "review"
src/visualizer/studio-templates.ts:10       | "quorum"
src/visualizer/studio-templates.ts:11       | "compensation"
src/visualizer/studio-templates.ts:12       | "zh-dev-team";
src/visualizer/studio-templates.ts:13
src/visualizer/studio-templates.ts:14     export type StudioAuthoringTemplate = {
src/visualizer/studio-templates.ts:15       id: StudioTemplateId;
src/visualizer/studio-templates.ts:16       title: string;
src/visualizer/studio-templates.ts:17       description: string;
src/visualizer/studio-templates.ts:18       systemSource: string;
src/visualizer/studio-templates.ts:19     };
src/visualizer/studio-templates.ts:20
src/visualizer/studio-templates.ts:21     function baseMetadata(systemId: string, entryRoleId: string): string[] {
src/visualizer/studio-templates.ts:22       return [
src/visualizer/studio-templates.ts:23         "flowchart TD",
src/visualizer/studio-templates.ts:24         `%% system.id=${systemId}`,
src/visualizer/studio-templates.ts:25         "%% system.version=1.0.0",
src/visualizer/studio-templates.ts:26         "%% law.global=law.minimal.base",
src/visualizer/studio-templates.ts:27         `%% entry.role=${entryRoleId}`
src/visualizer/studio-templates.ts:28       ];
src/visualizer/studio-templates.ts:29     }
src/visualizer/studio-templates.ts:30
src/visualizer/studio-templates.ts:31     const templates: StudioAuthoringTemplate[] = [
src/visualizer/studio-templates.ts:32       {
src/visualizer/studio-templates.ts:33         id: "debate",
src/visualizer/studio-templates.ts:34         title: "Debate",
src/visualizer/studio-templates.ts:35         description: "Parallel proposer and critic roles join into a final synthesis role.",
src/visualizer/studio-templates.ts:36         systemSource: [
src/visualizer/studio-templates.ts:37           ...baseMetadata("studio.template.debate", "moderator"),
src/visualizer/studio-templates.ts:38           "%% role.mode.moderator=parallel_split",
src/visualizer/studio-templates.ts:39           "%% route.order.moderator=proposer,critic",
src/visualizer/studio-templates.ts:40           "%% join.mode.synthesizer=all_of",
src/visualizer/studio-templates.ts:41           "%% join.sources.synthesizer=proposer,critic",
src/visualizer/studio-templates.ts:42           "input -->|START| a[Role:moderator]",
```

```
src/visualizer/studio-templates.ts:43           "a[Role:moderator] -->|PROPOSE| b[Role:proposer]",
src/visualizer/studio-templates.ts:44           "a[Role:moderator] -->|CRITIQUE| c[Role:critic]",
src/visualizer/studio-templates.ts:45           "b[Role:proposer] -->|DONE| d[Role:synthesizer]",
src/visualizer/studio-templates.ts:46           "c[Role:critic] -->|DONE| d[Role:synthesizer]",
src/visualizer/studio-templates.ts:47           "d[Role:synthesizer] -->|DONE| output",
src/visualizer/studio-templates.ts:48           ""
src/visualizer/studio-templates.ts:49         ].join("\n")
src/visualizer/studio-templates.ts:50       },
src/visualizer/studio-templates.ts:51       {
src/visualizer/studio-templates.ts:52         id: "consultation",
src/visualizer/studio-templates.ts:53         title: "Consultation",
src/visualizer/studio-templates.ts:54         description: "A coordinator fans out to specialists and joins their findings.",
src/visualizer/studio-templates.ts:55         systemSource: [
src/visualizer/studio-templates.ts:56           ...baseMetadata("studio.template.consultation", "coordinator"),
src/visualizer/studio-templates.ts:57           "%% role.mode.coordinator=parallel_split",
src/visualizer/studio-templates.ts:58           "%% route.order.coordinator=specialist_a,specialist_b",
src/visualizer/studio-templates.ts:59           "%% join.mode.summary=all_of",
src/visualizer/studio-templates.ts:60           "%% join.sources.summary=specialist_a,specialist_b",
src/visualizer/studio-templates.ts:61           "input -->|START| a[Role:coordinator]",
src/visualizer/studio-templates.ts:62           "a[Role:coordinator] -->|ASK_A| b[Role:specialist_a]",
src/visualizer/studio-templates.ts:63           "a[Role:coordinator] -->|ASK_B| c[Role:specialist_b]",
src/visualizer/studio-templates.ts:64           "b[Role:specialist_a] -->|DONE| d[Role:summary]",
src/visualizer/studio-templates.ts:65           "c[Role:specialist_b] -->|DONE| d[Role:summary]",
src/visualizer/studio-templates.ts:66           "d[Role:summary] -->|DONE| output",
src/visualizer/studio-templates.ts:67           ""
src/visualizer/studio-templates.ts:68         ].join("\n")
src/visualizer/studio-templates.ts:69       },
src/visualizer/studio-templates.ts:70       {
src/visualizer/studio-templates.ts:71         id: "review",
src/visualizer/studio-templates.ts:72         title: "Review",
src/visualizer/studio-templates.ts:73         description: "A writer role uses runtime-native human review with rework feedback projected back into the same role.",
src/visualizer/studio-templates.ts:74         systemSource: [
src/visualizer/studio-templates.ts:75           ...baseMetadata("studio.template.review", "writer"),
src/visualizer/studio-templates.ts:76           "%% review.mode.writer=required",
src/visualizer/studio-templates.ts:77           "%% review.timeout.writer=86400",
src/visualizer/studio-templates.ts:78           "%% review.timeout.action.writer=pause",
src/visualizer/studio-templates.ts:79           "%% review.rework.target.writer=writer",
src/visualizer/studio-templates.ts:80           "%% review.rework.max.writer=2",
src/visualizer/studio-templates.ts:81           "%% review.terminate.scope.writer=branch",
src/visualizer/studio-templates.ts:82           "%% context.map.writer.task=global.task",
src/visualizer/studio-templates.ts:83           "%% context.map.writer.review_comment=global.human_review.current.comment?",
src/visualizer/studio-templates.ts:84           "%% context.map.writer.review_round=global.human_review.current.round?",
src/visualizer/studio-templates.ts:85           "%% context.map.writer.previous_output=global.human_review.current.previous_output.content?",
src/visualizer/studio-templates.ts:86           "input -->|START| a[Role:writer]",
src/visualizer/studio-templates.ts:87           "a[Role:writer] -->|DRAFT_READY| output",
src/visualizer/studio-templates.ts:88           ""
src/visualizer/studio-templates.ts:89         ].join("\n")
src/visualizer/studio-templates.ts:90       },
src/visualizer/studio-templates.ts:91       {
src/visualizer/studio-templates.ts:92         id: "quorum",
```

```
src/visualizer/studio-templates.ts:93         title: "Quorum",
src/visualizer/studio-templates.ts:94         description: "A coordinator fans out to three specialists and a decision role proceeds after a quorum is reached.",
src/visualizer/studio-templates.ts:95         systemSource: [
src/visualizer/studio-templates.ts:96           ...baseMetadata("studio.template.quorum", "coordinator"),
src/visualizer/studio-templates.ts:97           "%% role.mode.coordinator=parallel_split",
src/visualizer/studio-templates.ts:98           "%% route.order.coordinator=specialist_a,specialist_b,specialist_c",
src/visualizer/studio-templates.ts:99           "%% join.mode.decision=quorum_of",
src/visualizer/studio-templates.ts:100          "%% join.sources.decision=specialist_a,specialist_b,specialist_c",
src/visualizer/studio-templates.ts:101          "%% join.min.decision=2",
src/visualizer/studio-templates.ts:102          "%% context.map.decision.task=global.task",
src/visualizer/studio-templates.ts:103          "input -->|CASE_RECEIVED| a[Role:coordinator]",
src/visualizer/studio-templates.ts:104          "a[Role:coordinator] -->|ASK_A| b[Role:specialist_a]",
src/visualizer/studio-templates.ts:105          "a[Role:coordinator] -->|ASK_B| c[Role:specialist_b]",
src/visualizer/studio-templates.ts:106          "a[Role:coordinator] -->|ASK_C| d[Role:specialist_c]",
src/visualizer/studio-templates.ts:107          "b[Role:specialist_a] -->|A_READY| e[Role:decision]",
src/visualizer/studio-templates.ts:108          "c[Role:specialist_b] -->|B_READY| e[Role:decision]",
src/visualizer/studio-templates.ts:109          "d[Role:specialist_c] -->|C_READY| e[Role:decision]",
src/visualizer/studio-templates.ts:110          "e[Role:decision] -->|DECISION_READY| output",
src/visualizer/studio-templates.ts:111          ""
src/visualizer/studio-templates.ts:112        ].join("\n")
src/visualizer/studio-templates.ts:113      },
src/visualizer/studio-templates.ts:114      {
src/visualizer/studio-templates.ts:115        id: "compensation",
src/visualizer/studio-templates.ts:116        title: "Compensation",
src/visualizer/studio-templates.ts:117        description: "An execution role routes runtime failures through typed ERROR* compensation edges.",
src/visualizer/studio-templates.ts:118        systemSource: [
src/visualizer/studio-templates.ts:119          ...baseMetadata("studio.template.compensation", "worker"),
src/visualizer/studio-templates.ts:120          "%% exec.bind.worker=profile.worker",
src/visualizer/studio-templates.ts:121          "%% exec.bind.error-handler-base=profile.error-handler",
src/visualizer/studio-templates.ts:122          "input -->|START| a[Role:worker]",
src/visualizer/studio-templates.ts:123          "a[Role:worker] -->|WORK_DONE| c[Role:finalizer]",
src/visualizer/studio-templates.ts:124          "a[Role:worker] -->|ERROR.TOOL_EXECUTION_TIMEOUT| b[Role:error-handler-base]",
src/visualizer/studio-templates.ts:125          "a[Role:worker] -->|ERROR| b[Role:error-handler-base]",
src/visualizer/studio-templates.ts:126          "b[Role:error-handler-base] -->|COMPENSATED| c[Role:finalizer]",
src/visualizer/studio-templates.ts:127          "b[Role:error-handler-base] -->|ESCALATED| output",
src/visualizer/studio-templates.ts:128          "b[Role:error-handler-base] -->|ABORTED| output",
src/visualizer/studio-templates.ts:129          "c[Role:finalizer] -->|DONE| output",
src/visualizer/studio-templates.ts:130          ""
src/visualizer/studio-templates.ts:131        ].join("\n")
src/visualizer/studio-templates.ts:132      },
src/visualizer/studio-templates.ts:133      {
src/visualizer/studio-templates.ts:134        id: "zh-dev-team",
src/visualizer/studio-templates.ts:135        title: "Chinese Dev Team",
src/visualizer/studio-templates.ts:136        description: "A Chinese multi-agent delivery team splits implementation work, joins plans, and pauses for human review.",
src/visualizer/studio-templates.ts:137        systemSource: [
src/visualizer/studio-templates.ts:138          ...baseMetadata("studio.template.zh.dev.team", "product-manager"),
src/visualizer/studio-templates.ts:139          "%% role.mode.solution-architect=parallel_split",
src/visualizer/studio-templates.ts:140          "%% route.order.solution-architect=backend-engineer,frontend-engineer,qa-engineer",
src/visualizer/studio-templates.ts:141          "%% join.mode.delivery-lead=all_of",
src/visualizer/studio-templates.ts:142          "%% join.sources.delivery-lead=backend-engineer,frontend-engineer,qa-engineer",
```

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src/visualizer/studio-templates.ts:143          "%% review.mode.delivery-lead=required",
src/visualizer/studio-templates.ts:144          "%% review.timeout.delivery-lead=86400",
src/visualizer/studio-templates.ts:145          "%% review.timeout.action.delivery-lead=pause",
src/visualizer/studio-templates.ts:146          "%% review.rework.target.delivery-lead=delivery-lead",
src/visualizer/studio-templates.ts:147          "%% review.rework.max.delivery-lead=2",
src/visualizer/studio-templates.ts:148          "%% review.terminate.scope.delivery-lead=branch",
src/visualizer/studio-templates.ts:149          "%% context.map.delivery-lead.backend_plan=source(backend-engineer).content",
src/visualizer/studio-templates.ts:150          "%% context.map.delivery-lead.frontend_plan=source(frontend-engineer).content",
src/visualizer/studio-templates.ts:151          "%% context.map.delivery-lead.qa_plan=source(qa-engineer).content",
src/visualizer/studio-templates.ts:152          "%% context.map.delivery-lead.project_goal=global.task",
src/visualizer/studio-templates.ts:153          "%% context.map.delivery-lead.review_comment=global.human_review.current.comment?",
src/visualizer/studio-templates.ts:154          "input -->|PROJECT_REQUEST| a[Role:product-manager]",
src/visualizer/studio-templates.ts:155          "a[Role:product-manager] -->|REQUIREMENTS_READY| b[Role:solution-architect]",
src/visualizer/studio-templates.ts:156          "b[Role:solution-architect] -->|START_BACKEND| c[Role:backend-engineer]",
src/visualizer/studio-templates.ts:157          "b[Role:solution-architect] -->|START_FRONTEND| d[Role:frontend-engineer]",
src/visualizer/studio-templates.ts:158          "b[Role:solution-architect] -->|START_QA| e[Role:qa-engineer]",
src/visualizer/studio-templates.ts:159          "c[Role:backend-engineer] -->|BACKEND_READY| f[Role:delivery-lead]",
src/visualizer/studio-templates.ts:160          "d[Role:frontend-engineer] -->|FRONTEND_READY| f[Role:delivery-lead]",
src/visualizer/studio-templates.ts:161          "e[Role:qa-engineer] -->|QA_READY| f[Role:delivery-lead]",
src/visualizer/studio-templates.ts:162          "f[Role:delivery-lead] -->|PLAN_READY| output",
src/visualizer/studio-templates.ts:163          ""
src/visualizer/studio-templates.ts:164        ].join("\n")
src/visualizer/studio-templates.ts:165      }
src/visualizer/studio-templates.ts:166    ];
src/visualizer/studio-templates.ts:167
src/visualizer/studio-templates.ts:168    export function listStudioAuthoringTemplates(): StudioAuthoringTemplate[] {
src/visualizer/studio-templates.ts:169      return templates.map((template) => ({ ...template }));
src/visualizer/studio-templates.ts:170    }
src/visualizer/studio-templates.ts:171
src/visualizer/studio-templates.ts:172    export function createStudioAuthoringFromTemplate(args: {
src/visualizer/studio-templates.ts:173      templateId: StudioTemplateId;
src/visualizer/studio-templates.ts:174      workdir: string;
src/visualizer/studio-templates.ts:175      systemPath: string;
src/visualizer/studio-templates.ts:176    }): StudioAuthoringDocument {
src/visualizer/studio-templates.ts:177      const template = templates.find((item) => item.id === args.templateId);
src/visualizer/studio-templates.ts:178      if (!template) {
src/visualizer/studio-templates.ts:179        throw new Error(`Unknown Studio template: ${args.templateId}`);
src/visualizer/studio-templates.ts:180      }
src/visualizer/studio-templates.ts:181      return importMermaidToAuthoring({
src/visualizer/studio-templates.ts:182        workdir: args.workdir,
src/visualizer/studio-templates.ts:183        systemPath: args.systemPath,
src/visualizer/studio-templates.ts:184        systemSource: template.systemSource
src/visualizer/studio-templates.ts:185      });
src/visualizer/studio-templates.ts:186    }
src/visualizer/studio-templates.ts:187
src/visualizer/studio-templates.ts:188    export function createStudioAuthoringFromMermaidDraft(args: {
src/visualizer/studio-templates.ts:189      workdir: string;
src/visualizer/studio-templates.ts:190      systemPath: string;
src/visualizer/studio-templates.ts:191      systemSource: string;
src/visualizer/studio-templates.ts:192    }): StudioAuthoringDocument {
```

```
src/visualizer/studio-templates.ts:193      return importMermaidToAuthoring(args);
src/visualizer/studio-templates.ts:194    }
src/visualizer/studio-templates.ts:195
src/visualizer/topology-order.ts:1        import type { GraphViewModelEdge, GraphViewModelNode } from "./studio-contracts.js";
src/visualizer/topology-order.ts:2
src/visualizer/topology-order.ts:3        type TopologyEdgeOrderArgs = {
src/visualizer/topology-order.ts:4          nodes: readonly GraphViewModelNode[];
src/visualizer/topology-order.ts:5          edges: readonly GraphViewModelEdge[];
src/visualizer/topology-order.ts:6          entryRoleId?: string;
src/visualizer/topology-order.ts:7        };
src/visualizer/topology-order.ts:8
src/visualizer/topology-order.ts:9        export function topologyComponentIds(args: TopologyEdgeOrderArgs): ReadonlyMap<string, string> {
src/visualizer/topology-order.ts:10         const nodeIds = args.nodes.map((node) => node.id).sort();
src/visualizer/topology-order.ts:11         const nodeSet = new Set(nodeIds);
src/visualizer/topology-order.ts:12         const outgoing = new Map(nodeIds.map((id) => [id, [] as string[]]));
src/visualizer/topology-order.ts:13         for (const edge of args.edges) {
src/visualizer/topology-order.ts:14           if (nodeSet.has(edge.source) && nodeSet.has(edge.target)) outgoing.get(edge.source)?.push(edge.target);
src/visualizer/topology-order.ts:15         }
src/visualizer/topology-order.ts:16         for (const targets of outgoing.values()) targets.sort();
src/visualizer/topology-order.ts:17         let index = 0;
src/visualizer/topology-order.ts:18         const indices = new Map<string, number>();
src/visualizer/topology-order.ts:19         const lowLinks = new Map<string, number>();
src/visualizer/topology-order.ts:20         const stack: string[] = [];
src/visualizer/topology-order.ts:21         const onStack = new Set<string>();
src/visualizer/topology-order.ts:22         const components: string[][] = [];
src/visualizer/topology-order.ts:23         const visit = (id: string): void => {
src/visualizer/topology-order.ts:24           indices.set(id, index);
src/visualizer/topology-order.ts:25           lowLinks.set(id, index);
src/visualizer/topology-order.ts:26           index += 1;
src/visualizer/topology-order.ts:27           stack.push(id);
src/visualizer/topology-order.ts:28           onStack.add(id);
src/visualizer/topology-order.ts:29           for (const target of outgoing.get(id) ?? []) {
src/visualizer/topology-order.ts:30             if (!indices.has(target)) {
src/visualizer/topology-order.ts:31               visit(target);
src/visualizer/topology-order.ts:32               lowLinks.set(id, Math.min(lowLinks.get(id)!, lowLinks.get(target)!));
src/visualizer/topology-order.ts:33             } else if (onStack.has(target)) {
src/visualizer/topology-order.ts:34               lowLinks.set(id, Math.min(lowLinks.get(id)!, indices.get(target)!));
src/visualizer/topology-order.ts:35             }
src/visualizer/topology-order.ts:36           }
src/visualizer/topology-order.ts:37           if (lowLinks.get(id) !== indices.get(id)) return;
src/visualizer/topology-order.ts:38           const component: string[] = [];
src/visualizer/topology-order.ts:39           let member = "";
src/visualizer/topology-order.ts:40           do {
src/visualizer/topology-order.ts:41             member = stack.pop()!;
src/visualizer/topology-order.ts:42             onStack.delete(member);
src/visualizer/topology-order.ts:43             component.push(member);
src/visualizer/topology-order.ts:44           } while (member !== id);
src/visualizer/topology-order.ts:45           components.push(component.sort());
src/visualizer/topology-order.ts:46         };
src/visualizer/topology-order.ts:47         for (const id of nodeIds) if (!indices.has(id)) visit(id);
```

```
src/visualizer/topology-order.ts:48         components.sort((left, right) => (left[0] ?? "").localeCompare(right[0] ?? ""));
src/visualizer/topology-order.ts:49         const result = new Map<string, string>();
src/visualizer/topology-order.ts:50         components.forEach((component, componentIndex) => {
src/visualizer/topology-order.ts:51           const cyclic = component.length > 1 || args.edges.some((edge) => edge.source === component[0] && edge.target === component[0]);
src/visualizer/topology-order.ts:52           if (cyclic) component.forEach((id) => result.set(id, `SCC-${componentIndex + 1}`));
src/visualizer/topology-order.ts:53         });
src/visualizer/topology-order.ts:54         return result;
src/visualizer/topology-order.ts:55       }
src/visualizer/topology-order.ts:56
src/visualizer/topology-order.ts:57       function stableEdgeKey(edge: GraphViewModelEdge): string {
src/visualizer/topology-order.ts:58         return `${edge.source}:${edge.target}:${edge.eventType}:${edge.id}`;
src/visualizer/topology-order.ts:59       }
src/visualizer/topology-order.ts:60
src/visualizer/topology-order.ts:61       /**
src/visualizer/topology-order.ts:62        * Compresses cycles before assigning one global order to every valid flow.
src/visualizer/topology-order.ts:63        * Forward handoffs are numbered first; cycle edges are numbered afterwards
src/visualizer/topology-order.ts:64        * with an `L` suffix so the main path remains easy to scan.
src/visualizer/topology-order.ts:65        */
src/visualizer/topology-order.ts:66       export function addTopologyFlowOrder(args: TopologyEdgeOrderArgs): GraphViewModelEdge[] {
src/visualizer/topology-order.ts:67         const nodeIds = args.nodes.map((node) => node.id);
src/visualizer/topology-order.ts:68         const nodeSet = new Set(nodeIds);
src/visualizer/topology-order.ts:69         const edges = args.edges.filter((edge) => nodeSet.has(edge.source) && nodeSet.has(edge.target));
src/visualizer/topology-order.ts:70         const outgoing = new Map(nodeIds.map((id) => [id, [] as GraphViewModelEdge[]]));
src/visualizer/topology-order.ts:71         for (const edge of edges) outgoing.get(edge.source)?.push(edge);
src/visualizer/topology-order.ts:72         for (const list of outgoing.values()) list.sort((a, b) => stableEdgeKey(a).localeCompare(stableEdgeKey(b)));
src/visualizer/topology-order.ts:73
src/visualizer/topology-order.ts:74         let index = 0;
src/visualizer/topology-order.ts:75         const indices = new Map<string, number>();
src/visualizer/topology-order.ts:76         const lowLinks = new Map<string, number>();
src/visualizer/topology-order.ts:77         const stack: string[] = [];
src/visualizer/topology-order.ts:78         const onStack = new Set<string>();
src/visualizer/topology-order.ts:79         const components: string[][] = [];
src/visualizer/topology-order.ts:80         const visit = (id: string): void => {
src/visualizer/topology-order.ts:81           indices.set(id, index);
src/visualizer/topology-order.ts:82           lowLinks.set(id, index);
src/visualizer/topology-order.ts:83           index += 1;
src/visualizer/topology-order.ts:84           stack.push(id);
src/visualizer/topology-order.ts:85           onStack.add(id);
src/visualizer/topology-order.ts:86           for (const edge of outgoing.get(id) ?? []) {
src/visualizer/topology-order.ts:87             if (!indices.has(edge.target)) {
src/visualizer/topology-order.ts:88               visit(edge.target);
src/visualizer/topology-order.ts:89               lowLinks.set(id, Math.min(lowLinks.get(id)!, lowLinks.get(edge.target)!));
src/visualizer/topology-order.ts:90             } else if (onStack.has(edge.target)) {
src/visualizer/topology-order.ts:91               lowLinks.set(id, Math.min(lowLinks.get(id)!, indices.get(edge.target)!));
src/visualizer/topology-order.ts:92             }
src/visualizer/topology-order.ts:93           }
src/visualizer/topology-order.ts:94           if (lowLinks.get(id) !== indices.get(id)) return;
src/visualizer/topology-order.ts:95           const component: string[] = [];
src/visualizer/topology-order.ts:96           let member = "";
src/visualizer/topology-order.ts:97           do {
```

```
src/visualizer/topology-order.ts:98             member = stack.pop()!;
src/visualizer/topology-order.ts:99             onStack.delete(member);
src/visualizer/topology-order.ts:100            component.push(member);
src/visualizer/topology-order.ts:101          } while (member !== id);
src/visualizer/topology-order.ts:102          components.push(component.sort());
src/visualizer/topology-order.ts:103        };
src/visualizer/topology-order.ts:104        for (const id of nodeIds.slice().sort()) if (!indices.has(id)) visit(id);
src/visualizer/topology-order.ts:105
src/visualizer/topology-order.ts:106        const componentByNode = new Map<string, number>();
src/visualizer/topology-order.ts:107        components.forEach((component, componentIndex) => component.forEach((id) => componentByNode.set(id, componentIndex)));
src/visualizer/topology-order.ts:108        const componentEdges = new Map<number, Set<number>>();
src/visualizer/topology-order.ts:109        const indegree = new Map<number, number>();
src/visualizer/topology-order.ts:110        components.forEach((_, componentIndex) => {
src/visualizer/topology-order.ts:111          componentEdges.set(componentIndex, new Set());
src/visualizer/topology-order.ts:112          indegree.set(componentIndex, 0);
src/visualizer/topology-order.ts:113        });
src/visualizer/topology-order.ts:114        for (const edge of edges) {
src/visualizer/topology-order.ts:115          const source = componentByNode.get(edge.source)!;
src/visualizer/topology-order.ts:116          const target = componentByNode.get(edge.target)!;
src/visualizer/topology-order.ts:117          if (source === target || componentEdges.get(source)!.has(target)) continue;
src/visualizer/topology-order.ts:118          componentEdges.get(source)!.add(target);
src/visualizer/topology-order.ts:119          indegree.set(target, indegree.get(target)! + 1);
src/visualizer/topology-order.ts:120        }
src/visualizer/topology-order.ts:121
src/visualizer/topology-order.ts:122        const componentName = (componentIndex: number): string => components[componentIndex]?.[0] ?? "";
src/visualizer/topology-order.ts:123        const entryComponent = componentByNode.get(args.entryRoleId ?? "input");
src/visualizer/topology-order.ts:124        const componentDistance = new Map<number, number>();
src/visualizer/topology-order.ts:125        if (entryComponent !== undefined) {
src/visualizer/topology-order.ts:126          componentDistance.set(entryComponent, 0);
src/visualizer/topology-order.ts:127          const pending = [entryComponent];
src/visualizer/topology-order.ts:128          for (let cursor = 0; cursor < pending.length; cursor += 1) {
src/visualizer/topology-order.ts:129            const current = pending[cursor]!;
src/visualizer/topology-order.ts:130            const distance = componentDistance.get(current) ?? 0;
src/visualizer/topology-order.ts:131            for (const target of componentEdges.get(current) ?? []) {
src/visualizer/topology-order.ts:132              if (componentDistance.has(target)) continue;
src/visualizer/topology-order.ts:133              componentDistance.set(target, distance + 1);
src/visualizer/topology-order.ts:134              pending.push(target);
src/visualizer/topology-order.ts:135            }
src/visualizer/topology-order.ts:136          }
src/visualizer/topology-order.ts:137        }
src/visualizer/topology-order.ts:138        const compareReadyComponents = (left: number, right: number): number => {
src/visualizer/topology-order.ts:139          const leftDistance = componentDistance.get(left);
src/visualizer/topology-order.ts:140          const rightDistance = componentDistance.get(right);
src/visualizer/topology-order.ts:141          if (leftDistance === undefined && rightDistance !== undefined) return 1;
src/visualizer/topology-order.ts:142          if (leftDistance !== undefined && rightDistance === undefined) return -1;
src/visualizer/topology-order.ts:143          if (leftDistance !== undefined && rightDistance !== undefined && leftDistance !== rightDistance) {
src/visualizer/topology-order.ts:144            return leftDistance - rightDistance;
src/visualizer/topology-order.ts:145          }
src/visualizer/topology-order.ts:146          return componentName(left).localeCompare(componentName(right));
src/visualizer/topology-order.ts:147        };
```

```
src/visualizer/topology-order.ts:148       const queue = components.map((_, componentIndex) => componentIndex)
src/visualizer/topology-order.ts:149         .filter((componentIndex) => indegree.get(componentIndex) === 0)
src/visualizer/topology-order.ts:150         .sort(compareReadyComponents);
src/visualizer/topology-order.ts:151       const orderedComponents: number[] = [];
src/visualizer/topology-order.ts:152       for (let cursor = 0; cursor < queue.length; cursor += 1) {
src/visualizer/topology-order.ts:153         const current = queue[cursor]!;
src/visualizer/topology-order.ts:154         orderedComponents.push(current);
src/visualizer/topology-order.ts:155         for (const target of componentEdges.get(current) ?? []) {
src/visualizer/topology-order.ts:156           indegree.set(target, indegree.get(target)! - 1);
src/visualizer/topology-order.ts:157           if (indegree.get(target) === 0) {
src/visualizer/topology-order.ts:158             queue.push(target);
src/visualizer/topology-order.ts:159             queue.sort(compareReadyComponents);
src/visualizer/topology-order.ts:160           }
src/visualizer/topology-order.ts:161         }
src/visualizer/topology-order.ts:162       }
src/visualizer/topology-order.ts:163
src/visualizer/topology-order.ts:164       // Defensive fallback for malformed disconnected condensation graphs. A
src/visualizer/topology-order.ts:165       // valid condensation DAG is fully emitted by Kahn's algorithm above.
src/visualizer/topology-order.ts:166       for (const componentIndex of components.map((_, index) => index)) {
src/visualizer/topology-order.ts:167         if (!orderedComponents.includes(componentIndex)) orderedComponents.push(componentIndex);
src/visualizer/topology-order.ts:168       }
src/visualizer/topology-order.ts:169       const orderedComponentIndex = new Map(orderedComponents.map((component, index) => [component, index]));
src/visualizer/topology-order.ts:170       const forwardEdges: GraphViewModelEdge[] = [];
src/visualizer/topology-order.ts:171       const cycleEdges: GraphViewModelEdge[] = [];
src/visualizer/topology-order.ts:172       for (const componentIndex of orderedComponents) {
src/visualizer/topology-order.ts:173         const componentEdgesForSource = edges
src/visualizer/topology-order.ts:174           .filter((edge) => componentByNode.get(edge.source) === componentIndex)
src/visualizer/topology-order.ts:175           .sort((left, right) => stableEdgeKey(left).localeCompare(stableEdgeKey(right)));
src/visualizer/topology-order.ts:176         for (const edge of componentEdgesForSource) {
src/visualizer/topology-order.ts:177           if (componentByNode.get(edge.target) === componentIndex) cycleEdges.push(edge);
src/visualizer/topology-order.ts:178           else forwardEdges.push(edge);
src/visualizer/topology-order.ts:179         }
src/visualizer/topology-order.ts:180       }
src/visualizer/topology-order.ts:181       cycleEdges.sort((left, right) => {
src/visualizer/topology-order.ts:182         const leftComponent = orderedComponentIndex.get(componentByNode.get(left.source)! ) ?? 0;
src/visualizer/topology-order.ts:183         const rightComponent = orderedComponentIndex.get(componentByNode.get(right.source)! ) ?? 0;
src/visualizer/topology-order.ts:184         return leftComponent - rightComponent || stableEdgeKey(left).localeCompare(stableEdgeKey(right));
src/visualizer/topology-order.ts:185       });
src/visualizer/topology-order.ts:186       const orderByEdge = new Map<GraphViewModelEdge, string>();
src/visualizer/topology-order.ts:187       [...forwardEdges, ...cycleEdges].forEach((edge, index) => {
src/visualizer/topology-order.ts:188         const component = componentByNode.get(edge.source);
src/visualizer/topology-order.ts:189         const isCycle = component !== undefined && component === componentByNode.get(edge.target);
src/visualizer/topology-order.ts:190         orderByEdge.set(edge, `${index + 1}${isCycle ? "L" : ""}`);
src/visualizer/topology-order.ts:191       });
src/visualizer/topology-order.ts:192       return args.edges.map((edge) => {
src/visualizer/topology-order.ts:193         const topologyOrder = orderByEdge.get(edge);
src/visualizer/topology-order.ts:194         return topologyOrder ? { ...edge, topologyOrder } : edge;
src/visualizer/topology-order.ts:195       });
src/visualizer/topology-order.ts:196     }
src/visualizer/topology-order.ts:197
```